
INEQUALITIES

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1. ABSOLUTE VALUE & INEQUALITIES

1.1 Absolute Value

A. Definition

1.1: Absolute Value

The value of a number without regard to its sign is called its absolute value.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Example 1.2

Find the exact value of $y = |7| + |-3| + \left| -3\frac{1}{4} \right| + |-\pi|$

$$\begin{aligned} |7| &= 7, & |-3| &= 3, & \left| -3\frac{1}{4} \right| &= 3\frac{1}{4}, & |-\pi| &= \pi \\ y &= 7 + 3 + 3\frac{1}{4} + \pi = 13\frac{1}{4} + \pi \end{aligned}$$

1.3: Property

$$|x| = |-x|$$

$$|7 - 5| = 2 = |5 - 7|$$

Example 1.4

Given that e is an irrational number with approximate 2.71, and π is an irrational number with approximate value 3.14, find the exact value of $|e - \pi|$

$$|e - \pi| = -(e - \pi) = \pi - e$$

Example 1.5

Find the value of $|x - y|$ if:

- A. x and y are positive integers such that $x > y$
- B. x and y are positive integers such that $x < y$
- C. x and y are negative integers such that $x > y$
- D. x and y are negative integers such that $x < y$

$$\text{Part A \& Part C: } |x - y| = x - y$$

$$\text{Part B \& Part D: } |x - y| = y - x$$

$$x = -3, y = -5 \Rightarrow -3 - (-5) = -3 + 5 = 2 = +ve$$

$$x = -3, y = -2 \Rightarrow -3 - (-2) = -3 + 2 = -1 = -ve$$

B. Expressions

Example 1.6: Calculating Values by Substitution

Prove that

Example 1.7: Simplification

Prove that

Example 1.8

Nesting means absolute value inside absolute value. Like brackets (or parentheses), these are solved inside out. Start from the innermost expression, and keep simplifying.

Simplify

A. $|-3 + |2 - 3|| = |-3 + 1| = |-2| = 2$

B. $\left|1 - \left|2 - \left|3 - \left|4 - \left|5 - \left|6 - 7\right|\right|\right|\right|\right|\right|$

$$\left|-\frac{1}{3} + \left|-\frac{2}{3} + \frac{1}{4}\right|\right| = \left|-\frac{1}{3} + \frac{3}{12}\right| = \left|-\frac{1}{12}\right| = \frac{1}{12}$$

$$\left|1 - \left|2 - \left|3 - \left|4 - \left|5 - 1\right|\right|\right|\right|\right| = \left|1 - \left|2 - \left|3 - \left|4 - 4\right|\right|\right|\right| = \left|1 - \left|2 - \left|3 - 0\right|\right|\right| = \left|1 - \left|2 - 3\right|\right| = \left|1 - |-1|\right| = 0$$

C. Distance Interpretation

1.9: Distance Interpretation

Distance between x and y on the Number Line:

= Absolute Value of Difference of x and y

We can convert this into a formula by writing:

$$\text{Distance} = |x - y| = |y - x|$$

- The order of the two numbers is not important, since we take the absolute value. Both the expressions give the same answer.
- The absolute value is important because distance can never be negative.

Example 1.10

Find the distance between 3 and -4 on the number line.

Graphical Method

Draw a number line.



Count the distance from 0 to 3, which is

3 Units right from zero

Count the distance from 0 to -4 , which is:

4 Units left from zero

And hence, the total distance is:

$$3 + 4 = 7$$

Another calculate the same distance is to calculate the distance of zero from both points.

$$\text{Total Distance} = D(3,0) \text{ and } D(0,-4) = 3 + 4 = 7$$

Algebraic Method

Distance between two points on the Number Line

$$\begin{aligned} &= \text{Absolute Value of Difference of the two numbers} \\ &= |3 - (-4)| = |3 + 4| = |7| = 7 \end{aligned}$$

Equivalently, we can also do:

$$\text{Distance} = |-4 - 3| = |-7| = 7$$

Example 1.11

In each part, find the distance between the numbers on the number line:

- A. 1, 3
- B. -1, 3
- C. -1, -3
- D. 1, -3

Part A

We want to find the distance between 1 and 3 on the number line (graphed alongside).



Substitute $x = 1, y = 3$ in $|x - y|$

$$|3 - 1| = |2| = 2$$

Substitute $x = 1, y = 3$ in $|y - x|$

$$|1 - 3| = |-2| = 2$$

Part B

Substitute $x = -1, y = 3$ in $|x - y|$

$$|-1 - 3| = |-4| = 4$$



Substitute $x = -1, y = 3$ in $|y - x|$

$$|3 - (-1)| = |3 + 1| = 4$$

Part C

Substitute $x = -1, y = -3$ in $|x - y|$

$$|-1 - (-3)| = |-1 + 3| = |2| = 2$$



Part D

Substitute $x = 1, y = -3$ in $|x - y|$

$$|1 - (-3)| = |1 + 3| = 4$$



Example 1.12

Find the distance between the given points on the number line:

- A. 2, 5
- B. -3, 7
- C. 4, -5
- D. -1, -8
- E. $\frac{1}{2}, \frac{1}{3}$
- F. $\frac{2}{3}, -\frac{1}{5}$
- G. $-\frac{1}{4}, \frac{1}{3}$

H. $-\frac{1}{3}, -\frac{1}{5}$

$$\begin{aligned} \left| \frac{1}{2} - \frac{1}{3} \right| &= \left| \frac{3}{6} - \frac{2}{6} \right| = \left| \frac{1}{6} \right| = \frac{1}{6} \\ \left| \frac{2}{3} - \left(-\frac{1}{5} \right) \right| &= \left| \frac{10}{15} + \frac{3}{15} \right| = \left| \frac{13}{15} \right| = \frac{13}{15} \\ \left| -\frac{1}{4} - \frac{1}{3} \right| &= \left| -\frac{3}{12} - \frac{4}{12} \right| = \left| -\frac{7}{12} \right| = \frac{7}{12} \end{aligned}$$

Example 1.13: Horizontal and Vertical Distance on the Co-ordinate Plane

Given points $A(3,4)$, $B(3,9)$ and $C(5,4)$. Find the sum of the distances from A to B , and A to C .

$$D(A, B) = |4 - 9| = |-5| = 5$$

$$D(A, C) = |3 - 5| = |-2| = 2$$

$$D(A, B) + D(A, C) = 5 + 2$$

Example 1.14: Creating Expressions

- The number x is 2 units away from 1 on the number line. Draw a number line and use it to find the possible values of x . Use the formula for distance on the number to get an absolute value expression for x .
- The maximum number of guests that I expect at my party is 6. The minimum number of guests that I expect is 2. Represent the max, min and average number of guests on a number line. Use the average that you found to create an absolute value expression that gives the maximum and minimum number of guests.

Part A



Distance on the number line between two points x and y is given by

$$|x - y|$$

Substitute $y = 1$ in the above expression:

$$|x - 1| = 2$$

Part B

$$\text{Average} = \frac{6 + 2}{2} = \frac{8}{2} = 4$$



$$|g - 4| = 2$$

D. Equations

1.15: Removing the absolute value sign

Every time, you remove an absolute value sign, you must solve two separate equations.

$$|x| = n \Leftrightarrow x = \pm n$$

For example, we can say that:

$$|x| = 6 \Leftrightarrow x = \pm 6$$

Example 1.16

Solve the following equations algebraically. Also solve them graphically.

- A. $|x| = 5$
- B. $|a| = \frac{3}{4}$
- C. $|b| = 0.4$
- D. $|c| = 0$
- E. $|e| = -1$

Part A

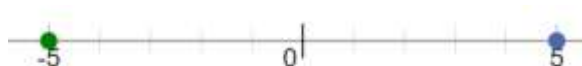
Algebraically

$$|x| = 5 \Rightarrow x = \pm 5 \Rightarrow x \in \{-5, +5\}$$

Graphically

The absolute value of a number represents its distance from the origin. Hence, we want numbers that are a distance of 5 units away from the origin.

And this is given by:



Part E

Algebraically

The absolute value of a number can never be negative.

Hence, the equation has no solutions.

Graphically

Absolute value represents the distance from the origin.

And distance can never be negative.

Hence, the equation has no solution.

Example 1.17

The absolute value of a number is double of the number itself. Identify the number.

$$|x| = 2x$$

The only number that will satisfy is

$$x = 0$$

Example 1.18: Absolute Value Expressions

If you have an expression inside the absolute value sign, the strategy remains the same.

Solve the equation. Check your answers by substituting the values.

Integers

- A. $|x + 3| = 4$
- B. $|a + 2| = 5$
- C. $|y + 1| = 4$
- D. $|b - 2| = 4$
- E. $|c + 3| = 10$
- F. $|z - 3| = 5$

Decimals

- G. $|s + 1.2| = 5.4$
- H. $|t - 0.3| = 0.7$

Fractions

$$\begin{array}{l} \text{I. } \left| p + \frac{1}{2} \right| = \frac{3}{4} \\ \text{J. } \left| s - \frac{2}{3} \right| = \frac{1}{5} \end{array}$$

Integers

$$x + 3 = \pm 4 \Rightarrow x = \pm 4 - 3 \Rightarrow x \in \{-7, 1\}$$

belongs to

Substitute the values to check:

$$x = -7 \Rightarrow LHS = |x + 3| = |-7 + 3| = |-4| = 4 = RHS$$

$$x = 1 \Rightarrow LHS = |1 + 3| = |1 + 3| = |4| = 4 = RHS$$

$$|a + 2| = 5 \Rightarrow a + 2 = \pm 5 \Rightarrow a \pm 5 - 2 \Rightarrow a \in \{-7, 3\}$$

$$|y + 1| = 4 \Rightarrow y + 1 = \pm 4 \Rightarrow y = \pm 4 - 1 \Rightarrow y \in \{-5, 3\}$$

$$|b - 2| = 4 \Rightarrow$$

$$y + 1 = \pm 4 \Rightarrow y = \pm 4 - 1 \Rightarrow y = \{3, -5\}$$

Decimals

$$s + 1.2 = \pm 5.4 \Rightarrow s \in \{-6.6, 4.2\}$$

Fractions

Example 1.19: Isolating Absolute Values

In solving linear equations, we isolate the variable to find its value. Similarly, if we are able to isolate the absolute value on one side of the equation, then we can apply the above property.

Find the values of z that satisfy $\left| z - \frac{1}{2} \right| - \frac{1}{3} = 0$

$$\left| z - \frac{1}{2} \right| = \frac{1}{3} \Rightarrow z - \frac{1}{2} = \pm \frac{1}{3} \Rightarrow z \in \left\{ \frac{1}{6}, \frac{5}{6} \right\}$$

Example 1.20: Variable Terms outside the Absolute Value

The equations we have seen so far have had the variable inside the absolute value. The next example shows how to handle variables outside the absolute value sign. The general idea remains the same: Isolate the absolute value expression on one side of the equation.

Solve

A. $x + |x + 1| = 3$

B. $2x + |3x - 5| = 4$

Part A

Isolate the absolute value sign:

$$|x + 1| = 3 - x$$

Apply the property $|x| = a \Rightarrow x = \pm a$:

$$x + 1 = \pm(3 - x) \Rightarrow x = \pm(3 - x) - 1$$

Case I: Positive Sign

$$x = 3 - x - 1 \Rightarrow x = 1 \Rightarrow \text{Works}$$

Case II: Negative Sign

$$x = -(3 - x) - 1 \Rightarrow x = -3 + x - 1 \Rightarrow 0 = -4 \Rightarrow x = \phi (\text{No solutions})$$

Hence, the final answer is:

$$x = 1$$

Checking for Validity of Solutions

In these kinds of questions, it is important to go back and check by substituting the candidate solutions in the original equation to see if they satisfy it.

In the question, we got two candidate solutions, but only one satisfied the original equation.

Part B

$$|3x - 5| = 4 - 2x \Rightarrow 3x - 5 = \pm(4 - 2x)$$

Case I

$$3x - 5 = 4 - 2x \Rightarrow 5x = 9 \Rightarrow x = \frac{9}{5}$$

Case II

$$3x - 5 = -(4 - 2x) \Rightarrow 3x - 5 = -4 + 2x \Rightarrow x = 1$$

Example 1.21: Two Absolute Value Expressions

Put one absolute value expression on one side, and the other absolute expression on the other side

Consider the equation

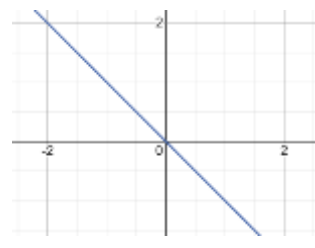
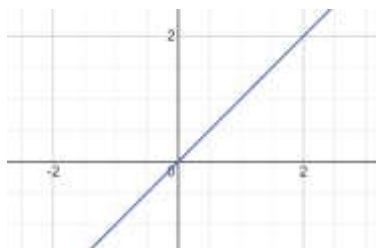
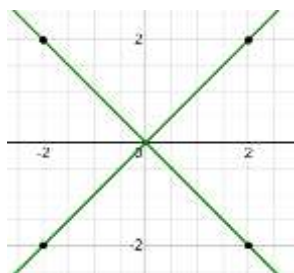
$$|y| = |x|$$

- Consider the four points: $(2, 2)$, $(-2, 2)$, $(2, -2)$, $(-2, -2)$ and show that these points satisfy the equation.
- Consider the four points: (n, n) , $(-n, n)$, $(n, -n)$, $(-n, -n)$ and show that these points satisfy the equation.
- Hence, graph the equation. Break the graph of the equation into two linear graphs, and write the equations of those graphs.

Part A

	x	y	$ x $	$ y $
	2	2	2	2
	-2	2	2	2
	2	-2	2	2
	-2	-2	2	2

Part B



We can break the solution into two parts:

$$y = x \text{ OR } y = -x$$

And the above solutions can be combined into a single solution by writing

$$y = \pm x$$

$$|y| = |x| \Leftrightarrow y = \pm x$$

Example 1.22

Solve

- A. $|2x + 4| = |3x - 5|$
- B. $|3x + 2| = |5x - 7|$
- C. $\left|\frac{3}{2}x + \frac{5}{7}\right| = \left|\frac{2}{5}x - \frac{5}{3}\right|$

Part A

$$\text{Case I: } 2x + 4 = +(3x - 5) \Rightarrow x = 9$$

$$\text{Case II: } 2x + 4 = -(3x - 5) \Rightarrow 5x = 1 \Rightarrow x = \frac{1}{5}$$

Part B

$$\text{Case I: } 3x + 2 = 5x - 7 \Rightarrow 2x = 9 \Rightarrow x = \frac{9}{2}$$

$$\text{Case II: } 3x + 2 = -5x + 7 \Rightarrow 8x = 5 \Rightarrow x = \frac{5}{8}$$

Part C

Case I: Take the positive sign

$$\frac{3}{2}x + \frac{5}{7} = +\left(\frac{2}{5}x - \frac{5}{3}\right)$$

Collect like terms:

$$\frac{3}{2}x - \frac{2}{5}x = -\frac{5}{3} - \frac{5}{7}$$

Take the LCM:

$$\frac{15}{10}x - \frac{4}{10}x = -\frac{35}{21} - \frac{15}{21}$$

Simplify:

$$\frac{11}{10}x = -\frac{50}{21}$$

Divide both sides $\frac{11}{10}$:

$$x = -\frac{50}{21} \div \frac{11}{10} = -\frac{50}{21} \times \frac{10}{11} = -\frac{500}{231}$$

Case II: Take the negative sign

$$\frac{3}{2}x + \frac{5}{7} = -\left(\frac{2}{5}x - \frac{5}{3}\right)$$

$$\frac{3}{2}x + \frac{5}{7} = -\frac{2}{5}x + \frac{5}{3}$$

$$\frac{3}{2}x + \frac{2}{5}x = \frac{5}{3} - \frac{5}{7}$$

$$\frac{15}{10}x + \frac{4}{10}x = \frac{35}{21} - \frac{15}{21}$$

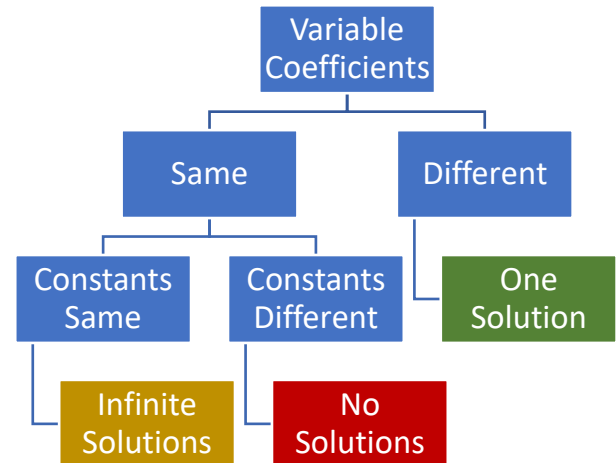
$$\frac{19}{10}x = +\frac{20}{21}$$

$$x = \frac{20}{21} \times \frac{10}{19} = \frac{200}{399}$$

E. No Solutions and Infinite Solutions

When the coefficients of x on both sides of an equation are the same, we get:

- No Solutions if the constants are different
- Infinite Solutions if the constants are the same



Example 1.23

Solve

$$|x + 4| - |x - 3| = 0$$

Rewrite this so the LHS has one absolute expression, and the RHS has one absolute value expression:

$$|x + 4| = |x - 3|$$

Use the property that $|y| = |x| \Leftrightarrow y = \pm x$:

$$x + 4 = \pm(x - 3)$$

Case I: Take the positive sign

$$x + 4 = +(x - 3) \Rightarrow x + 4 = x - 3 \Rightarrow +4 = -3 \Rightarrow \text{No Solutions} \Rightarrow x = \phi$$

Case II: Take the negative sign

$$x + 4 = -(x - 3) \Rightarrow x + 4 = -x + 3 \Rightarrow x = -\frac{1}{2} \Rightarrow \text{Works}$$

Example 1.24

Solve

$$|2x + 4| - |-2x - 4| = 0$$

$$|2x + 4| = |-2x - 4|$$

Use the property that $|y| = |x| \Leftrightarrow y = \pm x$:

$$2x + 4 = \pm(-2x - 4)$$

Case I: Take the positive sign

$$2x + 4 = +(-2x - 4) \Rightarrow 2x + 4 = -2x - 4 \Rightarrow 4x = -8 \Rightarrow x = -2 \Rightarrow \text{Works}$$

Case II: Take the negative sign

$$2x + 4 = -(-2x - 4) \Rightarrow 2x + 4 = 2x + 4 \Rightarrow 4 = 4 \Rightarrow \text{Infinite Solutions} \Rightarrow x \in \mathbb{R}$$

Example 1.25

$$|x + 5| = x + 5$$

Case I: $x + 5$ is $+ve$ or zero:

$$x + 5 = x + 5 \Rightarrow \text{Infinite Solutions}$$

$$x + 5 \geq 0 \Rightarrow x \geq -5$$

Case I: $x + 5$ is $-ve$ or zero:

$$-(x + 5) = x + 5 \Rightarrow -x - 5 = x + 5 \Rightarrow -10 = 2x \Rightarrow x = -5$$

$$x + 5 \leq 0 \Rightarrow x \leq -5$$

Hence, the final solution is the interval:

$$x \geq -5$$

F. Nested Absolute Value Equations

We have already seen nested value expressions. These are solved inside out. However, nested value equations are solved outside in.

We remove the outermost absolute value sign, to get two equations. Since, these equations are themselves absolute value equations, they need to be further solved, each with two cases.

Example 1.26

Solve

$$|x + |2x + 2|| = 5$$

Remove the Outermost sign

Use the property that $|y| = k \Rightarrow y = \pm k$

$$x + |2x + 2| = \pm 5$$

Take the Positive Sign

$$|2x + 2| = 5 - x \Rightarrow 2x + 2 = \pm(5 - x)$$

$$\text{Case IA: } 2x + 2 = 5 - x \Rightarrow x = \mathbf{1}$$

$$\text{Case IB: } 2x + 2 = x - 5 \Rightarrow x = \mathbf{-7}$$

Take the Negative Sign

$$x + |2x + 2| = -5 \Rightarrow |2x + 2| = -5 - x$$

$$\text{Case IIA: } 2x + 2 = -5 - x \Rightarrow x = \mathbf{-\frac{7}{3}}$$

$$\text{Case IIB: } 2x + 2 = 5 + x \Rightarrow x = \mathbf{3}$$

$$\text{Final Solution: } x = \{1, -7\}$$

G. Prime Solutions

Example 1.27

If sixteen is subtracted from the square of an integer, the absolute value of the difference is a prime.

Find all such numbers.

Requires proof that these are the only solutions

Set up the equation

$$|x^2 - 16| = p, \quad p \in \text{Prime Numbers}$$

Factoring the LHS

Factor the LHS as a difference of squares:

$$|x^2 - 16| = p \Rightarrow |(x + 4)(x - 4)| = p \Rightarrow (x + 4)(x - 4) = \pm p$$

Now, the RHS is a prime number, and can only be factored as shown in the table below:

	-p				+p	
(x + 4)	-1	1	p	-p	1	p
(x - 4)	p	-p	-1	1	p	1
Case	I	II	III	IV	V	VI

Case I, IV and V

In all situations, $x + 4$ is greater than $x - 4$.

A negative number cannot be more than a positive number, but this is precisely what happens in Case I and Case IV:

$$\text{Case I: } \underbrace{-1}_{\substack{\text{Negative} \\ \text{Number}}} > \underbrace{p}_{\substack{\text{Positive} \\ \text{Prime}}} \Rightarrow \text{Contradiction}$$

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$$\text{Case IV: } \underbrace{-p}_{\substack{\text{Negative} \\ \text{Prime}}} > \underbrace{1}_{\substack{\text{Positive} \\ \text{Integer}}} \Rightarrow \text{Contradiction}$$

In Case V, the smallest value that p can take is 2, but this is more than 1:

$$\text{Case V: } 1 > p \Rightarrow \text{Contradiction}$$

In general, if we have

$$x + 4 > x - 4 \Rightarrow \text{Contradiction}$$

Hence, we do not obtain any solutions from Cases I, IV and V.

Case VI

This case gives an integer solution, but the difference is not prime. So, it is not valid:

$$x - 4 = 1 \Rightarrow x = 5 \Rightarrow x^2 - 16 = 9 \Rightarrow \text{Not Prime} \Rightarrow \text{Not Valid}$$

Case II and III

These are the only two cases that result in valid solutions:

$$x + 4 = 1 \Rightarrow x = -3 \Rightarrow x^2 - 16 = -7 \Rightarrow |x^2 - 16| = 7 \Rightarrow \text{Valid}$$

$$x - 4 = -1 \Rightarrow x = 3 \Rightarrow x^2 - 16 = -7 \Rightarrow |x^2 - 16| = 7 \Rightarrow \text{Valid}$$

$$\text{Final Solution Set} = \{-3, 3\}$$

1.2 Quadratic Absolute Value

A. Graphs

B. Expressions

C. Equations

D. Range

1.3 Number Line

A. Representing Inequalities on a Number Line

We now look at representing inequalities on a number line. This is a key skill that is needed for inequality questions.

1.28: Inequalities on a Number Line

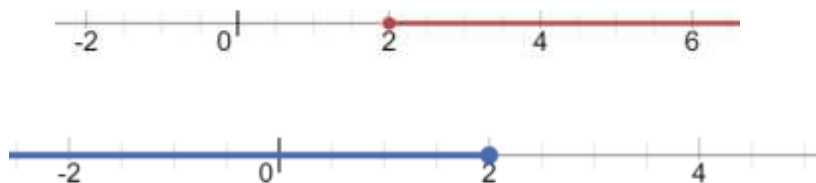
The region that satisfies the inequality is given a different color.

Example 1.29

Graph the inequalities below on a number line.

A. $x \geq 2$

B. $x \leq 2$



1.30: Endpoints

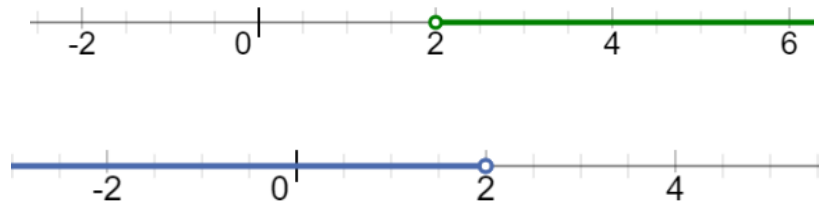
If the endpoints are included, then the circle is filled (as in the example above).

If the endpoints are not included, then the circle is empty (as in the example below).

Example 1.31

Graph the inequalities below on a number line.

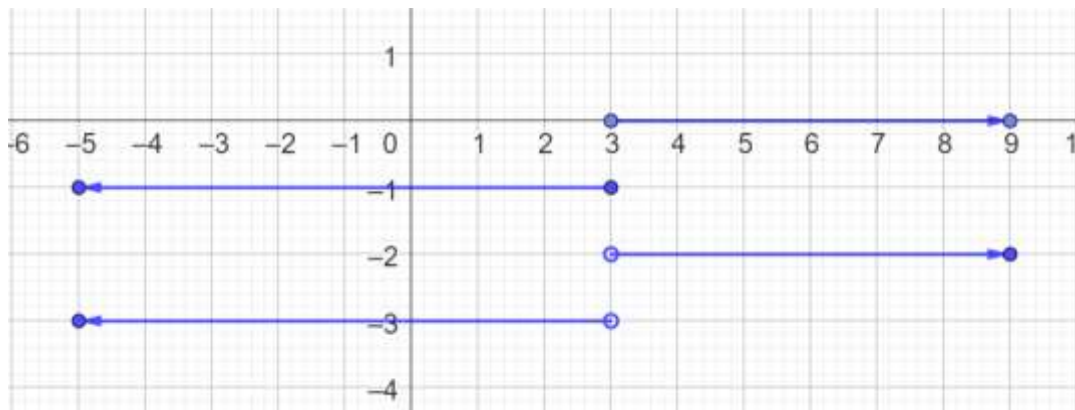
- A. $x > 2$
- B. $x < 2$



Example 1.32

Graph the following inequalities:

- A. $x \geq 3$
- B. $x \leq 3$
- C. $x > 3$
- D. $x < 3$



B. Compound Inequalities and Endpoints

A compound inequality has two inequalities put together. Consider the following inequalities in terms of n :

1.33: Endpoints

A compound inequality has four cases with respect to its endpoints.

$$\begin{array}{l} \underbrace{x < n < y}_{\text{Endpoints not included}} \\ \underbrace{x \leq n \leq y}_{\text{Endpoints included}} \\ \underbrace{x \leq n < y}_{\text{Left endpoint included, Right endpoint excluded}} \end{array}$$

$$\underbrace{x < n \leq y}_{\substack{\text{Left endpoint excluded} \\ \text{Right endpoint included}}}$$

Example 1.34

Graph the following inequalities:

$$\underbrace{3 < n < 5}_{\text{Endpoints not included}}$$



$$\underbrace{3 \leq n \leq 5}_{\text{Endpoints included}}$$



$$\underbrace{3 \leq n < 5}_{\substack{\text{Left endpoint included} \\ \text{Right endpoint excluded}}}$$



$$\underbrace{3 < n \leq 5}_{\substack{\text{Left endpoint excluded} \\ \text{Right endpoint included}}}$$



C. Continuous Numbers

1.35: Continuous Numbers

When we represent numbers on an interval on the number line, we are referring to continuous numbers. This means that all numbers in between are included.

Example 1.36

How many numbers does the number line alongside represent?



It represents an infinite number of numbers because it includes numbers like

3.1, 3.01, 3.001, ...

And there is no limit to which you can take the decimal point.

1.37: Infinite Number of Numbers

Between any two numbers on the real number line lie an infinite number of numbers.

Example 1.38

Find some numbers between:

$$4, 4.001$$

$$4.0005, 4.00051, 4.000501, \dots$$

Example 1.39 (Calculator Allowed)

$$\sqrt{32}, \sqrt{56}$$

$$\sqrt{32} \approx 5.65$$

$$\sqrt{56} \approx 7.48$$

$$6, 6.1, 6.01, 6.001, 6.0001, \dots$$

Example 1.40 (Calculator Allowed)

$$\sqrt[3]{-3}, \sqrt[3]{-9}$$

$$\sqrt[3]{-3} \approx -1.44$$

$$\sqrt[3]{-9} \approx -2.08$$

$$-1.7, -1.71, -1.701, -1.7001, \dots$$

D. Interval Notation

Interval Notation is an algebraic way to write the solution sets that we have been graphing on the number line so far.

1.41: Interval Notation

Interval Notation is written as:

$$(Start\ Point, End\ Point)$$

And it includes all the numbers in between.

For example:

$$(3, 4)$$

Means all the real numbers between 3 and 4.

1.42: Round Bracket Notation

Round brackets mean that the endpoint is not included.

They are equivalent to empty circles on the number line.

$$(3, 4)$$

Does not include either 3 or 4, but includes all the numbers in between.

This is equivalent to two empty circles on the number line, and the number line version is shown alongside.



Example 1.43

Write $(3, 4)$ as an inequality.

$$3 < x < 4$$

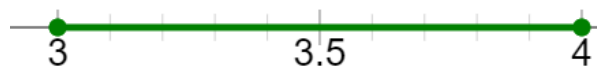
1.44: Square Bracket Notation

Square brackets mean that the endpoint is included.
They are equivalent to filled circles on the number line.

$$[3,4]$$

Includes 3 and 4, and all the numbers in between.

This is equivalent to two empty circles on the number line, and the number line version is shown alongside.



Example 1.45

Write $[3,4]$ as an inequality.

$$3 \leq x \leq 4$$

1.46: Summary of Notation

Meaning	Number Line	Interval Notation	Inequality Notation
Endpoint Not Included	Empty Circle	Round Bracket	$< OR >$
Endpoint Included	Filled Circle	Square Bracket	$\leq OR \geq$

$$\left(\underset{\text{Start Point}}{x}, \underset{\text{End Point}}{y} \right),$$

Endpoints not included

$$\left[\underset{\text{Start Point}}{x}, \underset{\text{End Point}}{y} \right],$$

Endpoints are included

$$\left(\underset{\text{Start Point}}{x}, \underset{\text{End Point}}{y} \right],$$

*Left Endpoint Excluded
Right Endpoint Included*

$$\left[\underset{\text{Start Point}}{x}, \underset{\text{End Point}}{y} \right),$$

*Left Endpoint Included
Right Endpoint Excluded*

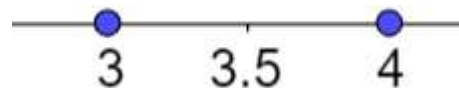
1.47: Set Notation

The notation

$$\{ \}$$

Stands for a set. It includes only those elements which are specified in the set, not the elements in between.

$\{3,4\}$, which is graphed alongside, includes only the discrete numbers
3 and 4



Example 1.48

Graph the inequality below on a number line. And also write it in interval notation.

- $3 < n < 5$
Endpoints not included
- $3 \leq n \leq 5$
Endpoints included
- $3 \leq n < 5$
*Left endpoint included
Right endpoint excluded*
- $3 < n \leq 5$
*Left endpoint excluded
Right endpoint included*

Part A

Interval from 3 to 5, but not want the endpoints

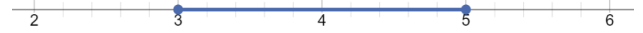
(3,5)



Part B

Interval from 3 to 5, with the endpoints:

[3,5]



Part C

Interval from 3 to 5.

- Use a square bracket on the left to include the left endpoint.
- Use a round bracket on the right to exclude the right endpoint.

[3,5)



Part D

We want the interval from 3 to 5.

- Use a square bracket on the right to include the left endpoint.
- Use a round bracket on the left to exclude the right endpoint.

(3,5]



Example 1.49

Write the following inequalities in Interval Notation.

- A. $\frac{1}{7} < n < \frac{3}{4}$
- B. $\frac{1}{7} \leq n < \frac{3}{4}$
- C. $\frac{1}{7} \leq n \leq \frac{3}{4}$
- D. $\frac{1}{7} < n \leq \frac{3}{4}$

$\left(\frac{1}{7}, \frac{3}{4}\right)$
 $\left[\frac{1}{7}, \frac{3}{4}\right)$
 $\left[\frac{1}{7}, \frac{3}{4}\right]$
 $\left(\frac{1}{7}, \frac{3}{4}\right]$

E. Converting from Interval to Inequality Notation

Example 1.50

Write the below in inequality notation:

- A. I will meet you at a time t after 3 pm, and before 5 pm.
- B. I will meet you at a time t between 3 and 5 pm.
- C. I will meet you at a time t from 3 to 5 pm.

Statement A: $x > 3$ AND $x < 5$

Statement B: $3 < x < 5$

Statement C: $3 \leq x \leq 5$

Example 1.51

Convert the following into inequality notation.

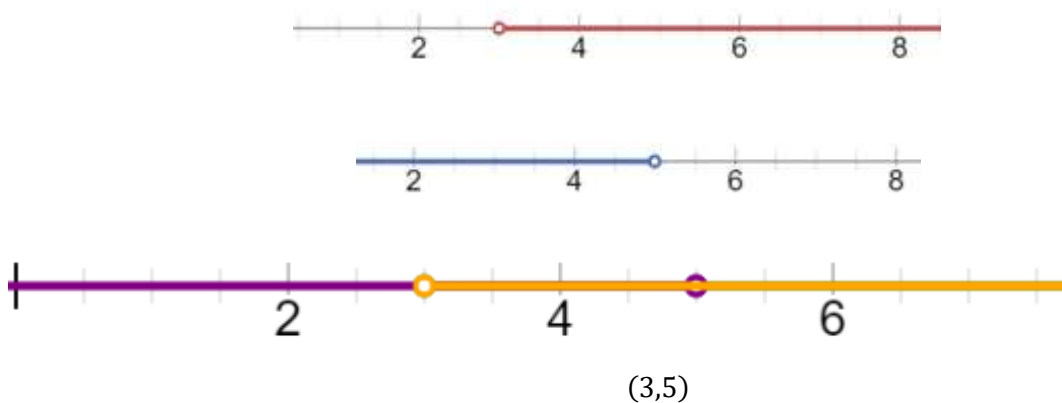
- A. (3,5)
- B. [3,5)
- C. (3,5]
- D. [3,5]

$$\begin{aligned}x > 3 \text{ AND } x < 5 &\Rightarrow 3 < x < 5 \\x \geq 3 \text{ AND } x < 5 &\Rightarrow 3 \leq x < 5 \\x > 3 \text{ AND } x \leq 5 &\Rightarrow 3 < x \leq 5 \\x \geq 3 \text{ AND } x \leq 5 &\Rightarrow 3 \leq x \leq 5\end{aligned}$$

Example 1.52

Find $A \cap B$ given that:

$$A = \{x: x > 3, \quad x \in \mathbb{R}\}, \quad B = \{x: x < 5, \quad x \in \mathbb{R}\}$$



In the above question, if $x \in \mathbb{R}$ had not been a part of the question statement, which set of numbers would have you taken for x ?

You would still have taken real numbers.

Example 1.53

Find $A \cap B$ given that:

$$A = \{x: x > 3, \quad x \in \mathbb{Z}\}, \quad B = \{x: x < 5, \quad x \in \mathbb{Z}\}$$

$$\begin{aligned}A &= \{4, 5, 6, \dots\} \\B &= \{\dots, 2, 3, 4\} \\A \cap B &= \{4\}\end{aligned}$$

F. Concept of Infinity

If we have a number which is greater than three, this is written in inequality notation as

$$x > 3$$

In interval notation, there is no higher bound for the value of x , and hence we write:

$$(3, \infty)$$

Infinity will always take a round bracket, since no number, howsoever large, will include infinity.

Example 1.54

Write the following inequalities in interval notation.

- A. $x > 5$
- B. $x < -3$

- C. $x \geq 0$
- D. $x \leq 9$

$$(5, \infty)$$
$$(-\infty, -3)$$
$$[0, \infty)$$
$$(-\infty, 9]$$

Example 1.55

Write the following inequalities in interval notation.

- A. $-2 < x < 5$
- B. $-8 < x \leq -3$
- C. $5 \leq x \leq 6$

$$(-2, 5)$$
$$(-8, -3]$$
$$[5, 6]$$

G. Types of Inequalities

Sign	Meaning	Sign	Meaning
$>$	Strictly greater than	$<$	Strictly less than
$\geq$ or $\geq$	Greater than or equal to	$\leq$ or $\leq$	Less than or equal to

H. Endpoints

The maximum number of mistakes are made in terms of including or not including endpoints.
The value at the end of the range of an inequality is called its endpoint. If an inequality takes the value at the end of its range, the endpoint is included. If not, the endpoint is excluded.

Example 1.56

Write the following statements into inequalities.

- A. The passing marks(m) in SSC board for the English subject are 35 or more marks. The maximum marks in the subject are 100. Marks are given in increments of 1.
- B. I need more than three hundred balloons(b) for that child's party.
- C. I need more than three hundred rupees(r) in my bank account to not pay a fine. The bank calculates my bank balance in rupees and paise, but not fractions of a paise.

SSC Board does not give negative marks to students, and hence, the number of marks must be a whole number:

$$35 \leq m \leq 100, \quad m \in \mathbb{W}$$

You cannot have half a balloon in the given context; hence the variable must take a value from the set of integers:

$$b > 300, \quad b \in \mathbb{N}$$

The money in my bank account can be modelling using a number from the set of Rational Number.

$$r > 300, \quad r \in \mathbb{Q}$$

Example 1.57

Write the following statements into interval notation:

- A. The passing marks(m) in SSC board for the English subject is 35 or more marks. The maximum marks in the subject are 100.
- B. I need more than three hundred balloons(b) for that child's party.
- C. I need more than three hundred rupees(r) in my bank account to not pay a fine.

$$\begin{aligned} 35 \leq m \leq 100, & \quad m \in \mathbb{W} \Rightarrow m \in \{35, 36, 37, \dots, 100\} \\ b > 300, & \quad b \in \mathbb{N} \Leftrightarrow \{301, 302, 303, \dots\} \\ r > 300, & \quad r \in \mathbb{Q} \Leftrightarrow \{300.01, 300.02, \dots\} \end{aligned}$$

I. Set Theory Notation

This is the most formal. It uses concepts from set theory to write the solution set as a valid set.

Example 1.58

Write the following statements into interval notation:

- A. The passing marks(m) in SSC board for the English subject is 35 or more marks. The maximum marks in the subject are 100.
- B. I need more than three hundred balloons(b) for that child's party.
- C. I need more than three hundred rupees(r) in my bank account to not pay a fine.

$$\left\{ \begin{array}{l} m \downarrow \text{such that } 35 \leq m \leq 100, \quad m \in \mathbb{W} \\ b \downarrow \text{such that } b > 300, \quad b \in \mathbb{N} \\ r \downarrow \text{such that } r > 300, \quad r \in \mathbb{R} \end{array} \right\}$$

J. Number Line

The number line is a visual notation. It is useful in solving questions since it provides an easy way to find the intersection, or the union of two solutions.

Example 1.59

x is a real number greater than five. Write this in the different notation formats.

$$\underbrace{x > 5}_{\text{Inequality Notation}} \Leftrightarrow \underbrace{(5, +\infty)}_{\text{Interval Notation}} \Leftrightarrow \underbrace{\{x \mid x > 5, \quad x \in \mathbb{R}\}}_{\text{Set Notation}}$$

Example 1.60: Notations: Number Line, Intervals, and Inequality

Write each of the following using the above three notations.

- A. x is a positive integer greater than 5.
- B. y is an odd integer less than 3.
- C. z is a real number less than or equal to 7.3
- D. p is a real number greater than or equal to 6
- E. s is a real number between 3 and 7
- F. w is a real number from 2 and 9
- G. r is a real number greater than 5 and less than or equal to 12
 $\{5 < r \leq 12, r \in \mathbb{R}\} \Rightarrow (5, 12]$

	Number Line		Interval Notation	Set Theory Notation
A			$[6, 7, 8 \dots)$	$\{x > 5, x \in I^+\}$
6 is included in the valid region for the inequality. Hence, we use a square bracket.				
B			$(\dots - 3, -1, 0, 1]$	$\{y < 3, y \in 2n + 1, n \in I\}$
C			$(-\infty, 7.3]$	$\{z \leq 7.3, z \in \mathbb{R}\}$
D			$[6, \infty)$	$\{p \geq 6, z \in \mathbb{R}\}$
6 is included in the valid region for the inequality. Hence, we use a square bracket. Infinity can be never be included. That is why we use a round bracket.				
E			$(3, 7)$	$\{3 < s < 7, s \in \mathbb{R}\}$
3 and 7 are not included in the valid region for the inequality. Hence, we use a round bracket both sides				
F			$[2, 9]$	$\{2 \leq s \leq 9, s \in \mathbb{R}\}$
Both the endpoints (2 and 9) are included in the solution set. Hence, we use square brackets.				

Example 1.61: Notations: Number Line, Intervals, and Inequality

The product of Hari's age in years on his last birthday and his age now in complete months is 1800. Hari's current age is: (NMTC Sub-Junior/Screening 2004/19, Modified)

$$y = \text{years}, \quad x = \text{months after his birthday} \Rightarrow \text{Age in Months} = 12y + x$$

$$y(12y + x) = 1800 \Rightarrow \underbrace{12y^2}_{\text{Larger}} + \underbrace{xy}_{\text{Smaller}} = 1800$$

$$12y^2 < 1800 \Rightarrow y^2 < 150 \Rightarrow y < 13 \Rightarrow y = 12, 12y^2 = 1728$$

$$1728 + 12x = 1800 \Rightarrow 12x = 72 \Rightarrow x = 6$$

1.4 Set Operations on Intervals

A. Union and Intersection

1.62: Union

The union sign in interval notation converts into OR in inequality notation

$$(a, b) \cup (c, d) \Leftrightarrow a < x < b \text{ OR } c < x < d$$

1.63: Intersection

The intersection sign in interval notation converts into AND in inequality notation

$$(a, b) \cap (c, d) \Leftrightarrow a < x < b \text{ AND } c < x < d$$

Example 1.64

Convert the following into inequality notation.

- A. $(-\infty, -3)$
- B. $(4, \infty)$
- C. $(-\infty, -3) \cup (4, \infty)$
- D. $(-\infty, -3) \cap (4, \infty)$

Part A



We might want to convert the interval into:

$$-\infty < x < -3$$

But, this has the same meaning as:

$$x < -3$$

And that is how we will write it.

Part B



$$x > 4$$

Part C

$$x < -3 \text{ OR } x > 4$$

Part D

$$x < -3 \text{ AND } x > 4 \Rightarrow x = \phi$$

Example 1.65

Convert the following into inequality notation.

A. $(-\infty, 5) \cup \{7\}$

$$x < 5 \text{ OR } x = 7$$

Example 1.66

Convert the following into inequality notation.

A. $(-\infty, 5) \cup \{3, 8\}$

$$x < 5 \text{ OR } x = 3 \text{ OR } x = 8$$

But note that 3 is already a part of the first inequality, and hence it simplifies to:

$$x < 5 \text{ OR } x = 8$$

B. Logical Operators

Example 1.67

Hriday can have a class with his Philosophy Teacher on either a Tuesday or a Wednesday. Santosh can have a class with his Philosophy Teacher on either a Monday or a Tuesday.

- List the days when Hriday can have a class.
- List the days when Santosh can have a class.
- Hriday and Santosh want to have a class together with their philosophy teacher. What are the possible days on which they can have it?
- If the Philosophy Teacher wants to have a class with a single student, what are the different days on which he can have the class? (Note: He can have two single back to back classes)

Part A

$$\{\text{Tuesday, Wednesday}\} \Rightarrow \text{Condition I}$$

Part B

$$\{\text{Monday, Tuesday}\} \Rightarrow \text{Condition II}$$

Part C

To have a class with Hriday and Santosh, we must meet both of their conditions.
This is an AND condition.
Hence, the only day available is

Tuesday

Part D

To have a class with either Hriday, or Santosh, we can meet any one of the conditions.
This is an OR condition.
Hence, the days available are:

$$\{\text{Monday, Tuesday, Wednesday}\}$$

C. Revision: OR Conditions

The Logical OR means that we will accept either of the conditions.
A logical OR means we take a union of the given conditions.

Example 1.68: Logical AND (Intersection) and Logical OR (Union)

The following conditions are mentioned:

either on the first or last day of the week, on a weekend.
Condition I *Condition II*

Note that Anshul's week starts on Monday.

Write the days of the week on which Anshul can visit his parents, if

- A. Either Condition I OR Condition II must be met
- B. Condition I AND Condition II must be met

Part A

Our answer will be the **union** of Condition I, and Condition II:

$$\underbrace{\text{Monday, Sunday}}_{\substack{\text{First Day} \quad \text{Last Day} \\ \text{Condition I}}} \cup \underbrace{\text{Saturday} + \text{Sunday}}_{\text{Condition II}} \Rightarrow \underbrace{\text{Monday, Saturday, Sunday}}_{\text{Final List}}$$

Part B

Our answer will be the **intersection** of Condition I, and Condition II:

$$\underbrace{\text{Monday, Sunday}}_{\substack{\text{First Day} \quad \text{Last Day} \\ \text{Condition I}}} \cap \underbrace{\text{Saturday} + \text{Sunday}}_{\text{Condition II}} \Rightarrow \underbrace{\text{Sunday}}_{\text{Final List}}$$

D. AND in language, which means OR

It is not necessary that the word used in the language translates directly into the logical condition.

Example 1.69: Using AND to represent a Logical OR

Anshul is available to visit his parents on the first day of the week. And, he is also available to visit his parents on the last day of the week. Anshul's week starts on Tuesday. Write the days of the week on which Anshul can visit his parents.

We still need to find the **union**.

Anshul can visit on the first day (Tuesday), or on the last day (Saturday).

E. Revision: AND Conditions

An AND condition means that all the mentioned conditions have to be satisfied simultaneously.

Example 1.70: Logical AND

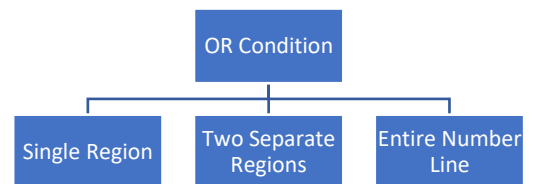
Priyesh is a vegetarian who does not like pulses. What food will he like?
Condition I *Condition II*

We need to meet both the conditions. Therefore, a menu which is suitable for him must be vegetarian, and without pulses.
Meet both Conditions

F. Merging Inequalities

When we get an OR condition, we find either of the conditions acceptable. In terms of set theory, we will find the union of the regions that satisfy each condition.

Depending upon the kind of question which is asked, the final answer may be a single region, two separate regions, or the entire number line.

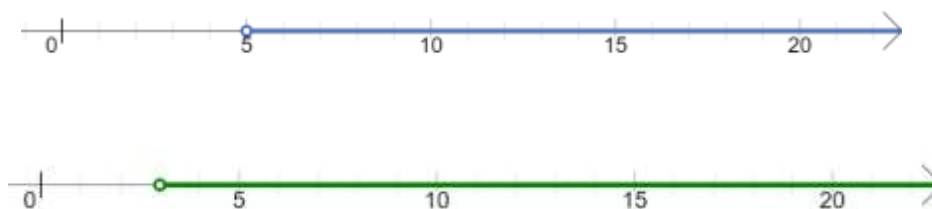


G. Single Region

Example 1.71

Combine into a single inequality:

$$x > 5 \text{ OR } x > 3$$



We can collapse the set of inequalities to a single inequality



And we can write the above using inequality notation, and interval notation:

$$x > 5 \text{ OR } x > 3 \Rightarrow x > 3 \Rightarrow (3, \infty)$$

Example 1.72

Combine into a single inequality:

- A. $x > -3$ OR $x > 2$
- B. $x > \pi$ OR $x > 3$
- C. $x > 1$ OR $x \geq 9$
- D. $x \geq -2$ OR $x > 12$
- E. $x \geq -5$ OR $x \geq 1$

$$x > -3 \text{ OR } x > 2 \Rightarrow x > -3 \Rightarrow x \in (-3, \infty)$$

$$x > \pi \text{ OR } x > 3 \Rightarrow x > 3$$

$$x > 1 \text{ OR } x \geq 9 \Rightarrow x > 1$$

$$x \geq -2 \text{ OR } x > 12 \Rightarrow x \geq -2$$

$$x \geq -5 \text{ OR } x \geq 1 \Rightarrow x \geq -5$$

H. Two Separate Regions

Example 1.73

Determine the interval which satisfies each part below:

- A. $x < 3$ OR $x \geq 5$

This inequality cannot be converted into a single inequality. It comprises two separate regions, and the same is reflected in both inequality as well as interval notation.



We want an OR, which means we want the union of the above two intervals.

$$(-\infty, 3) \cup [5, \infty)$$

Example 1.74

Determine the interval which satisfies:

- A. $x \leq 2$ OR $x > 5$
- B.

$$(-\infty, 2] \cup (5, \infty)$$

Example 1.75: Entire Number Line

$$x < 3 \text{ OR } x \geq 1$$



Note that the red inequality (representing $x < 3$) overlaps the blue inequality (representing $x \geq 1$).

Hence, when we combine the two inequalities into a single area, we get the entire number line.

$$(-\infty, 3) \cup [1, \infty) \Rightarrow (-\infty, \infty)$$

A compact way of writing the above solution is to state x can the value of any real number, that is, any value on the number line.

$$x \in \mathbb{R}$$

I. AND Conditions

Example 1.76

$$x > 2 \text{ AND } x < 3$$



Which can be written in interval notation as:

$$(2, 3)$$

Example 1.77

$$x \geq 2 \text{ AND } x < 3$$

$$x > 2 \text{ AND } x \leq 3$$

$$x \geq 2 \text{ AND } x \leq 3$$

$$x \geq 2 \text{ AND } x < 3 \Rightarrow [2, 3)$$

$$x > 2 \text{ AND } x \leq 3 \Rightarrow (2, 3]$$

$$x \geq 2 \text{ AND } x \leq 3 \Rightarrow [2, 3]$$

Example 1.78: No Solution

$$x < 3 \text{ AND } x > 5$$

$$x < 3 \text{ AND } x > 5 \Rightarrow (-\infty, 3) \cap [5, \infty) \Rightarrow \underbrace{\phi}_{\text{Null Set}} \Rightarrow \text{No Solution}$$

$$-9x + 2 > 18 \Rightarrow -9x > 16 \Rightarrow x < \frac{-16}{9}$$

$$13x + 15 \leq -14 \Rightarrow 13x \leq -29 \Rightarrow x \leq -\frac{29}{13}$$

Example 1.79

Solve the following inequalities

- A. x is a odd number between 7 and 12.
- B. x is an number fulfilling any of the following conditions:
 - a. It is more than or equal to 5
 - b. It is less than 3.

$$\{7 < x < 12, x = 2n + 1, n \in I\}$$
$$x < 3 \text{ OR } x \geq 5 \Rightarrow (-\infty, 3) \cup [5, \infty)$$

1.5 One Step Inequalities

A. Addition

1.80: Types of Inequalities

There are four types of inequalities

$$x > y$$

$$x < y$$

$$x \geq y$$

$$x \leq y$$

We will state the properties for only the first type, but they apply to all four types of inequalities.

1.81: Adding the same number

You can add the same number to both sides of an inequality.

$$x > y \Rightarrow x + a > y + a$$

This is similar to solving an equation by adding the same number to both sides.

$$x = y \Rightarrow x + a = y + a$$

For example:

$$4 > 3 \Rightarrow 4 + 1 > 3 + 1 \Rightarrow 5 > 4$$

Example 1.82

Solve the following, and write your answer in interval notation.

A. $x - 7 = 4$

B. $x - 7 > 4$

C. $x - \frac{1}{2} > \frac{1}{3}$

Part A

Add 7 to both sides:

$$x - 7 + 7 = 4 + 7 \Rightarrow x = 11$$

Part B

Add 7 to both sides of the inequality:

$$x - 7 + 7 > 4 + 7 \Rightarrow x > 11$$



$$\underbrace{11}_{\text{Start Point}}, \underbrace{\infty}_{\text{End Point}} \Rightarrow x \in (11, \infty)$$

Part C

$$x - \frac{1}{2} + \frac{1}{2} > \frac{1}{3} + \frac{1}{2} \Rightarrow x > \frac{5}{6} \Rightarrow \left(\frac{5}{6}, \infty\right)$$

Example 1.83

Find all possible value of the number in each case, and write your answer in interval notation:

- A. When $\frac{2}{3}$ is subtracted from a number, the number is greater than $\frac{4}{3}$. Find all possible values that the number can take.
- B. When $\frac{1}{2}$ is subtracted from a number, the number is greater than or equal to $\frac{2}{5}$. Find all possible values that the number can take.
- C. A number is at least $2\frac{3}{4}$ less than $1\frac{2}{3}$. Find the range of values that the number can take.

$$x - \frac{2}{3} > \frac{4}{3} \Rightarrow x > \frac{6}{3} \Rightarrow x > 2 \Rightarrow x \in (2, \infty)$$
$$x - \frac{1}{2} \geq \frac{2}{5} \Rightarrow x \geq \frac{2}{5} + \frac{1}{2} \Rightarrow x \geq \frac{9}{10} \Rightarrow \left[\frac{9}{10}, \infty\right)$$

$$x < 1\frac{2}{3} - 2\frac{3}{4} = \frac{5}{3} - \frac{11}{4} = \frac{20 - 33}{12} = -\frac{13}{12} = -1\frac{1}{12} \Rightarrow \left(-\infty, -\frac{13}{12}\right)$$

B. Constants

1.84: Some Constants

- π is the ratio of the circumference of a circle to its diameter.
- e is a number such that e^x grows at the same rate as itself.
-

$$C = \pi D \Rightarrow \frac{C}{D} = \pi \Rightarrow C:D = \pi:1$$
$$e \approx 2.71$$

If want the answer in terms of the exact value of π or e , treat it like a variable and keep solving as usual.

1.85: Approximate Value

- The exact value of π or e is represented using the symbol itself, that is, π . The exact value cannot be represented using a real number.
- When we want an approximate value, we replace the constant with its approximation.

$$\pi \approx 3.14$$
$$\pi \approx \frac{22}{7}$$

Note that:

$$\frac{22}{7} = 3.\overline{142857} \neq 3.14$$

Example 1.86

- A. A number is greater than 4 by at least 3. Find the exact range of values that the number can take.
- B. A number is greater than π by at least e . Find the exact range of values that the number can take.

Part A

$$x - 3 \geq 4 \Rightarrow x \geq 7 \Rightarrow [7, \infty)$$

Part B

$$\pi \approx 3.14 \Rightarrow \pi \neq 3.14$$

Since the questions asks for exact values, we will have to work using π and e themselves.

$$x - e \geq \pi \Rightarrow x \geq \pi + e \Rightarrow [\pi + e, \infty)$$

C. Integer Solutions

Example 1.87

- A. If the number of customers coming in to Fred's Diner for lunch everyday were to decrease by 7, he would still get more than 20 customers. What is the number of customers that could be coming in to Fred's Diner currently?
- B. After a flyover was introduced, the number of cars at a traffic signal reduced by 12. What was the number of cars at the traffic signal before the flyover?
- C. When 5 is subtracted from an integer n , the resulting number is less than -5 . What is the value of n ?
- D. Last week, Alfred sold seven sheep, and now the number of sheep he has is less than nine. What was the original number of sheep Alfred had?
- E. Mustufa is dreaming that the number of homework problems he has to do magically reduced by 9. However, he still has at least 12 hard and at least 6 very hard problems. If the number of problems reduced is split between hard and very hard as far as possible, find the original number of hard and very hard problems that Mustufa was supposed to solve.

Part A

$$d - 7 > 20 \Rightarrow d > 27 \Rightarrow d \geq 28 \Rightarrow d \in \{28, 29, \dots\}$$

$d \in [28, \infty)$ is wrong because that includes numbers like 28.3 which are not possible for the number of customers.

Part B

$$\begin{aligned} \text{Current number of cars} &\geq 0 \\ O = \text{Original number of cars} &\geq 12 \Rightarrow O \in \{12, 13, \dots\} \end{aligned}$$

Part C

$$n - 5 < -5 \Rightarrow n < 0 \Rightarrow n \in \{\dots, -2, -1\}$$

Part D

$$s - 7 < 9 \Rightarrow s < 16 \Rightarrow s \leq 15 \Rightarrow s \in \{0, 1, 2, \dots, 15\}$$

Part E

$$\text{Hard Problems} = h, \quad \text{Very hard problems} = v$$

There are two cases:

$$\begin{aligned} h - 4 &\geq 12 \Rightarrow h \geq 16 \Rightarrow h \in \{16, 17, 18, \dots\} \\ v - 5 &\geq 6 \Rightarrow v \geq 11 \Rightarrow v \in \{11, 12, 13, \dots\} \end{aligned}$$

$$\begin{aligned} h - 5 &\geq 12 \Rightarrow h \geq 17 \Rightarrow h \in \{17, 18, 19, \dots\} \\ v - 4 &\geq 6 \Rightarrow v \geq 10 \Rightarrow v \in \{10, 11, 12, \dots\} \end{aligned}$$

D. Subtraction

1.88: Subtracting the same number

You can subtract the same number from both sides of an inequality.

$$x > y \Rightarrow x - a > y - a$$

This is similar to solving an equation by subtracting the same number to both sides.

$$x = y \Rightarrow x - a = y - a$$

For example:

$$4 > 3 \Rightarrow 4 - 1 > 3 - 1 \Rightarrow 3 > 2$$

Example 1.89

Solve the following:

- A. $x + 5 = 9$
- B. $x + 5 < 9$
- C. $x + \frac{3}{4} > 2$
- D. $x + 0.01 > 3$

Part A

Subtract 5 from both sides:

$$x - 5 + 5 = 9 - 5 \Rightarrow x = 4$$

Part B

Subtract 5 from both sides:

$$x - 5 + 5 < 9 - 5 \Rightarrow x < 4$$

Part C

Subtract $\frac{3}{4}$ from both sides:

$$x + \frac{3}{4} - \frac{3}{4} > 2 - \frac{3}{4} \Rightarrow x > \frac{5}{4}$$

Part D

Subtract 0.01 from both sides:

$$x + 0.01 - 0.01 > 3 - 0.01 \Rightarrow x > 2.99$$

Example 1.90

- A. When 5 is added to a number, it is less than 4. Find the range of possible values n that the number can take.
- B. If the quantity of milk m in a container is increased by 3.5 litres, the increased quantity is greater than or equal to 5 Litres. Find the range that the original quantity could have been.
- C. When a number x is increased by $\frac{2}{3}$, it becomes greater than $\frac{1}{5}$. Find the range of values that the number can take.
- D. If the number of gorillas, g (which is greater than 1), in a local forest population increases by 4, it will be still be 3 or more less than 27 gorillas counted in the last census taken by a forest ranger. Find the range of values that the gorillas can take.
- E. Amogh could not finish his homework on Monday, and had three problems left to do. On Tuesday, he could not finish two problems. On Wednesday, there was no homework, and on Thursday, he completed his homework. If he didn't touch the homework problems from Monday and Tuesday, and the total number of homework problems that he had pending at the end of the week was less than twelve, then find a range for the number of homework problems, p , that he could have had pending on Friday. (For this problem, consider that homework is only assigned Monday to Friday, not over the weekend).
- F. Jane has x red cars and y blue cars. She paired her red and blue cars. At least four blue cars did not have red cars to pair with (and all red cars were paired). Find, in terms of y , the set of values that x can take.

Part A

$$n + 5 < 4 \Rightarrow n < -1 \Rightarrow n \in (-\infty, -1)$$

Part B

$$m + 3.5 \geq 5 \Rightarrow m \geq 1.5 \Rightarrow m \in [1.5, \infty)$$

Part C

$$x + \frac{2}{3} > \frac{1}{5} \Rightarrow x > \frac{1}{5} - \frac{2}{3} = \frac{3-10}{15} = -\frac{7}{15} \Rightarrow x \in \left(-\frac{7}{15}, \infty\right)$$

Part D

$$g + 4 \leq 27 - 3 \Rightarrow g \leq 20 \Rightarrow g \in \{2, 3, \dots, 20\}$$

Part E

$$3 + 2 + p < 12 \Rightarrow p < 7 \Rightarrow p \in \{0, 1, \dots, 6\}$$

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Pending	3	2	0	0	p	0	0

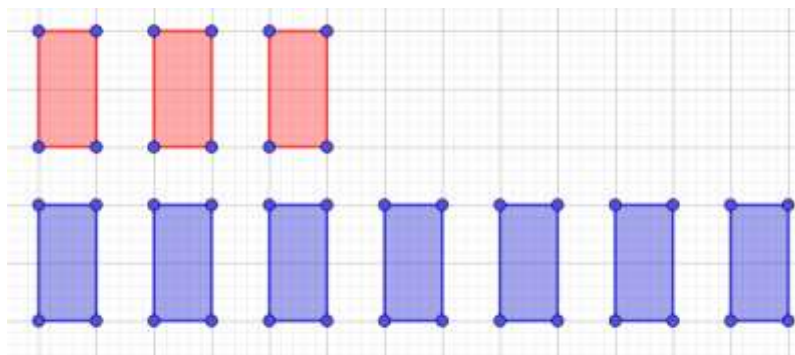
Part F

Pairs of Red and Blue Cars

All red cars were paired. In the diagram, we can add more pairs of red and blue cars without violating the condition of the question.

Unpaired Blue Cars

There were at least blue cars that were not paired. In the diagram, we can add more blue cars to the right, without violating the condition of the question.



Framing an Inequality

From the above, we can see that the number of blue cars is at least 4 more than the number of red cards. And, we can convert this into an inequality as follows:

$$\underbrace{x}_{\text{Blue}} + 4 \leq \underbrace{y}_{\text{Red}} \Rightarrow x \leq y - 4 \Rightarrow x \in (-\infty, y - 4]$$

E. Division

1.91: Dividing by the same number

For positive values of a :

$$x > y \Rightarrow \frac{x}{a} > \frac{y}{a}$$

For negative values of a :

$$x > y \Rightarrow \frac{x}{a} < \frac{y}{a}$$

Note: If the number that you are dividing by is negative, then the sign of the inequality will reverse.

$$2 < 3 \Rightarrow \frac{2}{5} < \frac{3}{5}$$

$$2 < 3 \Rightarrow -2 > -3$$

Example 1.92

Solve

- A. $3x < 4$
- B. $-5x \geq 9$
- C. $-2x > -3$
- D. $6z < 144$
- E. $4x \leq -4$
- F. $0.2x > 0.7$
- G. $-0.7x > -0.4$

H. $\frac{4}{3}x \geq \frac{3}{4}$
I. $-2.5a < 6.8$

Part A

$$x < \frac{4}{3} \Rightarrow x \in \left(-\infty, \frac{4}{3}\right)$$



Part B

$$x \leq -\frac{9}{5} \Rightarrow x \in \left(-\infty, -\frac{9}{5}\right]$$



Part C

$$x < \frac{3}{2} \Rightarrow x \in \left(-\infty, \frac{3}{2}\right)$$



Part D

$$6z < 144 \Rightarrow z < \frac{144}{6} \Rightarrow z < 24 \Rightarrow z \in (-\infty, 24)$$

Part E

$$x \leq -1 \Rightarrow x \in (-\infty, -1]$$



Part F

$$x > \frac{0.7}{0.2} = \frac{7}{2} \Rightarrow x \in (3.5, \infty)$$

Part G

$$x < \frac{-0.4}{-0.7} = \frac{4}{7} \Rightarrow x \in \left(-\infty, \frac{4}{7}\right)$$

Part H

$$x \geq \frac{3}{4} \times \frac{4}{3} = 1 \Rightarrow [1, \infty)$$

Part I

$$-2.5a < 6.8 \Rightarrow a > \frac{6.8}{2.5} \Rightarrow a > \frac{68}{25} \Rightarrow a \in \left(\frac{68}{25}, \infty\right)$$

Example 1.93

- Three times a number is less than 4. Find all possible values that the number can take.
- One-fifth of a number is more than two-fifth. Find all possible values that the number can take.
- Negative-two-fifth of a number is less than negative two-fifths. Find the values that the number can take.

$$3x < 4 \Rightarrow x < \frac{4}{3} \Rightarrow x \in \left(-\infty, \frac{4}{3}\right)$$

$$\frac{x}{5} > \frac{2}{5} \Rightarrow x > 2 \Rightarrow x \in (2, \infty)$$

Part C

$$-\frac{2}{5}n < -\frac{2}{5}$$

Divide both by $-\frac{2}{5}$, and reverse the sign of the inequality since we dividing by a negative number:

$$n > 1 \Rightarrow n \in (1, \infty)$$

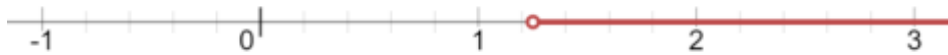
Example 1.94

Four times a positive number n is greater than or equal to 5.

- A. What is the range of the values that the number can take?
- B. Which integer values of n satisfy the condition above?
- C. How many integer values of n satisfy the condition above?

Part A

$$4n \geq 5 \Rightarrow n \geq \frac{5}{4} \Rightarrow n \geq 1\frac{1}{4} \Rightarrow n \in \left[1\frac{1}{4}, \infty\right)$$



Part B

$$n \in \{2, 3, 4, 5, \dots\}$$

Part C

Infinite

Example 1.95

An orchard has two trees, each with the same number of apples. The apples are all collected and divided into three piles. Each pile has more than 17 apples.

- A. Find the minimum number of apples that each tree can have.
- B. Find all possible values for the number of apples that the orchard can have.

Let the number of apples on a tree be t .

$$\frac{2t}{3} \geq 18 \Rightarrow t \geq 18 \times \frac{3}{2} \Rightarrow t \geq 27 \Rightarrow t = 27 \Rightarrow 2t = 54$$

Example 1.96

p is a prime number. When p is multiplied by the smallest prime number, and divided by the second smallest prime number, the answer is less than the third smallest prime number. Find the product of all values that p can take.

$$\frac{2p}{3} < 5 \Rightarrow p < 5 \times \frac{3}{2} \Rightarrow p < \frac{15}{2} \Rightarrow p < 7.5 \Rightarrow p \in \{2, 3, 5, 7\} \Rightarrow \text{Product} = 210$$

Example 1.97

Define a *factor* to be a natural number that divides another number, leaving no remainder. For example, the factors of 8 are 1, 2 and 4. A number n is divided by the factors of 6, in succession. That is, it is divided by the first factor, then the result is divided by the second factor, and so on, till the last factor. The final answer is less than -3 . Find the:

- A. range of values that n can take.
- B. largest integer value that n can take.
- C. largest integer value that n can take, if the final answer is an integer less than -3 , and other things remain the same.

$$\text{Factors of } 6 = \{1, 2, 3, 6\}$$

$$n \div 1 \div 2 \div 3 \div 6$$

Convert the division into multiplication by taking the reciprocal of each divisor:

$$n \times \frac{1}{1} \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{6} = \frac{n}{1 \times 2 \times 3 \times 6} = \frac{n}{36}$$

Part A

Now, we can frame the inequality:

$$\frac{n}{36} < -3 \Rightarrow n < -108 \Rightarrow n \in (-\infty, -108)$$



Part B

$$-109$$

Part C

$$\frac{n}{36} \leq -4 \Rightarrow n \leq -144 \Rightarrow n \in (-\infty, -144] \Rightarrow \text{Largest value is } n = -144$$

Example 1.98

Reema has to order pizzas for a party. She and eight of her friends will have three slices each. Each pizza has four slices. Reema is fine with ordering extra, but she should not fall short. Write and solve an inequality that represents a valid range for the number of pizzas p that Reema can order. (The pizza shop accepts orders for a pizzas in positive integers).

$$4p > 9 \times 3 \Rightarrow 4p > 27 \Rightarrow p > \frac{27}{4} \Rightarrow p > 6\frac{3}{4} \Rightarrow p = \{7, 8, 9, \dots\}$$

F. Multiplication

1.99: Multiplying by the same number

For positive values of a :

$$x > y \Rightarrow ax > ay$$

For negative values of a :

$$x > y \Rightarrow ax < ay$$

Note:

If the number that you are multiplying by is positive, then the sign of the inequality stays the same.

If the number that you are multiplying by is negative, then the sign of the inequality will reverse.

For example

$$2 < 3 \Rightarrow -2 > -3$$

Example 1.100

Solve:

A. $\frac{x}{3} > 4$

B. $\frac{x}{-2} < 3$

C. $\frac{x}{5} \geq -9$

D. $\frac{-x}{8} \leq 1$

E. $\frac{x}{0.1} > 0.2$

F. $\frac{x}{-\frac{1}{2}} < \frac{2}{3}$

$$\begin{aligned} \text{G. } \frac{\frac{x}{0.2}}{0.5} &\geq -\frac{0.3}{0.7} \\ \text{H. } \frac{\frac{x}{-1}}{\frac{-2}{\frac{2}{3}}} &\leq -\frac{3}{7} \end{aligned}$$

Part A

Multiply both sides by 3:

$$\frac{x}{3} > 4 \Rightarrow x > 12 \Rightarrow x \in (12, \infty)$$

Part B

Multiply both sides by -2 , and flip the inequality:

$$x > -6$$

Part C

Multiply both sides by 5:

$$x \geq -45$$

Part D

Multiply both sides by -8 , and flip the inequality:

$$x \geq -8$$

Part E

Multiply both sides by 0.1:

$$x > 0.02$$

Part F

Multiply both sides by $-\frac{1}{2}$:

$$x > \left(\frac{2}{3}\right)\left(-\frac{1}{2}\right) \Rightarrow x > -\frac{1}{3}$$

Part G

Multiply both sides by $\frac{0.2}{0.5}$:

$$x \geq -\frac{0.3}{0.7}\left(\frac{0.2}{0.5}\right) \Rightarrow x \geq -\left(\frac{3}{7}\right)\left(\frac{2}{5}\right) \Rightarrow x \geq -\frac{6}{35}$$

Part H

The denominator of the LHS simplifies to $\frac{-\frac{1}{2}}{\frac{2}{3}} = -\frac{1}{2} \div \frac{2}{3} = -\frac{1}{2} \times \frac{3}{2} = -\frac{3}{4}$ giving us:

$$\frac{\frac{x}{-3}}{-\frac{4}{4}} \leq -\frac{3}{7}$$

Multiply both sides of the inequality by $-\frac{3}{4}$, and flip the inequality:

$$-\frac{3}{4} \times \frac{x}{-\frac{3}{4}} \geq -\frac{3}{7}\left(-\frac{3}{4}\right) \Rightarrow x \geq \frac{9}{28}$$

1.101: Multiplying by the Reciprocal

For positive values of $\frac{a}{b}$:

$$\frac{a}{b}x > y \Rightarrow x > \frac{b}{a}y$$

For negative values of $\frac{a}{b}$:

$$\frac{a}{b}x > y \Rightarrow x < \frac{b}{a}y$$

G. Review

1.102: Summary of Inequality Properties

Consider the inequality $x > y$	
Adding/Subtracting the same number (Valid) $x \pm a > y \pm a$	
Multiplying/Dividing by a Negative number Reverses the direction of the inequality. $ax < ay, \quad a < 0$	Multiplying/Dividing by a Positive number Valid (and maintains the direction of the inequality) $ax > ay, \quad a > 0$

MCMC 1.103

Mark all options which must be true.

We flip the sign of an inequality when:

- A. We exchange the LHS and the RHS
- B. We have a negative sign on the LHS, or on the RHS.
- C. We multiply by a negative number.
- D. We add a negative number.
- E. We divide by a number.

MCQ 1.104

If we multiply both sides of an inequality by 2:

- A. It is valid, and both sides become larger.
- B. It is valid, and both sides become smaller
- C. It is valid, and the LHS and the RHS can become larger or smaller.
- D. It is not a valid operation. The inequality may not hold true.

Matrix Match 1.105

$$-\frac{2z}{0.2} > 0.3$$

Pick one option from Column I, and one option from Column II to give a set of steps that will solve the inequality. Give *all valid pairs*.

Column I	Column II
1. Flip the sign of the inequality	A. Multiply by 0.2, and divide by -2
2. Do not flip the sign of the inequality	B. Multiply by 0.2, and divide by 2
3. Change the inequality to $\geq$	C. Multiply by -10
4. Change the inequality to $\leq$	D. Divide by -10

1.6 Two Step Inequalities

A. Justification of Steps

Example 1.106

Solve $3x + \frac{4}{2} > 5x - \frac{1}{2}$. Justify each step using an appropriate property.

$3x > 5x - \frac{1}{2} - \frac{4}{2} = 5x - \frac{5}{2}$	Subtract $\frac{4}{2}$ from both sides of the inequality
--	--

$-2x > -\frac{5}{2}$	Subtract $5x$ from both sides of the inequality
$x < -\frac{5}{2} \div (-2) = -\frac{5}{2} \times \frac{-1}{2} = \frac{5}{4}$	Divide both sides of the inequality by -2 . Accordingly, reverse the sign of the inequality.
$x < \frac{5}{4}$	Simplify
$x \in \left(-\infty, \frac{5}{4}\right)$	Convert the inequality into interval notation

B. Basics

Example 1.107

Solve the following inequality:

$$3x + 4 > 6$$

$$3x > 2 \Rightarrow x > \frac{2}{3} \Rightarrow x \in \left(\frac{2}{3}, \infty\right)$$

Example 1.108

$$-65y + 19 < -2y + 41$$

$$-63y + 19 < 41 \Rightarrow -63y < 22 \Rightarrow y > -\frac{22}{63}$$

Example 1.109

Solve the following inequality:

$$2x + 7 \leq 8x - 5$$

$$-6x + 7 \leq -5 \Rightarrow -6x \leq -12 \Rightarrow x \geq 2 \Rightarrow x \in [2, \infty)$$

Example 1.110

Solve the following inequality:

$$4x + 3 > 5x + 2$$

$$4x + 3 > 5x + 2 \Rightarrow 4x > 5x - 1 \Rightarrow -x > -1 \Rightarrow x < 1 \Rightarrow x \in (-\infty, 1)$$

Example 1.111

Solve the following inequality:

$$2x + 5 \leq 9x - 7$$

$$12 \leq 7x \Rightarrow \frac{12}{7} \leq x \Rightarrow x \geq \frac{12}{7} \Rightarrow x \in \left[\frac{12}{7}, \infty\right)$$

C. Word Problems on Numbers

Example 1.112

When a number is increased by five, and then decreased by six, the result is more than seven. Find the value(s)

that the number can take.

$$x + 5 - 6 > 7 \Rightarrow x > 8 \Rightarrow x \in (8, \infty)$$

Example 1.113

When three times a number is decreased by seven, it is more than or equal to seven times the number increased by four. Find the values(s) that the number can take.

$$3x - 7 \geq 7x + 4 \Rightarrow 3x \geq 7x + 11 \Rightarrow -4x \geq 11 \Rightarrow x \leq \frac{11}{-4}$$

Example 1.114

Twice a number is more than the sum of the number and fourteen. What could the number be?

$$2x > x + 14 \Rightarrow 2x > 14 \Rightarrow x > 7 \Rightarrow x \in (14, \infty)$$



Example 1.115

Five times a number decreased by four is greater than four times a number increased by seven. Find the values(s) that the number can take.

$$5x - 4 > 4x + 7 \Rightarrow x > 11 \Rightarrow x \in (11, \infty)$$

Example 1.116

When four is subtracted from a number and the result is divided by four, the answer is greater than or equal to four. Find the values(s) that the number can take.

$$\frac{x - 4}{4} > 4 \Rightarrow x - 4 > 16 \Rightarrow x > 20 \Rightarrow x \in (20, \infty)$$

Example 1.117

When a number is divided by four and then four subtracted from the result, the answer is greater than or equal to four. Find the values(s) that the number can take.

$$\frac{x}{4} - 4 > 4 \Rightarrow \frac{x}{4} > 8 \Rightarrow x > 32 \Rightarrow x \in (32, \infty)$$

Example 1.118

Operation I: *Multiply by 2*

Operation II: *Add 3*

- A. Consider a number x . Perform Operation I followed by Operation II. The answer is less than 3. Find the values that x can take.
- B. Consider a number y . Perform Operation II followed by Operation I. The answer is more than or equal to 3. Find the values that y can take.

Part A

$$2x + 3 < 3 \Rightarrow 2x < 0 \Rightarrow x < 0 \Rightarrow x \in (-\infty, 0)$$

Part B

$$2(y + 3) \geq 3 \Rightarrow 2y + 6 \geq 3 \Rightarrow 2y \geq -3 \Rightarrow y \geq -\frac{3}{2} \Rightarrow y \in \left[-\frac{3}{2}, \infty\right)$$

Example 1.119

Alan takes a number, multiplies it by two, subtracts four, and then divides by three. Allen takes the same number than Alan originally took, multiplies it by three, adds four, and then divides by two. Alan's final answer is less than or equal to Allen's answer. Find the values that the number can be.

$$\underbrace{\frac{2n - 4}{3}}_{\text{Alan}} \leq \underbrace{\frac{3n + 4}{2}}_{\text{Allen}}$$

Multiply both sides by 6:

$$\begin{aligned} 2(2n - 4) &\leq 3(3n + 4) \Rightarrow 4n - 8 \leq 9n + 12 \\ 4n &\leq 9n + 20 \\ -5n &\leq 20 \\ n &\geq -4 \end{aligned}$$

D. Word Problems on Scenarios

Example 1.120

E. Integral Answers

Example 1.121

For a party, Shaun invites p people over. He wants three slices of pizza for every person (and also for himself). He also wants five slices extra. Every pizza has seven slices. What is the minimum (integral) number of pizzas that he should order? (Answer in terms of p .)

$$\begin{aligned} \text{No. of Slices Needed} &= \underbrace{3}_{\substack{\text{Slices per} \\ \text{Person}}} \times \underbrace{(p + 1)}_{\substack{\text{No. of} \\ \text{People}}} + \underbrace{5}_{\substack{\text{Extra} \\ \text{Slices}}} = 3p + 8 \\ \text{Slices per Pizza: } &7 \end{aligned}$$

$$\text{No. of Pizzas} = \frac{\text{Total Slices}}{\text{Slices per Pizza}} = \frac{3p + 8}{7}$$

To get the number of pizzas, we must round up the expression above:

$$\left\lceil \frac{3p + 8}{7} \right\rceil$$

F. Variables

Example 1.122

Jason has three large bags of rice, each with l kg of rice. He uses the rice to make lunch every day for a week, and at the end of the week, he has 1 kg rice left over. The next week, he gets five medium sized bags of rice, each with m kg of rice. He makes lunch from the rice from Monday to Friday, and he has two kg rice left over. The total quantity of rice used in the first week is more than the total quantity of rice used in the second week.

A. Find the value of m in terms of l .

B. Find the value of l in terms of m .

$$\underbrace{3l - 1}_{\text{First Week}} > \underbrace{5m - 2}_{\text{Second Week}} \Rightarrow 3l > 5m - 1 \Rightarrow l > \frac{5m - 1}{3}$$

$$\underbrace{3l - 1}_{\text{First Week}} > \underbrace{5m - 2}_{\text{Second Week}} \Rightarrow 3l + 1 > 5m \Rightarrow \frac{3l + 1}{5} > m \Rightarrow m < \frac{3l + 1}{5}$$

Example 1.123

G. Fractions and Decimals

Example 1.124

Solve the following inequalities

$$\frac{2}{3}x - \frac{1}{2} > \frac{3}{5}$$

$$\frac{2}{3}x > \frac{11}{10} \Rightarrow x > \frac{33}{20} \Rightarrow x \in \left(\frac{33}{20}, \infty \right)$$

Example 1.125

What is the greatest integer value for m such that $5m - 3 > 8m + 21$? (MathCounts 1998 State Countdown)

Add 3 both sides:

$$5m > 8m + 24$$

Subtract $8m$ both sides:

$$-3m > 24$$

Divide both sides by -3 , and flip the inequality:

$$m < \frac{24}{-3}$$

Example 1.126

Nine copies of a certain pamphlet cost less than \$10.00 while ten copies of the same pamphlet (at the same price) cost more than \$11.00. How much does one copy of this pamphlet cost? (Answer to two decimal places) (AMC 8 1985/18)

$$9x < 10 \Rightarrow x < \frac{10}{9} \Rightarrow x < 1\frac{1}{9} = 1.\bar{1}$$

$$10x > 11 \Rightarrow x > \frac{11}{10} = 1\frac{1}{10} = 1.10$$

$$1.10 < x < 1.\bar{1} \Rightarrow x = 1.11$$

H. No Solutions

Example 1.127: Inequalities with no solutions

Solve $3 + 4(3x + 1) > 4(3x + 2) + 2$

$$3 + 4(3x + 1) > 4(3x + 2) + 2 \Rightarrow 3 + 12x + 4 > 12x + 8 + 2 \Rightarrow 7 > 10 \Rightarrow \text{No Solutions}$$

$$x = \phi$$

I. Reciprocals

	Example	Property
Case I (Same Sign)	x, y are both + ve: $2 < 3 \Rightarrow \frac{1}{2} > \frac{1}{3}$	$x < y \Rightarrow \frac{1}{x} > \frac{1}{y}$ Reverse the inequality when taking reciprocals if both sides have the same sign.
	x, y are both - ve: $-3 < -2 \Rightarrow -\frac{1}{3} > -\frac{1}{2}$	
Case II (Opposite Sign)		

J. Squares

Let x and y be positive.

If both are positive:

$$x < y \Rightarrow x^2 < y^2$$

$$3 < 4 \Rightarrow 3^2 < 4^2 \Rightarrow 9 < 16$$

If both are negative (Sign of the inequality reverses):

$$-x < -y \Rightarrow x^2 < y^2$$

$$-4 < -3 \Rightarrow 16 > 9$$

One is negative and one is positive:

$$-3 < 4 \Rightarrow 3^2 < 4^2$$

$$-5 < 4 \Rightarrow 25 > 16$$

K. Square Roots (and Absolute Value)

$$\sqrt{x^2} = |x|$$

Take an example:

$$\text{Case I: } x = 5 \Rightarrow x^2 = 25 \Rightarrow \sqrt{x^2} = \sqrt{25} = 5$$

$$y = -5 \Rightarrow y^2 = 25 \Rightarrow \sqrt{y^2} = \sqrt{25} = 5$$

1.7 Compound Inequalities

A. Compound Inequalities

Compound inequalities are inequalities where two conditions are written in the same inequality.

The simpler versions can be solved by manipulating the entire inequality in one go. The more complex versions must be broken into two different inequalities, each of which must be solved separately, and the merged using an intersection.

Example 1.128: Simpler Version

A. $\frac{3}{5} < x < 4, x \in \mathbb{Z}$

We can solve the inequality by manipulating it as one entity.

$$\frac{3}{5} < x < 4 \Rightarrow \frac{3}{5} < \frac{5x}{5} < \frac{20}{5} \Rightarrow \underbrace{3 < 5x < 20}_{\text{Make denominators common}} \Rightarrow \underbrace{3 < 5x < 20}_{\text{Multiply by 5 throughout}} \Rightarrow x \in \{1, 2, 3\}$$

B. AND Conditions

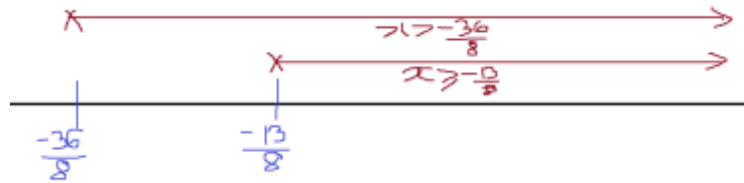
Example 1.129

Solve:

$$x \geq -\frac{13}{8} \text{ AND } x > -\frac{9}{2}$$

$$x \geq -\frac{13}{8}$$

$$x > -\frac{9}{2} \Rightarrow x > -\frac{36}{8}$$



$$x \geq -\frac{13}{8}$$

Example 1.130

$$4 - 3x \leq 13 \text{ AND } 6 + 2x > -2$$

$$-3x \leq 9 \Rightarrow x \geq -3$$

$$2x > -8 \Rightarrow x > -4$$

Final Answer:

$$x \geq -3$$

1.131: Tripartite Inequalities

$$a < b < c$$

$$a < b \text{ AND } b < c$$

Example 1.132

$$-5 < 2n - 1 \leq 5$$

$$-4 < 2n \leq 6$$

$$-2 < n \leq 3$$

Example 1.133

$$2x < x + 1 < 2x + 5$$

If you have variables in more than one part of the inequality, it is generally required to break the inequality into two parts:

$$\begin{aligned} 2x < x + 1 &\Rightarrow x < 1 \\ x + 1 < 2x + 5 &\Rightarrow -x < 4 \Rightarrow x > -4 \\ x < 1 \text{ AND } x > -4 &\Rightarrow -4 < x < 1 \Rightarrow (-4, 1) \end{aligned}$$

Example 1.134

Consider the compound inequality $3x + \frac{1}{2} < x - \frac{3}{4} < \frac{1}{2}x + \frac{2}{3}$

- Find the solution set that satisfies the inequality.
- What is the largest integer that satisfies the above inequality?
- What is the largest natural number that satisfies the above inequality?

Part A

$$\begin{aligned} \text{First Condition: } 3x + \frac{1}{2} < x - \frac{3}{4} &\Rightarrow 2x < -\frac{3}{4} - \frac{1}{2} = -\frac{5}{4} \Rightarrow x < -\frac{5}{8} \\ \text{Second Condition: } x - \frac{3}{4} < \frac{1}{2}x + \frac{2}{3} &\Rightarrow \frac{1}{2}x < \frac{3}{4} + \frac{2}{3} = \frac{17}{12} \Rightarrow x < \frac{17}{6} \\ \text{Merge the two Conditions: } x < -\frac{5}{8} \text{ AND } x < \frac{17}{6} &\Rightarrow \underbrace{x < -\frac{5}{8}}_{\text{Inequality Notation}} \Rightarrow \underbrace{x \in \left(-\infty, -\frac{5}{8}\right)}_{\text{Interval Notation}} \end{aligned}$$

Part B

We cannot have positive answers, nor zero. We look for the first integer that we will encounter to the left of $-\frac{5}{8}$ on the number line, which is -1 .

Part C

No natural numbers satisfy the inequality. Hence, there are no solutions.

Example 1.135

An electrical device supports a minimum temperature of -10°C and a maximum temperature of 40°C (both inclusive). Convert the given range into degrees Fahrenheit using the conversion $\frac{5}{9}(F - 32) = C$.

Write out the condition given in the question in inequality form:

$$-10 \leq C \leq 40$$

Substitute the expression for degrees Fahrenheit:

$$-10 \leq \frac{5}{9}(F - 32) \leq 40$$

Multiply through the inequality by $\frac{9}{5}$:

$$\left(\frac{9}{5}\right)(-10) \leq \left(\frac{9}{5}\right)\frac{5}{9}(F - 32) \leq \left(\frac{9}{5}\right)40$$

Simplify:

$$-18 \leq F - 32 \leq 72$$

Add 32 to all parts of the inequality:

$$-18 + 32 \leq F - 32 + 32 \leq 72 + 32$$

Simplify:

$$14 \leq F \leq 104$$

Example 1.136

A bakery needs to make dough for 100 pizzas. Since there is some estimation involved, it does not need to make exactly 100 pizzas. Rather, the final number of pizzas should be at least 90, or less than 110.

- Write a compound inequality that the number of pizzas must satisfy.
- Combine the two inequalities into a single inequality using absolute value.

The number of pizzas should be between 90 and 100 (both included).

$$\underbrace{90}_{Min} \leq p \leq \underbrace{110}_{Max}$$

Subtract 100 from all three parts of the inequality:

$$-10 \leq p - 100 \leq 10$$

Combine using absolute value:

$$|p - 100| \leq 10$$

The absolute value of the difference of the number of pizzas and 100 should be less than or equal to 10.

Example 1.137

$$\frac{3}{5}(4 - x) \leq -\frac{1}{3}(x + 2) \text{ AND } (x - 1) + (3x + 6) > 2(x - 9)$$

Multiply the first inequality by 15 both sides:

$$\begin{aligned} 9(4 - x) &\leq -5(x + 2) \\ 36 - 9x &\leq -5x - 10 \\ -4x &\leq -46 \\ x &\geq -\frac{46}{-4} \\ x &\geq \frac{23}{2} \end{aligned}$$

Simplify the second inequality:

$$\begin{aligned} 4x + 5 &> 2x - 18 \\ 2x &> -23 \\ x &> -\frac{23}{2} \end{aligned}$$

Example 1.138

One morning each member of Angela's family drank an 8-ounce mixture of coffee with milk. The amounts of coffee and milk varied from cup to cup, but were never zero. Angela drank a quarter of the total amount of milk and a sixth of the total amount of coffee. How many people are in the family?¹ (AMC 10 2000/22, AMC 12 2000/13)

Method I

Let the number of people in the family be p .

Angela drank a greater proportion of the milk ($\frac{1}{4}$ th) than she drank of the coffee ($\frac{1}{6}$ th).

If the milk had been shared equally, then Angela would have got:

$$\frac{1}{p} \text{ of the milk}$$

But she drank a greater proportion of the milk than the coffee:

¹ An alternate solution to this using Remainders can be found in the Number Theory Note on Mod Arithmetic.

$$\frac{1}{p} < \frac{1}{4}$$

Inequality I

But she drank a smaller proportion of the coffee than the milk:

$$\frac{1}{6} < \frac{1}{p}$$

Inequality II

Combine I and II to get:

$$\frac{1}{6} < \frac{1}{p} < \frac{1}{4} \Rightarrow 6 > p > 4$$

And since p is an integer, we must have:

$$p = 5$$

Method II

We can do this by introducing variables for each quantity under consideration. The resulting solution is lengthier and more cumbersome. Let

$$\text{Total amt of coffee} = c$$

$$\text{Total amt of milk} = m$$

$$\text{No. of people} = p$$

Note that Angela drank:

$$\frac{c}{6} + \frac{m}{4} = 8 \text{ ounces}$$

The total number of ounces consumed is then:

$$= 8p = \left(\frac{c}{6} + \frac{m}{4}\right)p = \frac{2cp + 3mp}{12}$$

And it is also:

$$c + m$$

Hence:

$$\frac{2cp + 3mp}{12} = c + m$$

Multiply by 12:

$$2cp + 3mp = 12c + 12m$$

Rearrange to get c terms on one side and m terms on the other side:

$$2cp - 12c = 12m - 3mp$$

Factor:

$$2c(p - 6) = 3m(4 - p)$$

Since $c > 0, m > 0$:

Both $p - 6$ and $4 - p$ must have the same sign

If both are positive:

$$p - 6 > 0 \Rightarrow p > 6$$

$$4 - p > 0 \Rightarrow p < 4 \Rightarrow \text{No Solution}$$

Hence, both must be negative:

$$p - 6 < 0 \Rightarrow p < 6$$

$$4 - p < 0 \Rightarrow 4 < p \Rightarrow p > 4$$

And since $p \in \mathbb{N}$:

$$p = 5$$

C. Multiple Conditions

Example 1.139

$$(x > 5 \text{ and } x < 10) \text{ and } (x > 7)$$

$$5 < x < 10 \text{ AND } x > 7$$

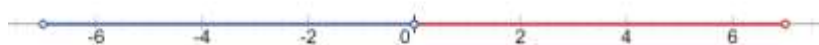


$$7 < x < 10 \Rightarrow (7, 10)$$

Example 1.140

$$(x > 0 \text{ and } x < 7) \text{ OR } (x < 0 \text{ and } x > -7)$$

$$(-7 < x < 0) \text{ OR } (0 < x < 7)$$



$$(-7, 0) \cup (0, 7)$$

1.8 Absolute Value Inequalities

A. Warmup

Before we introduce some rules, we solve some questions based only on logic. In the next heading, we generalize these questions to get properties that will be useful in solving inequalities.

Example 1.141

Match the column below. Solve the inequalities for integer values. Use this to find the number of solutions, and hence, solve them for the valid interval on the number line.

Inequality	Inequality Notation	Interval	Integral Solutions	No. of Integral Solutions
A. $ x < 3$	$-3 < x < 3$	I. $x \in (-2, 2)$	X. $\{-2, -1, 0, 1, 2\}$	3
B. $ x \leq 3$	$-3 \leq x \leq 3$	II. $x \in [-2, 2]$	Y. $\{-3, -2, -1, 0, 1, 2, 3\}$	5
C. $ x < 2$	$-2 < x < 2$	III. $x \in (-3, 3)$	Z. $\{-1, 0, 1\}$	7
D. $ x \leq 2$	$-2 \leq x \leq 2$	IV. $x \in [-3, 3]$		9

$$A - III - X - 5$$

The largest value of x which satisfies $|x| < 3$ is a number just less than three. Any number more than three, or three itself will have an absolute value more than 3.

$$|3.5| = 3.5 > 3$$

Similarly, any value of x which is less than -3 will not satisfy the inequality.

For example

$$|-3.5| = 3.5 > 3$$

Hence, the numbers which satisfy the inequality are the number between -3 and 3 . The endpoints are not included since the inequality is strict (it does have the equals sign).

Therefore, the interval that satisfies the inequality is

$$(-3, 3) \Rightarrow \text{All real numbers between } -3 \text{ and } 3 \text{ on the number line}$$

$$B - IV - Y - 7$$

The only difference between A and B is that B also includes the endpoints of the range, since it has an equals sign.

$$C - I - Z - 3$$

$$D - II - X - 5$$

Example 1.142

Match the column below. Solve the inequalities for integer values. Use this to find the number of solutions, and hence, solve them for the valid interval on the number line.

Inequality	Inequality Notation	Interval	Integral Solutions
A. $ x > 4$		I. $x \in (-\infty, -3) \cup (3, \infty)$	X. $\{-5, -4, -3\} \cup \{3, 4, 5 \dots\}$
B. $ x \geq 4$		II. $x \in (-\infty, -4) \cup (4, \infty)$	Y. $\{\dots - 6, -5, -4\}$ $\cup \{4, 5, 6 \dots\}$
C. $ x > 3$		III. $x \in (-\infty, -5) \cup (5, \infty)$	Z. $\{\dots - 7, -6, -5\}$ $\cup \{5, 6, 7 \dots\}$
D. $ x \geq 3$		IV. $x \in (-\infty, -3] \cup [3, \infty)$	
		V. $x \in (-\infty, -4] \cup [4, \infty)$	
		VI. $x \in (-\infty, -5] \cup [5, \infty)$	

Part A

$|x| > 4$ will take minimum value 5, if x is an integer. But if x is not restricted to be an integer, it will take values between 4 and 5 as well.

4.1 is a valid solution for A. But if we match A with III, we do not get 4.1 in our solutions.

$$A - II - Z$$

Part B

The $\geq$ sign lets us include the endpoints. Hence, in interval notation, the answer remains the same in terms of numbers, but the round brackets change to square brackets.

$$B - V - Y$$

Part C

$$C - I - Y$$

Part D

$$D - IV - X$$

B. Properties

Example 1.143

Find the solution set that satisfies

$$|x| < 5$$

To structure your thinking, break the problem down into three parts:

- A. Positive Numbers
- B. Zero
- C. Negative Numbers

Part A

Any number larger than 5 is not going to work.

$$|100| = 100 > 5 \Rightarrow \text{Does not work}$$

$$|6| = 6 > 5 \Rightarrow \text{Does not work}$$

In fact,

$$|5| = 5 \Rightarrow \text{Also does not work}$$

Because we want a number that is less than 5.

Hence, the positive values that will work are:

$$x < 5, \quad x \text{ is } +ve$$

Part B

Zero also works

Part C

Any number smaller than -5 is not going to work.

$$|-100| = 100 > 5 \Rightarrow \text{Does not work}$$

$$|-6| = 6 > 5 \Rightarrow \text{Does not work}$$

In fact,

$$|-5| = 5 \Rightarrow \text{Also does not work}$$

Because we want a number that is less than 5.

Hence, the negative values that will work are:

$$x > -5, \quad x \text{ is } -ve$$

Combine the above three to get:

$$x < 5, \quad x \text{ is } +ve$$

$$x = 0$$

$$x > -5, \quad x \text{ is } -ve$$

And these three can be written in one inequality as:

$$-5 < x < 5$$



$$|x| < 5 \Leftrightarrow -5 < x < 5 \Leftrightarrow x \in (-5, 5)$$

1.144: Less Than Absolute Value Inequalities

A less than absolute value inequality will convert into an AND inequality and *vice versa*.

Assume that c is a positive number.

Property		Absolute Value Notation	Inequality Notation	Interval Notation
I	A	$ x \leq c$	$-c \leq x \leq c$	$x \in [-c, c]$
	B	$ x < c$	$-c < x < c$	$x \in (-c, c)$

1.145: "Negative" Absolute Values

For a positive number c :

$$|x| < -c \Rightarrow x \in \phi$$

$$|x| \leq -c \Rightarrow x \in \phi$$

There are no solutions to the inequalities above, because an absolute value expression can never be negative.

Example 1.146

Find the solution set for the inequalities below. Write your answer in both inequality notation, and interval notation.

- A. $|x| < 4$
- B. $|x| \leq 3$
- C. $|x| < \frac{1}{2}$
- D. $|x| < -7$
- E. $|x| + 2 < 5$
- F. $\frac{1}{3}|x| + 2 > \frac{1}{2}|x| + 1$
- G. $|x| \leq -9$
- H. $|x + 1| \leq 4$

Part A

$$-4 < x < 4 \Leftrightarrow x \in (-4, 4)$$

Part B

$$|x| \leq 3 \Leftrightarrow -3 \leq x \leq 3 \Leftrightarrow [-3, 3]$$

Part C

$$x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Part D

$$|x| < -7 \Rightarrow x \in \phi$$

Part E

$$|x| < 3 \Leftrightarrow x \in (-3, 3)$$

Part F

$$\frac{1}{3}|x| > \frac{1}{2}|x| - 1 \Rightarrow 2|x| > 3|x| - 6 \Rightarrow -|x| > -6$$

$$|x| < 6 \Rightarrow x \in (-6, 6)$$

Part G

$$|x| \leq -9 \Rightarrow x \in \phi$$

Part I

$$\begin{aligned} |x + 1| &\leq 4 \\ -4 &< x + 1 < 4 \\ -5 &< x < 3 \end{aligned}$$

1.147: Absolute Values greater than a Negative Number

For a positive number c :

$$|x| > -c \Rightarrow x \in (-\infty, \infty) \Leftrightarrow x \in \mathbb{R}$$

$$|x| \geq -c \Rightarrow x \in (-\infty, \infty) \Leftrightarrow x \in \mathbb{R}$$

An absolute value expression can never be negative. Hence, all inputs are valid for the expression. Hence, the solution is the real number line.

Equality will never be achieved in the condition above, since the absolute value of an expression is never negative.

$$\therefore |x| = -c \Rightarrow x = \phi$$

Example 1.148

Solve

A. $|x| > -7$

B. $|x| \geq -9$

$$|x| > -7 \Rightarrow x \in (-\infty, \infty) \Leftrightarrow x \in \mathbb{R}$$

$$|x| \geq -9 \Rightarrow x \in (-\infty, \infty) \Leftrightarrow x \in \mathbb{R}$$

1.149: Absolute Values greater than a positive number

A greater than absolute value inequality will convert into an OR inequality and *vice versa*.

$$|x| > c \Rightarrow x < -c \text{ OR } x > c \Leftrightarrow x \in (-\infty, -c) \cup (c, \infty)$$

$$|x| \geq c \Rightarrow x \leq -c \text{ OR } x \geq c \Leftrightarrow x \in (-\infty, -c] \cup [c, \infty)$$

Example 1.150

Solve:

A. $|x| > 7$

B. $|x| \geq 9$

C. $|x + 8| > 1$

$$|x| > 7 \Leftrightarrow x < -7 \text{ OR } x > 7$$

$$|x| \geq 9 \Leftrightarrow x \leq -9 \text{ OR } x \geq 9$$

$$|x + 8| > 1$$

$$x + 8 > 1 \Rightarrow x > -7$$

$$x + 8 < -1 \Rightarrow x < -9$$

1.151: Summary Table

$$|x| < c \Rightarrow -c < x < c, \quad c > 0$$

$$|x| < -c \Rightarrow x \in \phi, \quad c > 0$$

$$|x| > -c \Rightarrow x \in \mathbb{R}, \quad c > 0$$

$$|x| > c \Rightarrow x > c \text{ OR } x < -c, \quad c > 0$$

Property	Example
I	$ x < 5 \Leftrightarrow -5 < x < 5 \Leftrightarrow x \in (-5, 5)$ $ x \leq \pi \Leftrightarrow -\pi \leq x \leq \pi \Leftrightarrow x \in [-\pi, \pi]$
II	$ x \leq -3 \Leftrightarrow x \in \phi$

	$ x \leq -e \Leftrightarrow x \in \phi$
III	$ x > -3 \Rightarrow x \in (-\infty, \infty) \Leftrightarrow x \in \mathbb{R}$
IV	$ x > 3 \Rightarrow x \in (-\infty, -3) \cup (3, \infty) \Leftrightarrow x < -3 \text{ OR } x > 3$

C. Solving Absolute Value Inequalities

Example 1.152

- A. $|x| > 4$
- B. $|x| < 4$
- C. $|2c - 5| < 9$
- D. $|3c + 2| < 4$
- E. $|x + 7| \leq 2$
- F. $|x - 1| \geq 3$
- G. $3|2 - 2x| - 7 \leq 29$
- H. $|2x + 1| - 5 < 0$
- I. $4 - 3|2x + 1| > -2$
- J. $|2x - 5| \leq 3x$

Part A

$$x > 4 \text{ OR } x < -4$$

Part B

$$x < 4 \text{ AND } x > -4 \Rightarrow -4 < x < 4$$

Part C

Recall that if $|x| < c \Leftrightarrow -c < x < c$:

$$-9 < 2c - 5 < 9$$

$$-4 < 2c < 14$$

$$-2 < c < 7$$

$$x \in (-2, 7)$$

Part D

$$-4 < 3c + 2 < 4$$

Subtract 2 from all parts:

$$-6 < 3c < 2$$

Divide all three parts by 3:

$$-2 < c < \frac{2}{3} \Rightarrow x \in \left(-2, \frac{2}{3}\right)$$

Part E

$$-2 \leq x + 7 \leq 2$$

$$-9 \leq x + 7 \leq -5$$

$$x \in [-5, -9]$$

Part F

$$x - 1 \geq 3 \Rightarrow x \geq 4 \Rightarrow x \in [4, \infty)$$

$$x - 1 \leq -3 \Rightarrow x \leq -2 \Rightarrow x \in (-\infty, -2]$$

$$(-\infty, -2] \cup [4, \infty)$$

Part G

Add 7 to both sides, and factor out 2 from the absolute value:

$$6|1 - x| \leq 36$$

$$|1 - x| \leq 6$$

$$-6 \leq 1 - x < 6$$

$$-7 \leq -x < 5$$

$$7 \geq x > -5$$

$$-5 \leq x \leq 7$$

Part H

$$|2x + 1| - 5 < 0$$

$$|2x + 1| < 5$$

$$-5 < 2x + 1 < 5$$

$$-6 < 2x < 4$$

$$-3 < x < 2$$

$$x \in (-3, 2)$$

Part I

$$-3|2x + 1| > -6 \Rightarrow |2x + 1| < 2$$

Apply Property I

$$-2 < 2x + 1 < 2$$

Subtract one throughout:

$$-3 < 2x < 1$$

Divide by 2 throughout:

$$-\frac{3}{2} < x < \frac{1}{2}$$

Write it in interval notation:

$$x \in \left(-\frac{3}{2}, \frac{1}{2}\right)$$

Part J

Use the property that $|y| \leq a \Leftrightarrow -a < y < a$:

$$-3x \leq 2x - 5 \leq 3x$$

First Part:

$$-3x \leq 2x - 5 \Rightarrow -5x \leq -5 \Rightarrow x \geq 1$$

Second Part:

$$2x - 5 \leq 3x \Rightarrow -x \leq 5 \Rightarrow x \geq -5$$

Both the inequalities need to be satisfied:

$$x \geq 1 \text{ AND } x \geq -5 \Rightarrow x \geq 1$$

D. Critical Points

A critical point of an expression is a value at which the expression changes its behavior. For example, consider the equation

$$y = x + 5$$

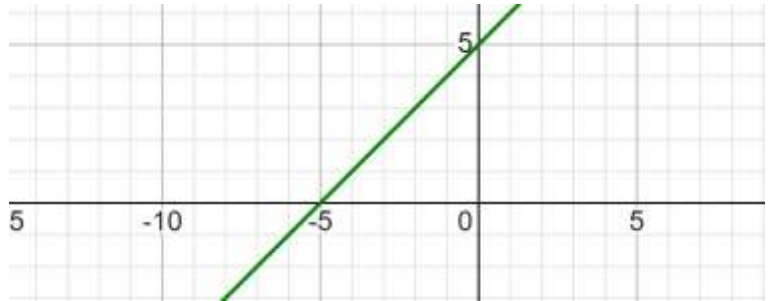
Which is graphed alongside.

From the graph, we see that:

$$x < -5 \Rightarrow y < 0$$

$$x > -5 \Rightarrow y > 0$$

$$x = -5 \Rightarrow y = 0$$



x	-8	-7	-6	-5	-4	-3	-2	-1	0
$x + 5$	-3	-2	-1	0	1	2	3	4	5
	<i>Negative Interval</i>			Critical Point	<i>Positive</i>				
	$(-\infty, -5)$ OR $x < -5$				$(-5, \infty)$ OR $x > -5$				

We are interested in finding the value of the critical point. Tabulating or graphing is not feasible for every expression.

A continuous function will not change sign from positive to negative, or negative to positive, without crossing the x -axis.

Hence, we find all points where the expression is zero. This lets us identify critical points.

Critical Points for Products

Critical Points for Products

$$abc = 0$$

abc is a product. Hence, if any of a , b or c become zero, the entire expression is zero.

Critical Points for Sum of Absolute Values

$$|x| + |y| + |z| = 0$$

Analyzing

$$x = 0, \quad y = 0, \quad z = 0$$

is useful since the behaviour of the expression changes

for $x > 0$, and $x < 0$

Example 1.153

$$3|2 - 2x| - 7 \leq 29$$

Add 7 to both sides, and factor out 2 from the absolute value:

$$6|1 - x| \leq 36$$

$$|1 - x| \leq 6$$

Apply casework on the absolute value:

$$1 - x \geq 0 \Rightarrow -x \geq -1 \Rightarrow x \leq 1$$

Case I: $x \leq 1$

$$1 - x \leq 6$$

$$-x \leq 5$$

$$x \geq -5$$

Case II: $x > 1$

$$x - 1 \leq 6$$

$$x \leq 7$$

Combine the results from the two cases to get:

$$-5 \leq x \leq 7$$

Example 1.154

Solve

$$4 - 3|2x + 1| > -2$$

Just like absolute value equations, here also we isolating the absolute value expression on side:

$$-3|2x + 1| > -6 \Rightarrow |2x + 1| < 2$$

Apply Property I

Break into two separate inequalities

$$\text{Case I: } 2x + 1 < 2 \Rightarrow 2x < 1 \Rightarrow x < \frac{1}{2}$$

$$\text{Case II: } 2x + 1 > -2 \Rightarrow 2x > -3 \Rightarrow x > -\frac{3}{2}$$

$$x \in \left(-\frac{3}{2}, \frac{1}{2}\right)$$

Solve the compound inequality directly

$$-2 < 2x + 1 < 2$$

Subtract one throughout:

$$-3 < 2x < 1$$

Divide by 2 throughout:

$$-\frac{3}{2} < x < \frac{1}{2}$$

Write it in interval notation:

$$x \in \left(-\frac{3}{2}, \frac{1}{2}\right)$$

Example 1.155

$$|2x - 5| \leq 3x$$

$$2x - 5 \geq 0 \Rightarrow x \geq \frac{5}{2}$$

Case I: $x \geq \frac{5}{2}$

$$|2x - 5| = 2x - 5$$

$$2x - 5 \leq 3x \Rightarrow -x \leq 5 \Rightarrow x \geq -5 \Rightarrow x \in [-5, \infty)$$

But since $x \geq \frac{5}{2}$:

$$x \in \left[\frac{5}{2}, \infty\right)$$

Case II: $x < \frac{5}{2}$

$$\begin{aligned} |2x - 5| &= 5 - 2x \\ 5 - 2x &\leq 3x \Rightarrow -5x \leq -5 \Rightarrow x \geq 1 \Rightarrow x \in [1, \infty) \end{aligned}$$

But since $x < \frac{5}{2}$:

$$x \in \left[1, \frac{5}{2}\right)$$

And then we can combine both cases:

$$x \in \left[1, \frac{5}{2}\right) \cup \left[\frac{5}{2}, \infty\right) \Rightarrow x \in [1, \infty)$$

Example 1.156

$$|2x - 5| \leq x - 4$$

$$2x - 5 \geq 0 \Rightarrow 2x \geq 5 \Rightarrow x \geq \frac{5}{2}$$

Case I: $x \geq \frac{5}{2}$

$$2x - 5 \leq x - 4 \Rightarrow x \leq 1 \Rightarrow \text{Contradiction}$$

Case I: $x < \frac{5}{2}$

$$5 - 2x \leq x - 4 \Rightarrow -3x < -9 \Rightarrow x > 3 \Rightarrow \text{Contradiction}$$

Hence, there are no solutions to the above inequality.

Example 1.157

$$13 - 2|3x - 4| \geq 3$$

Isolate the absolute value:

$$-2|3x - 4| \geq -10 \Rightarrow |3x - 4| \leq 5$$

Method I: Convert to a compound inequality

$$-5 \leq 3x - 4 \leq 5 \Rightarrow -1 \leq 3x \leq 9 \Rightarrow -\frac{1}{3} \leq x \leq 3$$

Method II: Critical Points

$$3x - 4 > 0 \Rightarrow 3x > 4 \Rightarrow x > \frac{4}{3}$$

$$\text{Case I: } x \geq \frac{4}{3} \Rightarrow 3x - 4 \leq 5 \Rightarrow 3x \leq 9 \Rightarrow x \leq 3 \Rightarrow x \in \left[\frac{4}{3}, 3\right]$$

$$\text{Case II: } x \leq \frac{4}{3} \Rightarrow 4 - 3x \leq 5 \Rightarrow -3x \leq 1 \Rightarrow x \geq -\frac{1}{3} \Rightarrow x \in \left[-\frac{1}{3}, \frac{4}{3}\right]$$

$$\left[-\frac{1}{3}, 3\right]$$

Example 1.158

Solve

$$|4x + 2| > x + 3$$

$$4x + 2 > 0 \Rightarrow 4x > -2 \Rightarrow x > -\frac{1}{2}$$

Case I: $x \geq -\frac{1}{2}$

$$4x + 2 \geq x + 3 \Rightarrow 3x \geq 1 \Rightarrow x \geq \frac{1}{3}$$

Case I: $x < -\frac{1}{2}$

$$-4x - 2 > x + 3$$

$$-5 > 5x$$

$$-1 > x$$

$$x < -1$$

Final Answer

$$x < -1 \text{ OR } x > \frac{1}{3} \Leftrightarrow (-\infty, -1) \cup \left[\frac{1}{3}, \infty\right)$$

Example 1.159

Find the critical points of

$$|x + 2| + |x - 3| = 0$$

First Critical Point

The first critical point is obtained by equating the first expression inside the absolute value to zero:

$$x + 2 = 0 \Rightarrow x = -2$$

-2 divides the real number line into two regions:

$$(-\infty, -2) \cup (-2, \infty)$$

We will check the behaviour of the expression $x + 2$ for a single value in each region. A single value is sufficient because the expression will not its behaviour sign in the same region.

Right of -2:

$$\text{When } x = 0: x + 2 = 0 + 2 = 2 = +ve$$

Hence, we can conclude:

$$|x + 2| = x + 2, \text{ when } x > -2$$

Left of -2:

$$\text{When } x = -3: x + 2 = -3 + 2 = -1 = -ve$$

Similarly, we can conclude:

$$|x + 2| = -(x + 2), \text{ when } x < -2$$

Conclusion in Piece Wise Notation

The same thing can be written in piece-wise notation as follows:

$$|x + 2| = \begin{cases} x + 2, & \text{when } x > -2 \\ -(x + 2), & \text{when } x < -2 \end{cases}$$

Second Critical Point

$$x - 3 = 0 \Rightarrow x = 3$$

$$x = 4 \Rightarrow x - 3 = 4 - 3 = 1 = +ve \Rightarrow |x - 3| = x - 3, x \in \{3, \infty\}$$

$$x = 0 \Rightarrow x - 3 = 0 - 3 = -3 = -ve \Rightarrow |x - 3| = -(x - 3), x \in \{\infty, 3\}$$

We don't need to write the piece wise notation here. It just expresses the same information in a more compact manner.

Example 1.160

Find the length of the largest interval that satisfies $(A \cup B)'$ over the domain $U \in (-3, 3)$ given that:

$$A: |3x + 4| < 2x + 5$$

$$B: |4x - 5| < 3x + 2$$

Condition A

Find the critical point

$$3x + 4 > 0 \Rightarrow x > -\frac{4}{3}$$

$$\text{Case I: } x > -\frac{4}{3}:$$

$$3x + 4 < 2x + 5 \Rightarrow x < 1$$

$$\text{Case II: } x < -\frac{4}{3}:$$

$$-(3x + 4) < 2x + 5$$

$$3x + 4 > -2x - 5$$

$$5x > -9$$

$$x > -\frac{9}{5}$$

$$A: \left(-\frac{9}{5}, 1\right)$$

Condition B

$$4x - 5 > 0 \Rightarrow x > \frac{5}{4}$$

$$\text{Case I: } x > \frac{5}{4}:$$

$$4x - 5 < 3x + 2$$

$$x < 7$$

$$x \leq 3$$

$$\text{Case II: } x < \frac{5}{4}:$$

$$-(4x - 5) < 3x + 2$$

$$4x - 5 > -3x - 2$$

$$7x > 3$$

$$x > \frac{3}{7}$$

$$B: \left(\frac{3}{7}, 3\right]$$

Combine the Conditions

$$(A \cup B) \in \left(-\frac{9}{5}, 3\right]$$

$$(A \cup B)' \in \left[-3, -\frac{9}{5}\right]$$

There is only a single interval, whose length is given by:

$$\left|-3 - \left(-\frac{9}{5}\right)\right| = \left|-\frac{15}{5} + \frac{9}{5}\right| = \left|-\frac{6}{5}\right| = 1.2$$

E. Two Absolute Value Expressions

If you have more than one absolute value expression, you will need to determine the range over which expression is positive or negative.

Then, you can will have to solve based on cases.

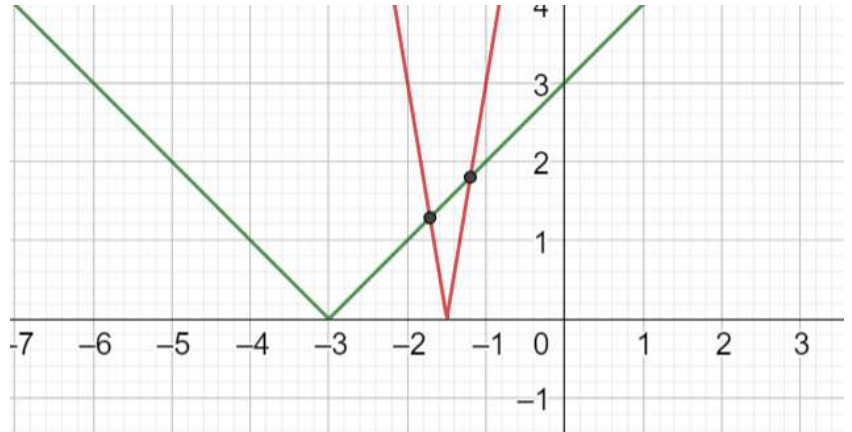
Example 1.161: Solving Using Inequalities

$$|x + 3| \geq |6x + 9|$$

Strategy

There are two ways of solving this.

- The first one, using inequalities, is more conceptual, but also has more involved calculations.
- The second one, uses equations instead of inequalities. But it is necessary to understand the method behind it first.



Finding the Critical points

$$|x + 3| \geq |6x + 9|$$

First, we find the regions where each absolute value expressions is greater than zero.

$$x + 3 \geq 0 \Rightarrow x \geq -3 \Rightarrow \text{Critical Point} = x = -3$$

$$6x + 9 \geq 0 \Rightarrow 6x \geq -9 \Rightarrow x \geq -\frac{9}{6} \Rightarrow x \geq -\frac{3}{2} \Rightarrow \text{Critical point: } x = -1\frac{1}{2}$$

Finding the Intervals

The critical points divide the number line into three intervals

$$(-\infty, -3) \cup \left(-3, -1\frac{1}{2}\right) \cup \left(-1\frac{1}{2}, \infty\right)$$

Solve the inequality for each Interval

Case I: $x < -3$

$$-(x + 3) \geq -(6x + 9) \Rightarrow x + 3 \leq 6x + 9 \Rightarrow -5x \leq 6 \Rightarrow x \geq -\frac{6}{5} \Rightarrow x = \phi$$

Case II: $-3 < x < -1\frac{1}{2} \Rightarrow -3 < x < -1\frac{7}{14}$

$$x + 3 \geq -(6x + 9) \Rightarrow x + 3 \geq -6x - 9 \Rightarrow 7x \geq -12 \Rightarrow x \geq -\frac{12}{7} = -\frac{60}{35}$$

Case III: $x > -1\frac{1}{2} \Rightarrow x > -1\frac{5}{10}$

$$x + 3 \geq 6x + 9 \Rightarrow -5x \geq 6 \Rightarrow x \leq -\frac{6}{5} = -\frac{42}{35}$$

Merge the Cases

$$\{\phi\} \text{ OR } \left[-\frac{60}{35}, \infty\right) \text{ OR } \left(-\infty, -\frac{42}{35}\right] \Rightarrow \left[-\frac{60}{35}, -\frac{42}{35}\right] = \left[-\frac{12}{7}, -\frac{7}{6}\right]$$

Example 1.162

Find the critical points of

- $|x + 3|$
- $|6x + 9|$

Find the Critical Points

$$x + 3 = 0 \Rightarrow x = -3 \Rightarrow \text{Critical Point: } x = -3$$

$$6x + 9 = 0 \Rightarrow x = -\frac{9}{6} = -\frac{3}{2} = -1\frac{1}{2} \Rightarrow \text{Critical points: } x = -1\frac{1}{2}$$

Find the positive and negative intervals

$$x + 3: \underbrace{(-\infty, -3)}_{\text{Negative}} \cup \underbrace{(-3, \infty)}_{\text{Positive}}, \quad 6x + 9: \underbrace{\left(-\infty, -1\frac{1}{2}\right)}_{\text{Negative}} \cup \underbrace{\left(-1\frac{1}{2}, \infty\right)}_{\text{Positive}}$$

Example 1.163: Solve using Critical Points

$$|x + 3| \geq |6x + 9|$$

$$|x + 3| - |6x + 9| \geq 0$$

Making the Cases

We want to see when the expressions inside the absolute value will be positive and negative.

		$x + 3$	
		Positive	Negative
$6x + 9$	Positive	$\underbrace{(x + 3) = (6x + 9)}_{\text{I}}$	$\underbrace{-(x + 3) = (6x + 9)}_{\text{II}}$
	Negative	$\underbrace{(x + 3) = -(6x + 9)}_{\text{III}}$	$\underbrace{-(x + 3) = -(6x + 9)}_{\text{IV}}$

Case IV is the same as Case I
Case II is the same as Case III

Hence, we only need to solve

Case I and Case II

Finding the Critical Points

To find the critical points, we solve the equations that we found:

$$x + 3 = 6x + 9 \Rightarrow -6 = 5x \Rightarrow x = -\frac{6}{5} = -\frac{42}{35}$$

$$x + 3 = -(6x + 9) \Rightarrow x + 3 = -6x - 9 \Rightarrow 7x = -12 \Rightarrow x = -\frac{12}{7} = -\frac{60}{35}$$

Critical Points are $-\frac{12}{7}$ and $-\frac{6}{5}$

Finding the Intervals and checking them

$$\underbrace{\left(-\infty, -\frac{12}{7}\right)}_{\text{Interval I Negative}} \cup \underbrace{\left(-\frac{12}{7}, -\frac{6}{5}\right)}_{\text{Interval II Positive}} \cup \underbrace{\left(-\frac{6}{5}, +\infty\right)}_{\text{Interval III Negative}}$$

Check Interval I by substituting -2 :

$$|x + 3| - |6x + 9| = |-2 + 3| - |6(-2) + 9| = |1| - |-3| = 1 - 3 = -2 = -ve$$

Check Interval II by substituting -1.5 :

$$|x + 3| - |6x + 9| = |-1.5 + 3| - |6(-1.5) + 9| = |1.5| - |0| = 1.5 = +ve$$

Check Interval III by substituting $x = 0$:

$$|x + 3| - |6x + 9| = |0 + 3| - |6(0) + 9| = |3| - |9| = -6 = -ve$$

Final Solution

Reject the negative intervals and keep the positive intervals. Since the original inequality was $\geq$

We will keep the endpoints.

$$\left[-\frac{12}{7}, -\frac{6}{5}\right]$$

F. Compound Inequalities

Solving absolute value compound inequalities is a lengthy process. You will need to:

- Rewrite the compound inequality as two separate inequalities

- Solve each inequality separately. Some of the inequalities may themselves have cases. Combine the solution set for these cases
- Combine the final solution set using an intersection of solution sets.

Example 1.164

Solve

$$2 < |x - 1| \leq 5$$

Approach

This is tricky because

- It is a compound inequality
- The middle term has absolute value
- The left-hand side has a $<$ sign, which the right-hand side has a $\leq$ sign. In other words, from one part of it, the endpoint is included, while from the other part, the endpoint is not included.

We have learnt how to handle each of these individually. But the chances of making a mistake go up as the number of concepts to track increases.

Break up the compound inequality: Inequality I

$$2 < |x - 1|$$

Reverse the sign of the inequality:

$$|x - 1| > 2$$

Apply Property IV:

$$\begin{aligned} \text{Case I: } x - 1 > 2 &\Rightarrow x > 3 \Rightarrow (3, \infty) \\ \text{Case II: } x - 1 < -2 &\Rightarrow x < -1 \Rightarrow (-\infty, -1) \\ &\underline{(-\infty, -1) \cup (3, \infty)} \\ &\text{From Case I and II} \end{aligned}$$

Break up the compound inequality: Inequality II

$$|x - 1| \leq 5 \Rightarrow -5 \leq x - 1 \leq 5 \Rightarrow -4 \leq x \leq 6 \Rightarrow x \in [-4, 6]$$

Combine the solution sets using AND condition

Combine the two solution ranges. The final solution needs to satisfy both.

$$\{(-\infty, -1) \cup (3, \infty)\} \cap [-4, 6] \Rightarrow [-4, -1) \cup (3, 6]$$

G. Nested Absolute Value

Example 1.165

Find the number of integer solutions to $||x| - 2020| < 5$. (IOQM 2021/5)

$$\begin{aligned} -5 &< |x| - 2020 < 5 \\ 2015 &< |x| < 2025 \end{aligned}$$

$$\text{Positive Solutions: } x \in \{2016, 2017, \dots, 2024\} \rightarrow 2024 - 2016 + 1$$

$$\text{Total Solutions} = 9 \times 2 = 18$$

Logic

$$||x| - 2020| < 5 \Rightarrow |x| - 2020 \in \{-4, -3, \dots, 3, 4\} \Rightarrow 9 \text{ Values}$$

$$|x| - 2020 = -4 \Rightarrow |x| = 2016 \Rightarrow x = \pm 2016$$

Similarly, each of the 9 values above has two solutions

$$\text{Total Solutions} = 9 \times 2 = 18$$

Table

x	2016	.	.	2019	2020	2021	.	.	2024
$ x - 2020$	-4			-1	0	1			4

2. ABSOLUTE VALUE TOPICS

2.1 Quadratics

A. Type I Equations

Before you do this section, you should know Quadratics.

Example 2.1

$$|x^2 + 5x + 5| = 1$$

- A. Solve by Algebraic Method
- B. Graph $y = |x^2 + 5x + 5|$

$$x^2 + 5x + 5 = \pm 1$$

Case I: $x^2 + 5x + 5 = 1$

$$(x + 4)(x + 1) = 0$$

$$x \in \{-1, -4\}$$

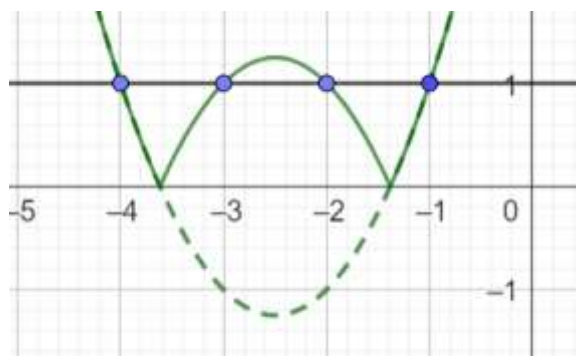
Case II: $x^2 + 5x + 5 = -1$

$$(x + 2)(x + 3) = 0$$

$$x \in \{-2, -3\}$$

Combining the cases gives us:

$$x \in \{-1, -2, -3, -4\}$$



Example 2.2

$$|x^2 + 5x + 5| = c, \quad c \in \mathbb{R}$$

Determine the number of solutions of the equation above by considering various cases in the graph of $y = |x^2 + 5x + 5|$

$x^2 + 5x + 5$ has vertex

$$x \text{ coordinate} = -\frac{b}{2a} = -\frac{5}{2}$$

$$y \text{ coordinate} = \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) + 5 = \frac{25}{4} - \frac{25}{2} + 5 = \frac{25 - 50 + 20}{4} = -\frac{5}{4}$$

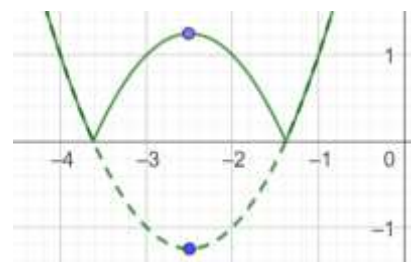
$$\text{Vertex} = (x, y) = \left(-\frac{5}{2}, -\frac{5}{4}\right)$$

Since we are taking the absolute value, we need to take the absolute value of the y coordinate:

$$\text{Vertex} = (x, y) = \left(-\frac{5}{2}, \frac{5}{4}\right)$$

$$c < 0 \Rightarrow \text{No Solutions}$$

$$c = 0 \Rightarrow 2 \text{ Solutions}$$



$$0 < c < \frac{5}{4}: 4 \text{ Solutions}$$

$$c = \frac{5}{4}: 3 \text{ Solutions}$$

$$c > \frac{5}{4}: 2 \text{ Solutions}$$

Example 2.3

Determine the number of real solutions of $y = ax^2 + bx + c$ given that

$$|ax^2 + bx + c| = d, \quad d > 0$$

has either zero real solutions, one real solution, or two real solutions depending on the value of d .

If the equation had been of the type graphed in the previous question, then we would have

$$\text{Number of Solutions} \in \{0, 2, 3, 4\}$$

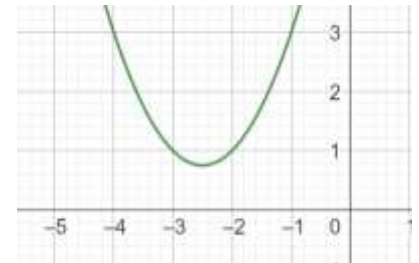
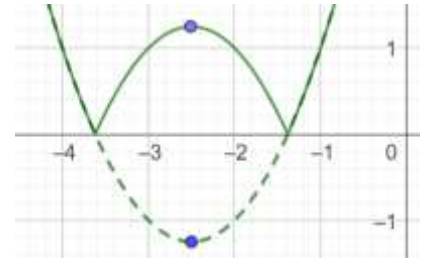
However, for the given equation, the possible number of solutions is:

$$\text{Number of Solutions} \in \{0, 1, 2\}$$

Hence, the graph of $y = ax^2 + bx + c$ must always be above the x axis or always be below the x axis.

Since the graph of y is always above the x axis, it has:

No Real Solutions



Example 2.4

Answer each part separately

Solve $|(x + 1)^2 + 5(x + 1) + 5| = 1$ by using:

- A. the algebraic method
- B. transformation of functions

Part A

$$|(x + 1)^2 + 5(x + 1) + 5| = 1$$

Use a change of variable. Let $y = x + 1$:

$$|y^2 + 5y + 5| = 1$$

This is the same quadratic solved earlier in the chapter (with a different variable):

$$y \in \{-1, -2, -3, -4\}$$

$$x = y - 1 \in \{-2, -3, -4, -5\}$$

Part B

Shift the graph of $|(x + 1)^2 + 5(x + 1) + 5|$ one unit to the right by substituting $x - 1$ in place of x :

$$|((x - 1) + 1)^2 + 5((x - 1) + 1) + 5| = 1$$

$$|x^2 + 5x + 5| = 1$$

And then solve as in Part A.

B. Type I Inequalities

Example 2.5

Solve

$$|x^2 + 8x + 5| > 2$$

We split into cases using the property that

$$|y| > 2 \Rightarrow y > 2 \text{ OR } y < -2$$

Case I: $x^2 + 8x + 5 > 2$

$$\begin{aligned} x^2 + 8x + 3 &> 0 \\ x &= \frac{-8 \pm \sqrt{64 - 12}}{2} = \frac{-8 \pm \sqrt{52}}{2} = -4 \pm \sqrt{13} \\ x &\in (-\infty, -4 - \sqrt{13}) \cup (-4 + \sqrt{13}, \infty) \end{aligned}$$

Case II: $x^2 + 8x + 5 < -2$

$$\begin{aligned} x^2 + 8x + 7 &< 0 \\ (x + 7)(x + 1) &< 0 \\ x &\in (-7, -1) \end{aligned}$$

The final answer is the union of the solutions from Case I and II:

$$(-\infty, -4 - \sqrt{13}) \cup (-7, -1) \cup (-4 + \sqrt{13}, \infty)$$

Note that:

$$\begin{aligned} -4 + \sqrt{13} &> -4 + 3 = -1 \\ -4 + \sqrt{13} &> -1 \end{aligned}$$

Example 2.6

Solve

$$|x^2 + \sqrt{47}x - \sqrt{53}| \leq -3$$

Absolute value always greater than zero

$$x \in \phi$$

Example 2.7

Solve

$$\left| \sqrt{\frac{23}{4}}x^2 + \sqrt{23}x + \sqrt{\frac{25}{4}} \right| > 0$$

Consider the discriminant of the quadratic expression:

$$D = b^2 - 4ac = (\sqrt{23})^2 - 4\left(\frac{\sqrt{23}}{2}\right)\left(\frac{\sqrt{25}}{2}\right) = 23 - \sqrt{23}\sqrt{25} < 0$$

Hence, this quadratic is always positive:

All real values

Note: The coefficients of the quadratic are such that it is difficult to solve directly.

Example 2.8

Solve

$$\left| \sqrt{\frac{17}{4}}x^2 + \sqrt{23}x + \sqrt{\frac{29}{4}} \right| > 0$$

This is valid for all real numbers except where:

$$\sqrt{\frac{17}{4}}x^2 + \sqrt{23}x + \sqrt{\frac{29}{4}} = 0$$

Applying the quadratic formula gives:

$$x = \frac{-\sqrt{23} \pm \sqrt{23 - \sqrt{17}\sqrt{29}}}{2\sqrt{17}}$$

Hence, the final answer is:

$$x \in \mathbb{R} - \left\{ \frac{-\sqrt{23} \pm \sqrt{23 - \sqrt{17}\sqrt{29}}}{2\sqrt{17}} \right\}$$

And we can confirm the discriminant is positive in a couple of different ways:

$$(\sqrt{17}\sqrt{29})^2 = 17 \times 29 = 493 < 529 = 23^2 \Rightarrow 23 - \sqrt{17}\sqrt{29} > 0$$

Substitute $a = 17, b = 29$ in the $AM - GM$ inequality $a + b \geq 2\sqrt{ab}$

$$17 + 29 \geq 2\sqrt{17} \cdot \sqrt{29}$$

$$23 \geq \sqrt{17} \cdot \sqrt{29}$$

$$23 - \sqrt{17} \cdot \sqrt{29} \geq 0$$

C. Type II Equations

Example 2.9

$$x^2 + 5|x| - 6 = 0$$

- Solve the equation above by factoring
- Considering cases for $|x|$, and solving algebraically.
- Uses the cases from Part B to graph the equation. Graph the “invalid” portions of the cases as well to see the complete behaviour.

Part A

Use the property that $|x|^2 = x^2$

$$(|x| + 6)(|x| - 1) = 0$$

$$|x| + 6 = 0 \Rightarrow |x| = -6 \Rightarrow \text{No Solution}$$

$$|x| - 1 = 0 \Rightarrow |x| = 1 \Rightarrow x = \pm 1$$

The final answer is:

$$x \in \{-1, +1\}$$

Part B

Case I: $x > 0 \Rightarrow |x| = x$

$$x^2 + 5x - 6 = 0 \Rightarrow (x + 6)(x - 1) = 0 \Rightarrow x \in \{-6, +1\} \Rightarrow x = +1$$

Case II: $x < 0 \Rightarrow |x| = -x$

$$x^2 - 5x - 6 = 0 \Rightarrow (x - 6)(x + 1) = 0 \Rightarrow x \in \{+6, -1\} \Rightarrow x = -1$$

The final answer is:

$$x \in \{-1, +1\}$$

Example 2.10

$$x^2 - 5|x| + 6 = 0$$

$$(|x| - 2)(|x| - 3) = 0$$

$$|x| = 2 \Rightarrow x = \pm 2$$

$$|x| = 3 \Rightarrow x = \pm 3$$

Example 2.11

$$x^2 - 5|x| - 6 = 0$$

$$(|x| - 6)(|x| + 1) = 0$$

$$|x| = 6 \Rightarrow x = \pm 6$$

$$|x| = -1 \Rightarrow \text{No Solution}$$

Example 2.12

$$x^2 + 5|x| - 2 = 0$$

Using the quadratic formula

$$|x| = \frac{-5 \pm \sqrt{5^2 - 4(-2)}}{2} = \frac{-5 \pm \sqrt{33}}{2}$$

$$|x| = -\frac{5 + \sqrt{33}}{2} \Rightarrow \text{No Solutions}$$

$$|x| = -\frac{5 - \sqrt{33}}{2} = \frac{\sqrt{33} - 5}{2} \Rightarrow x = \pm \frac{\sqrt{33} - 5}{2}$$

Example 2.13

$$x^2 + 5|x| + 3 = 0$$

Using the quadratic formula:

$$x = \frac{-5 \pm \sqrt{5^2 - 4(3)}}{2} = \frac{-5 \pm \sqrt{13}}{2}$$

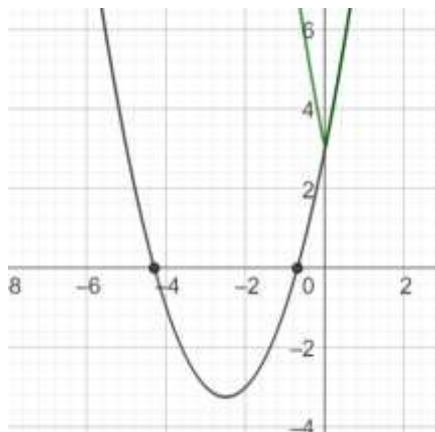
$$|x| = -\frac{5 + \sqrt{13}}{2} \Rightarrow \text{No Solutions}$$

$$|x| = -\frac{5 - \sqrt{13}}{2} \approx -1.4 \Rightarrow \text{No Solutions}$$

D. Type II Inequalities

2.14: Quadratic Inequalities with Absolute Value

$$x^2 + b|x| + c, \quad b > 0$$



Example 2.15

Solve

$$x^2 + 18|x| + 77 \leq 0$$

Always greater than zero because:

$$\begin{aligned} x^2 &\geq 0 \\ 18|x| &\geq 0 \\ 77 &> 0 \end{aligned}$$

No Solutions

Example 2.16

Solve

$$x^2 + 5\sqrt{3}|x| + 18 \geq 0$$

Always greater than zero.

All real numbers $\Rightarrow x \in \mathbb{R}$

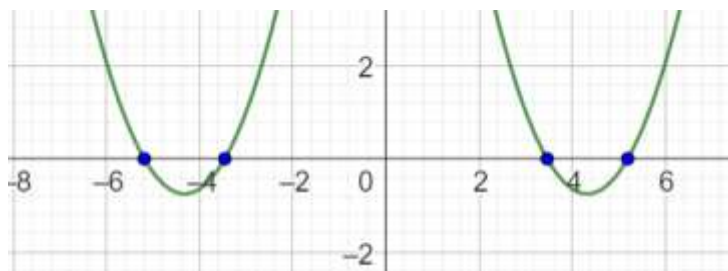
Example 2.17

Solve

$$x^2 - 5\sqrt{3}|x| + 18 \geq 0$$

$$\begin{aligned} (|x| - 2\sqrt{3})(|x| - 3\sqrt{3}) &= 0 \\ x &\in \{-3\sqrt{3}, -2\sqrt{3}, 2\sqrt{3}, 3\sqrt{3}\} \end{aligned}$$

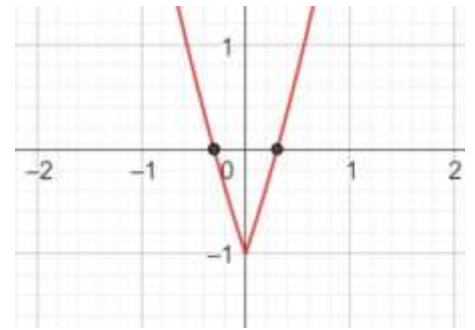
$$(-\infty, -3\sqrt{3}) \cup [-2\sqrt{3}, 2\sqrt{3}] \cup [3\sqrt{3}, \infty)$$



Example 2.18

$$x^2 + 3|x| - 1 > 0$$

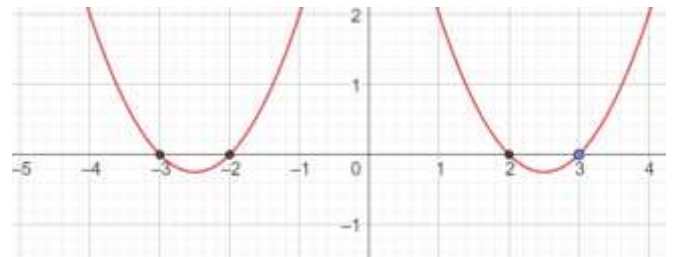
$$\left(-\infty, -\frac{\sqrt{13}-3}{2}\right) \cup \left(\frac{\sqrt{13}-3}{2}, \infty\right)$$



Example 2.19

$$x^2 - 5|x| + 6 > 0$$

$$\begin{aligned} (|x| - 2)(|x| - 3) &= 0 \\ |x| = 2 &\Rightarrow x = \pm 2 \\ |x| = 3 &\Rightarrow x = \pm 3 \end{aligned}$$



$$(-\infty, -3) \cup (-2, +2) \cup (3, \infty)$$

E. Mixing Type I and Type II

Example 2.20

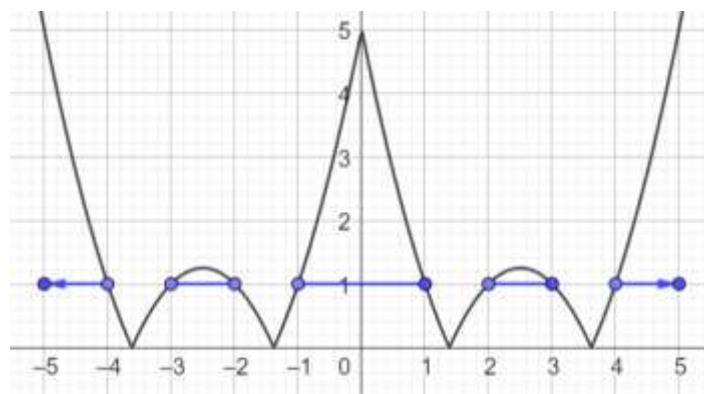
$$|x^2 - 5|x| + 5| \geq 1$$

Solve the corresponding equation $|x^2 - 5|x| + 5| = 1$ into cases:

$$\begin{aligned} x^2 - 5|x| + 5 = 1 &\Rightarrow (|x| - 4)(|x| - 1) = 0 \Rightarrow x \in \{-4, -1, 1, 4\} \\ x^2 - 5|x| + 5 = -1 &\Rightarrow (|x| - 2)(|x| - 3) = 0 \Rightarrow x \in \{-3, -2, 2, 3\} \end{aligned}$$

$$x \in \{-4, -3, -2, -1, 1, 2, 3, 4\}$$

Use the wavy curve method:



The final answer is:

$$(-\infty, -4] \cup [-3, -2] \cup [-1, 1] \cup [2, 3] \cup [4, \infty)$$

2.2 Logarithms

A. Equations

Before you do this section, you should know Logarithms.

2.21: Absolute Value Equations

$$\begin{aligned}|x| &= a, a \in \mathbb{R} \\ a > 0 &\Rightarrow x = \pm a \\ a = 0 &\Rightarrow x = 0 \\ a < 0 &\Rightarrow x \in \phi\end{aligned}$$

$$\begin{aligned}|x| &= 5 \Rightarrow x = \pm 5 \\ |x| &= 0 \Rightarrow x = 0 \\ |x| &= -5 \Rightarrow x \in \phi\end{aligned}$$

Example 2.22

Solve for x in each part.

Absolute Value in the Number

A. $\log_5 |x + 2| = 2$

Absolute Value in the Answer

B. $\log_3 x = y, |y| = 2$

Absolute Value in the Base

C. $\log_{|x|} 4 = 3$

Absolute Value Terms

D. $|\log_x 4| = 2$

Absolute Value in the Number

$$\log_5 |x + 2| = 2$$

Convert to exponential form

$$\begin{aligned}|x + 2| &= 5^2 = 25 \\ x + 2 &= \pm 25 \\ x &\in \{-27, 23\}\end{aligned}$$

Absolute Value in the Answer

$$\begin{aligned}\log_3 x &= y, |y| = 2 \\ x &= 3^y = 3^{\pm 2} \Rightarrow x \in \left\{\frac{1}{9}, 9\right\}\end{aligned}$$

Absolute Value in the Base

$$\begin{aligned}\log_{|x|} 4 &= 3 \\ |x|^3 &= 4\end{aligned}$$

Take cube roots:

$$\begin{aligned}|x| &= \sqrt[3]{4} \\ x &= \pm \sqrt[3]{4}\end{aligned}$$

Absolute Value Terms

$$\log_x 4 = \pm 2$$

Convert to exponential form:

$$x^{\pm 2} = 4$$

$$x^2 = 4 \Rightarrow x = \{+2, -2\} \Rightarrow x = 2$$

$$\frac{1}{x^2} = 4 \Rightarrow x^2 = \frac{1}{4} \Rightarrow x = \pm \frac{1}{2}$$

Since is the base, $x > 0$:

$$x = \frac{1}{2}$$

Example 2.23: Absolute Value

- A. The product of the solutions to $\log_4 |x + 1| = -7$ can be written in the form $\frac{a}{b}$ for integers a, b with no common factor other than 1. Find $b - a$.
- B. Find the sum of the solutions to $|\log_2 x| + |\log_3 x| + \dots + |\log_{2022} x| = -2022$

Part A

$$\log_4 |x| = -7$$

$$|x + 1| = 4^{-7} = \frac{1}{4^7} = \frac{1}{2^{14}}$$

$$x + 1 = \pm \frac{1}{2^{14}}$$

$$x = \pm \frac{1}{2^{14}} - 1$$

$$x = \frac{1}{2^{14}} - 1 = \frac{1 - 2^{14}}{2^{14}}$$

$$x = -\frac{1}{2^{14}} - 1 = -\frac{1 + 2^{14}}{2^{14}}$$

$$x \in \left\{ -\frac{1 + 2^{14}}{2^{14}}, \frac{1 - 2^{14}}{2^{14}} \right\}$$

$$\text{Product} = -\frac{(1 + 2^{14})}{2^{14}} \times \frac{(1 - 2^{14})}{2^{14}}$$

$$= \frac{1 - 2^{28}}{2^{28}} = \frac{2^{28} - 1}{2^{28}}$$

$$b - a = 2^{28} - (2^{28} - 1) = 1$$

Part B

$$LHS \geq 0$$

$$RHS < 0$$

Contradiction

$$x \in \{\emptyset\}$$

Sum of empty set is zero (by convention)

$$\text{Answer} = 0$$

Example 2.24: Absolute Value

If $|x - \log y| = x + \log y$ where x and $\log y$ are real, then

- A. $x = 0$
- B. $y = 1$
- C. $x = 0$ and $y = 1$
- D. $x(y - 1) = 0$
- E. None of these (AHSME 1972/8)

$$x - \log y = \pm(x + \log y)$$

Case I:

$$\begin{aligned} x - \log y &= x + \log y \\ -\log y &= \log y \\ 2 \log y &= 0 \\ \log y &= 0 \\ y &= 1 \end{aligned}$$

Case II:

$$\begin{aligned} x - \log y &= -x - \log y \\ x &= -x \\ 2x &= 0 \\ x &= 0 \end{aligned}$$

We combine the two cases to get:

$$\begin{aligned} y &= 1 \text{ OR } x = 0 \\ y - 1 &= 0 \text{ OR } x = 0 \end{aligned}$$

And multiply the two equations to get:

$$x(y - 1) = 0$$

B. Inequalities

Example 2.25

$$\log_3(|x^2 + 5x + 3|) > 1$$

$$|x^2 + 5x + 3| > 0$$

$$|x^2 + 5x + 3| > 3$$

$$x^2 + 5x + 3 > 3$$

$$x(x + 5) > 0$$

$$x \in (-\infty, -5) \cup (0, \infty)$$

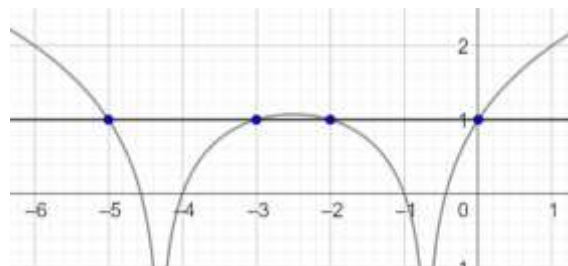
$$x^2 + 5x + 3 < -3$$

$$x^2 + 5x + 6 < 0$$

$$(x + 2)(x + 3) < 0$$

$$x \in (-3, -2)$$

$$(-\infty, -5) \cup (-2, -3) \cup (0, \infty)$$



Example 2.26

Given that $y = \ln|x|$ determine the solution set of

$$|y| > 1$$

Case I:

$$\text{Case I: } y > 1 \Rightarrow \ln|x| > 1 \Rightarrow |x| > e \Rightarrow x > e \text{ OR } x < -e$$

Case II:

$$y < -1 \Rightarrow \ln|x| < -1 \Rightarrow |x| < \frac{1}{e} \Rightarrow -\frac{1}{e} < x < \frac{1}{e}$$

Final answer is the union of the above:

$$(-\infty, -e) \cup \left(-\frac{1}{e}, \frac{1}{e}\right) \cup (e, \infty)$$

2.3 Trigonometry

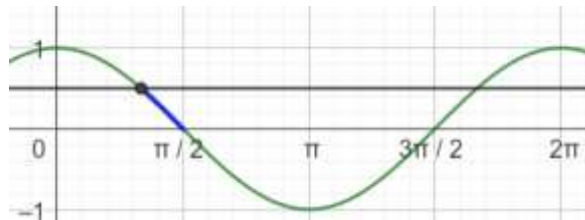
A. Inequalities versus Equalities

Example 2.27

Explain why the “solution below” is not correct:

$$0 < \theta < \frac{\pi}{2} : \cos \theta > \frac{1}{2} \Rightarrow \theta > \cos^{-1} \frac{1}{2} \Rightarrow \theta > \frac{\pi}{3}$$

As $\theta \uparrow$, $\cos \theta \downarrow$



B. Basic Inequalities

Example 2.28

Solve in radians:

$$\sin \theta \cos \theta > \frac{1}{4}$$

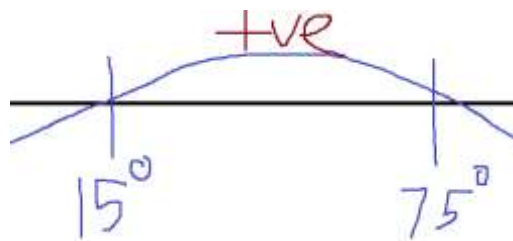
Use the double angle identity:

$$\sin 2\theta > \frac{1}{2} \Rightarrow \left(\sin 2\theta - \frac{1}{2}\right) > 0$$

Note that $\sin 2\theta$ has period $\frac{360}{2} = \frac{360}{2} = 180^\circ$

Find the roots over $[0, 180^\circ]$:

$$\begin{aligned}\sin 2\theta - \frac{1}{2} &= 0 \\ \sin 2\theta &= \frac{1}{2} \\ 2\theta &\in \{30^\circ, 150^\circ\} \\ \theta &\in \{15^\circ, 75^\circ\}\end{aligned}$$



The solution in the interval $[0, 180^\circ]$

$$(15^\circ, 75^\circ)$$

The general solution is:

$$(15^\circ, 75^\circ) + 180n, \quad n \in \mathbb{Z}$$

Convert to radians:

$$\left(\frac{\pi}{12}, \frac{5\pi}{12}\right) + \pi n, \quad n \in \mathbb{Z}$$

C. Absolute Value Inequalities

Example 2.29

Solve in radians:

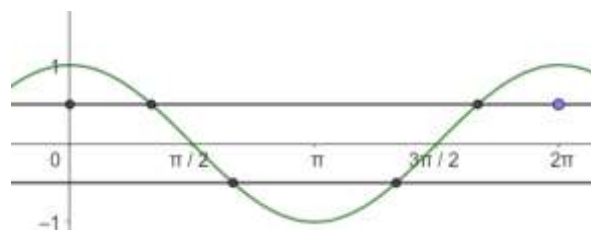
$$|\cos \theta| > \frac{1}{2}$$

$$\cos \theta > \frac{1}{2} \text{ OR } \cos \theta < -\frac{1}{2}$$

$$(-60^\circ, 60^\circ) \cup (120^\circ, 240^\circ) + 360n, \quad n \in \mathbb{Z}$$

Convert to radians:

$$\left(-\frac{\pi}{3}, \frac{\pi}{3}\right) \cup \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right) + 2\pi n, \quad n \in \mathbb{Z}$$



Example 2.30

Solve in radians:

$$|\sin \pi \theta| |\cos \pi \theta| > \frac{1}{4}$$

Use the property $|a||b| = |ab|$:

$$|\sin \pi \theta \cos \pi \theta| > \frac{1}{4}$$

Use the double angle identity $\sin 2\theta = 2 \sin \theta \cos \theta$:

$$|\sin 2\pi \theta| > \frac{1}{2}$$

The principal solutions for $0 \leq 2\pi \theta < 2\pi \Rightarrow 0 \leq \theta < 1$ are

$$\sin 2\pi \theta > \frac{1}{2} \Rightarrow \sin 2\pi \theta = \frac{1}{2} \Rightarrow 2\pi \theta \in \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\} \Rightarrow \theta \in \left\{\frac{1}{12}, \frac{5}{12}\right\} \Rightarrow \theta \in \left(\frac{1}{12}, \frac{5}{12}\right)$$

$$\sin 2\pi\theta < -\frac{1}{2} \Rightarrow \sin 2\pi\theta = -\frac{1}{2} \Rightarrow 2\pi\theta \in \left\{\frac{7\pi}{6}, \frac{11\pi}{6}\right\} \Rightarrow \theta \in \left\{\frac{7}{12}, \frac{11}{12}\right\} \Rightarrow \theta \in \left(\frac{7}{12}, \frac{11}{12}\right)$$

The final solution is:

$$\left(\frac{1}{12}, \frac{5}{12}\right) + \frac{n}{2}, \quad n \in \mathbb{Z}$$

D. Quadratic Inequalities

Example 2.31

Solve in radians:

$$2 \sin 2\theta - 2 \sin \theta - 2 \cos \theta > -1$$

Factor the Inequality

Collate all terms on the LHS, and substitute $\sin 2\theta = 2 \sin \theta \cos \theta$

$$4 \sin \theta \cos \theta - 2 \sin \theta - 2 \cos \theta + 1 > 0$$

$$2 \sin \theta (2 \cos \theta - 1) - 1(2 \cos \theta - 1) > 0$$

$$(2 \cos \theta - 1)(2 \sin \theta - 1) > 0$$

Principal Solutions

Note that both $\sin \theta$ and $\cos \theta$ have period 360° . Hence, we will find principal solutions, and then extend to the real number line.

Find the roots of the expression on the LHS. Solve in degrees first:

$$2 \sin \theta - 1 = 0 \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow x \in \{30^\circ, 150^\circ\}$$

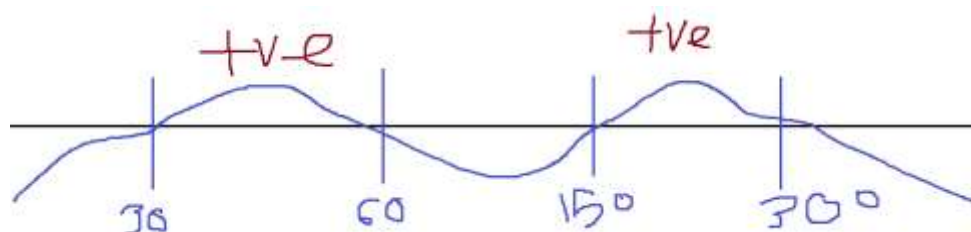
$$2 \cos \theta - 1 = 0 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow x \in \{60^\circ, 300^\circ\}$$

The roots are:

$$\{30^\circ, 60^\circ, 150^\circ, 300^\circ\}$$

Check a point in the interval $[0, 30^\circ]$. Use $\theta = 0$:

$$4 \sin 0 \cos 0 - 2 \sin 0 - 2 \cos 0 + 1 = 4(0) \cos 0 - 2(0) - 2(1) + 1 = -2 + 1 = -1 < 0$$



$$(30^\circ, 60^\circ) \cup (150^\circ, 300^\circ)$$

General Solution

Extend the principal solution to a general solution by using the periodicity (360°) of the trigonometric functions involved:

$$\{(30^\circ, 60^\circ) \cup (150^\circ, 300^\circ)\} + 360n^\circ, \quad n \in \mathbb{Z}$$

Convert to radians:

$$\left\{ \left(\frac{\pi}{6}, \frac{\pi}{3} \right) \cup \left(\frac{5\pi}{6}, \frac{5\pi}{3} \right) \right\} + 2\pi k, \quad k \in \mathbb{Z}$$

Example 2.32

$$6 \cos^2 \theta + 5|\sin \theta| > 7$$

Use the Pythagorean Identity:

$$6(1 - \sin^2 \theta) + 5|\sin \theta| > 7$$

Substitute $x = \sin \theta \Rightarrow x^2 = \sin^2 \theta$:

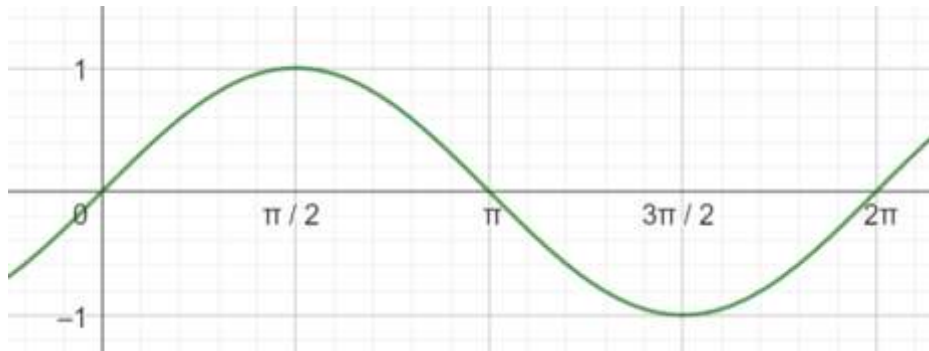
$$6(1 - x^2) + 5|x| > 7$$

$$6x^2 - 5|x| + 1 < 0$$

Solve the corresponding equation:

$$(3|x| - 1)(2|x| - 1) = 0 \Rightarrow \text{Roots or Critical Points are } x \in \left\{ -\frac{1}{2}, -\frac{1}{3}, \frac{1}{3}, \frac{1}{2} \right\}$$

$$x \in \left(-\frac{1}{2}, -\frac{1}{3} \right) \cup \left(\frac{1}{3}, \frac{1}{2} \right)$$



$x = \sin \theta$ will be in the range $\left(\frac{1}{3}, \frac{1}{2} \right)$ in the first quadrant:

$$x \in \left(\frac{1}{3}, \frac{1}{2} \right) \Rightarrow \theta \in \underbrace{\left(\sin^{-1} \frac{1}{3}, \frac{\pi}{6} \right)}_{\text{Quadrant I Solution}} \cup \underbrace{\left(\pi - \sin^{-1} \frac{1}{3}, \frac{5\pi}{6} \right)}_{\text{Quadrant II Solution}}$$

$$x \in \left(-\frac{1}{2}, -\frac{1}{3} \right) \Rightarrow \theta \in \underbrace{\left(\sin^{-1} \left(-\frac{1}{3} \right), \frac{7\pi}{6} \right)}_{\text{Quadrant III Solution}} \cup \underbrace{\left(\pi - \sin^{-1} \left(-\frac{1}{3} \right), \frac{11\pi}{6} \right)}_{\text{Quadrant IV Solution}}$$

Note: Just for reference the roots can be calculated as follows:

$$x = |\sin \theta| = \frac{1}{3}$$

$$\text{Positive Case: } \sin \theta = \frac{1}{3} \Rightarrow \theta = \left\{ \sin^{-1} \frac{1}{3}, \pi - \sin^{-1} \frac{1}{3} \right\}$$

$$\text{Negative Case: } \sin \theta = -\frac{1}{3} \Rightarrow \theta = \left\{ \sin^{-1} \left(-\frac{1}{3} \right), \pi - \sin^{-1} -\frac{1}{3} \right\}$$

$$x = |\sin \theta| = \frac{1}{2}$$

$$x = \sin \theta = \frac{1}{2} \Rightarrow \theta = \left\{ \frac{\pi}{6}, \frac{5\pi}{6} \right\}$$

$$\sin \theta = -\frac{1}{2} \Rightarrow \theta = \left\{ \frac{7\pi}{6}, \frac{11\pi}{6} \right\}$$

E. Strange Bases

Example 2.33

Solve the inequality

$$\frac{1}{2} < |\log_{\cos x} \sin x| < 2$$

Domain of the Expression

Determine the domain of the expression. The input for the log function must be positive:

$$\sin x > 0 \Rightarrow x \in (0, \pi)$$

$$\cos x > 0 \Rightarrow \left(0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$$

Take the intersection of the above two to get the principal solution and then add $2\pi k$ to get the general solution:

$$\left(0, \frac{\pi}{2}\right) + 2\pi k, \quad k \in \mathbb{Z}$$

Determine $\log_{\cos x} \sin x < 0$

Consider the graphs of $\sin x$ and $\cos x$ in the domain $\left(0, \frac{\pi}{2}\right)$:

$$x \in \left(0, \frac{\pi}{2}\right) \Rightarrow \cos x \in (0, 1)$$

Since $\cos x \in (0, 1)$

$$\log_{\cos x} \sin x < 0 \text{ when } \sin x > 1 \Rightarrow x \in \phi$$

Alternatively solve the inequality $\log_{\cos x} \sin x < 0$ by considering the critical points:

$$\log_{\cos x} \sin x = 0 \Rightarrow \sin x = 1 \Rightarrow x = \frac{\pi}{2} \Rightarrow \text{Not in domain} \Rightarrow x \in \phi$$

Since $\log_{\cos x} \sin x > 0$ for all values of x , we can remove the absolute value:

$$\frac{1}{2} < \log_{\cos x} \sin x < 2$$

Split the Inequality: Part I

$$\log_{\cos x} \sin x > \frac{1}{2}$$

Consider the critical points:

$$\begin{aligned}\log_{\cos x} \sin x &= \frac{1}{2} \\ \sin x &= \sqrt{\cos x} \\ \sin^2 x &= \cos x \\ 1 - \cos^2 x &= \cos x\end{aligned}$$

Substitute $y = \cos x$:

$$\begin{aligned}y^2 + y - 1 &= 0 \\ \cos x = y &\in \left\{ -\frac{1 - \sqrt{5}}{2}, -\frac{1 + \sqrt{5}}{2} \right\}\end{aligned}$$

$-\frac{1 + \sqrt{5}}{2}$ is not valid. Let:

$$\begin{aligned}-\frac{1 - \sqrt{5}}{2} &= \frac{\sqrt{5} - 1}{2} \approx 0.61 \text{ is valid} \\ \cos x = \frac{\sqrt{5} - 1}{2} &\Rightarrow x = \cos^{-1} \frac{\sqrt{5} - 1}{2}\end{aligned}$$

(And we only need solutions in the first quadrant since $x \in (0, \frac{\pi}{2})$)

Split the Inequality: Part II

$$\log_{\cos x} \sin x < 2$$

Consider the critical points:

$$\begin{aligned}\log_{\cos x} \sin x &= 2 \\ \sin x &= \cos^2 x \\ 1 - \sin^2 x &= \sin x\end{aligned}$$

Substitute $y = \sin x$:

$$\begin{aligned}y^2 + y - 1 &= 0 \\ \sin x = y &\in \left\{ -\frac{1 - \sqrt{5}}{2}, -\frac{1 + \sqrt{5}}{2} \right\}\end{aligned}$$

$-\frac{1 + \sqrt{5}}{2}$ is not valid. Let:

$$\begin{aligned}-\frac{1 - \sqrt{5}}{2} &= \frac{\sqrt{5} - 1}{2} \approx 0.61 \text{ is valid} \\ \sin x = \frac{\sqrt{5} - 1}{2} &\Rightarrow x = \sin^{-1} \frac{\sqrt{5} - 1}{2}\end{aligned}$$

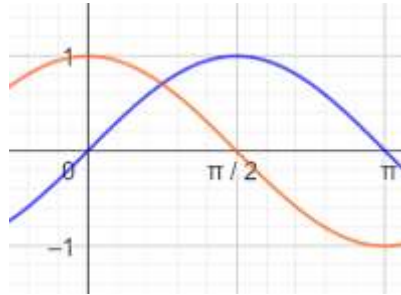
(And we only need solutions in the first quadrant since $x \in (0, \frac{\pi}{2})$)

Consider

$$\begin{aligned}x = 0.001 &\Rightarrow \sin x \approx 0.001, \cos x \approx 0.999 \\ \log_{0.999} 0.001 &= \frac{\ln 0.001}{\ln 0.999} = \frac{\text{Some Number}}{\text{Some Number with Small Absolute Value}} \rightarrow \text{Some Large Number}\end{aligned}$$

$$x = \frac{\pi}{2} - 0.001 \Rightarrow \sin x \approx 0.999, \cos x \approx 0.001$$

$$\log_{0.001} 0.999 \rightarrow \text{Some positive number close to zero}$$



$$\left(\sin^{-1} \frac{\sqrt{5}-1}{2}, \cos^{-1} \frac{\sqrt{5}-1}{2} \right)$$

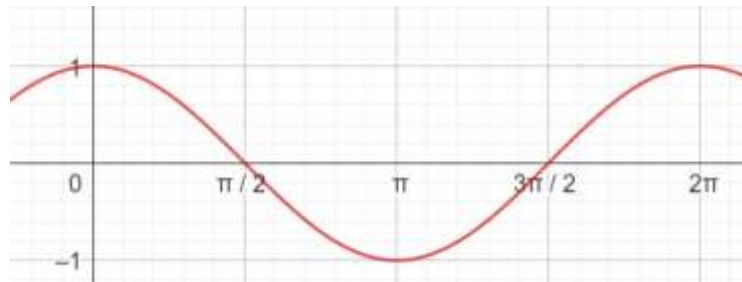
F. Nested Functions

Example 2.34

Solve the inequality

$$|\sin(\cos x)| > \frac{1}{2}$$

Graph $f(x) = \cos x$:



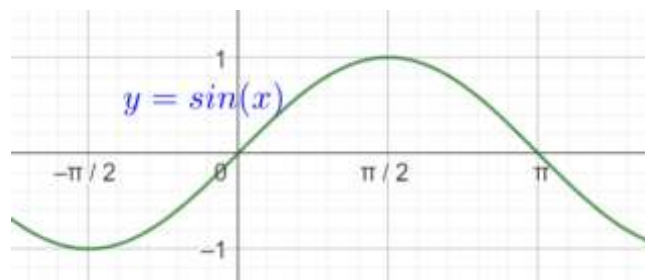
Graph $f(x) = \sin(\cos x)$:

$\cos x$ has *maximum* = 1 at 0, and *minimum* = -1 at π :

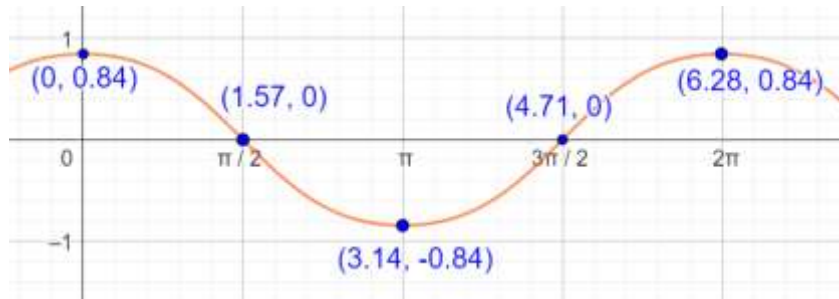
$$\sin(\cos 0) = \sin 1 \approx \sin 57^\circ \approx \sin 60^\circ = \frac{\sqrt{3}}{2} \approx \frac{1.7}{2} = 0.85$$

$$\sin(\cos \pi) = \sin -1 \approx \sin -57^\circ \approx \sin -60^\circ = -\frac{\sqrt{3}}{2} \approx -\frac{1.7}{2} = -0.85$$

$$\sin\left(\cos \frac{\pi}{2}\right) = \sin\left(\cos \frac{3\pi}{2}\right) = \sin 0 \approx 0$$



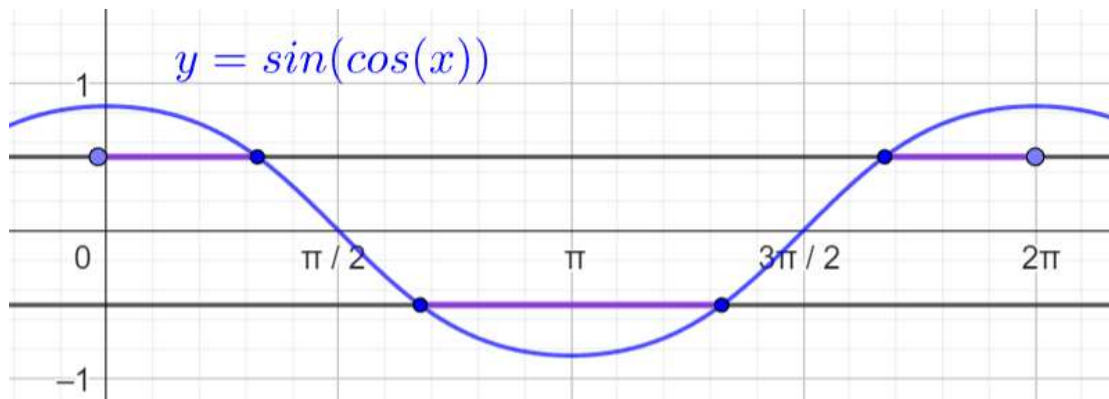
$\sin x$ is decreasing and continuous from $\sin 1$ to $\sin -1$
 $\sin x$ is increasing and continuous from $\sin -1$ to $\sin 1$



Split the given inequality into cases by removing the absolute value:

$$\sin(\cos x) > \frac{1}{2} \text{ OR } \sin(\cos x) < -\frac{1}{2}$$

The regions that we want are marked in violet on the graph.



Case I: $\sin(\cos x) > \frac{1}{2}$

Solve the corresponding equation to find the critical points:

$$\begin{aligned} \sin(\cos x) &= \frac{1}{2} \Rightarrow \cos x \in \left\{ \frac{\pi}{6}, \frac{5\pi}{6} \right\} \\ \frac{5\pi}{6} &\approx \frac{5(3)}{6} = 2.5 \Rightarrow \text{Not a valid value of } \cos x \end{aligned}$$

Also, from the graph, we know that we are looking for a solution in the first and the fourth quadrant.

$$\cos x = \frac{\pi}{6} \Rightarrow x \in \left\{ \cos^{-1}\left(\frac{\pi}{6}\right), -\cos^{-1}\left(\frac{\pi}{6}\right) \right\}$$

Since this gives solutions which are in the principal domain, add 2π to the second solution:

$$x \in \left\{ \cos^{-1}\left(\frac{\pi}{6}\right), 2\pi - \cos^{-1}\left(\frac{\pi}{6}\right) \right\}$$

The principal solution from Case I is:

$$\left[0, \cos^{-1}\left(\frac{\pi}{6}\right)\right) \cup \left(2\pi - \cos^{-1}\left(\frac{\pi}{6}\right), 2\pi\right)$$

The general solution from Case I is:

$$\left[0, \cos^{-1}\left(\frac{\pi}{6}\right)\right) \cup \left(2\pi - \cos^{-1}\left(\frac{\pi}{6}\right), 2\pi\right) + 2\pi k, \quad k \in \mathbb{Z}$$

Case II: $\sin(\cos x) < -\frac{1}{2}$

Solve the corresponding equation to find the critical points:

$$\begin{aligned}\sin(\cos x) &= -\frac{1}{2} \Rightarrow \cos x \in \left\{\frac{7\pi}{6}, \frac{11\pi}{6}\right\} \\ \frac{7\pi}{6} &\approx \frac{7(3)}{6} \approx 3 \Rightarrow \text{Not a valid value of } \cos x\end{aligned}$$

$$\frac{11\pi}{6} = \frac{11\pi}{6} - 2\pi = -\frac{\pi}{6}$$

Also, from the graph, we know that we are looking for a solution in the first and the fourth quadrant.

$$\cos x = -\frac{\pi}{6} \Rightarrow x \in \left\{\cos^{-1}\left(-\frac{\pi}{6}\right), \right\}$$

The principal solution from Case I is:

$$\left(\cos^{-1}\left(-\frac{\pi}{6}\right), 2\pi - \cos^{-1}\left(-\frac{\pi}{6}\right)\right)$$

The general solution from Case I is:

$$\left(\cos^{-1}\left(-\frac{\pi}{6}\right), 2\pi - \cos^{-1}\left(-\frac{\pi}{6}\right)\right) + 2\pi k, \quad k \in \mathbb{Z}$$

Combine Case I and Case II:

Combine the cases to get:

$$\left[0, \cos^{-1}\left(\frac{\pi}{6}\right)\right) \cup \left(\cos^{-1}\left(-\frac{\pi}{6}\right), 2\pi - \cos^{-1}\left(-\frac{\pi}{6}\right)\right) \cup \left(2\pi - \cos^{-1}\left(\frac{\pi}{6}\right), 2\pi\right) + 2\pi k, \quad k \in \mathbb{Z}$$

3. “NAMED” INEQUALITIES

3.1 Some Basics

A. Reciprocals in Inequalities

Example 3.1

Let $p > x > y > q > 0$.

- A. Compare $\frac{p}{q}$ with $\frac{x}{y}$.
- B. Compare $\frac{q}{p}$ with $\frac{y}{x}$.
- C. Does your answer change if $p < x < y < q < 0$.

$$\frac{p}{q} > \frac{x}{y}$$
$$\frac{q}{p} < \frac{y}{x}$$

Answer does not change.

B. Creating Numerators

Example 3.2

Given $\frac{p}{q} > \frac{x}{y}$, show that

$$\frac{p-q}{q} > \frac{x-y}{y}$$

Subtract one from both sides of the inequality.

$$\frac{p}{q} - 1 > \frac{x}{y} - 1$$

This simplifies to the desired results when we combine fractions:

$$\frac{p-q}{q} > \frac{x-y}{y}$$

C. Moving all terms to one side

Example 3.3

If $p > q$ and $x > y$, then show that

$$px + qy > py + qx$$

$$(p-q)(x-y) > 0 \Rightarrow px - py - qx + qy > 0 \Rightarrow px + qy > py + qx$$

D. Positive Numbers

Example 3.4

If $a < b$, when can we say that

$$ax < bx$$

If x is positive.

Example 3.5

Show that for positive real numbers p, q and k :

$$\frac{p}{q} < \frac{p+k}{q+k} \text{ if and only if } p < q$$

$$pq + pk < qp + qk \Leftrightarrow (p - q)k < 0$$

Since k is positive, this is only possible if $p < q$.

E. Applications

Example 3.6

The number of elements in the set (JEE Main, July 22, 2021-II)

$$\{n \in \{1, 2, 3, \dots, 100\}, 11^n > 10^n + 9^n\}$$

Divide both sides of the given inequality by 11^n :

$$1 > \underbrace{\left(\frac{10}{11}\right)^n + \left(\frac{9}{11}\right)^n}_{\text{Inequality I}}$$

As n increases, $\left(\frac{10}{11}\right)^n + \left(\frac{9}{11}\right)^n$ decreases. Hence, the inequality is satisfied for some value of n , and once it is satisfied, then all larger values of n also satisfy it.

The smallest value of n such that Inequality I is satisfied is:

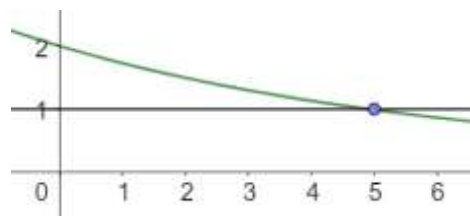
$$n = 1: 11 - 9 = 2 < 10$$

$$n = 2: 121 - 81 = 40 < 100$$

$$n = 3: 1331 - 729 < 1000$$

$$n = 4: 14641 - 6561 < 10,000$$

$$n = 5: 161051 - 59049 = 1,02,002 > 1,00,000$$



$$\{5, 6, 7, \dots, 100\} \Rightarrow 100 - 5 + 1 = 96 \text{ Values}$$

Example 3.7

$$a^n > c_1 b_1^n + c_2 b_2^n + \dots + c_x b_x^n$$

Where

$$c_1, c_2, \dots, c_x \text{ are constants}$$

$$a > b_i, 1 \leq i \leq x$$

Divide both sides by a^n :

$$1 > c_1 \left(\frac{b_1}{a}\right)^n + c_2 \left(\frac{b_2}{a}\right)^n + \dots + c_x \left(\frac{b_x}{a}\right)^n$$

The above will be achieved at some value of n and will hold for all values of n thereafter.

3.2 $AM \geq GM$ (In)equality

A. Trivial Inequality

3.8: Trivial Inequality

The square of any real quantity is greater than or equal to zero:

$$x^2 \geq 0$$

The trivial inequality forms the basis for proving many, many inequalities and solving many questions.

Example 3.9

What is the least possible value of $(xy - 1)^2 + (x + y)^2$ for real numbers x and y ? (AMC 12A 2021/7, AMC 10A 2021/9)

$$x^2y^2 - 2xy + 1 + x^2 + 2xy + y^2 = x^2 + y^2 + x^2y^2 + 1$$

$$\text{Minimum of } x^2 = 0 \Rightarrow \text{Achieved at } x = 0$$

$$\text{Minimum of } y^2 = 0 \Rightarrow \text{Achieved at } y = 0$$

$$\text{At } x = y = 0 \Rightarrow x^2y^2 = 0$$

By the trivial inequality, we know that this is the best we can do for the first three terms. Hence, the fourth term, which is a constant, is also the minimum.

Example 3.10

Identify the mistake in the “proof” below:

$$4 = 5 \Rightarrow \text{Multiply both sides by } 0 \Rightarrow 0 = 0$$

You cannot begin the proof by assuming the truth of what needs to be proved.

Example 3.11

Prove that:

A. $a^2 + b^2 \geq 2ab$

B. $a^2 + b^2 + c^2 \geq 2ab + 2bc - 2ac$

C. $a^2 + b^2 + c^2 \geq ab + bc + ac$

Part A

Subtract $2ab$ from both sides of the given inequality:

$$\Leftrightarrow a^2 - 2ab + b^2 \geq 0$$

$$\Leftrightarrow (a - b)^2 \geq 0$$

Each step above is reversible, and hence the original equality holds.

Part B:

Expand, and move the required terms to RHS:

$$(a - b + c)^2 \geq 0 \Leftrightarrow a^2 + b^2 + c^2 \geq 2ab + 2bc - 2ac$$

Part C:

$$(a - b)^2 + (b - c)^2 + (c - a)^2 \geq 0 \Leftrightarrow a^2 + b^2 + c^2 \geq ab + bc + ac$$

3.12: Sum of a positive number and its reciprocal is always greater than 2.

For positive real numbers a and b :

$$\frac{a}{b} + \frac{b}{a} \geq 2$$

Note that

$$\frac{a}{b} \text{ has reciprocal } \frac{b}{a}$$

Hence:

Sum of a positive number and its reciprocal is always greater than 2.

Method I: Working with the LHS

Add fractions:

$$\frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab}$$

Subtract 2 and add 2:

$$\frac{a^2 + b^2}{ab} - 2 + 2 = \frac{a^2 - 2ab + b^2}{ab} + 2$$

Complete the square:

$$\frac{(a - b)^2}{ab} + 2$$

The above expression has minimum value the square is zero: $a = b$

$$\frac{(a - a)^2}{ab} + 2 = 0 + 2 = 2$$

Method II: Working with the Inequality

When working with the inequality, we must ensure that all steps are reversible.

We indicate this by using

$$\Leftrightarrow \text{ instead of } \Rightarrow$$

Add fractions:

$$\frac{a}{b} + \frac{b}{a} \geq 2 \Leftrightarrow \frac{a^2 + b^2}{ab} \geq 2$$

Clear the denominator. Note that we can multiply both sides by a , and b only because both are positive:

$$\Leftrightarrow a^2 + b^2 \geq 2ab$$

Collate all terms on one side and factor:

$$\begin{aligned} \Leftrightarrow a^2 - 2ab + b^2 &\geq 0 \\ \Leftrightarrow (a - b)^2 &\geq 0 \end{aligned}$$

And the above is true by the trivial inequality.

(All steps are reversible, and hence the equality we started with is true.)

In the above, where was the condition that a and b are both positive used?

In one step, we multiplied by ab .

If $ab < 0$, then the inequality will flip.

In the above, can a and b be both negative?

Yes.

Example 3.13

If $a \geq b \geq 1$, what is the largest possible value of $\log_a \left(\frac{a}{b}\right) + \log_b \left(\frac{b}{a}\right)$. (AMC 12A 2003/24)

Use the quotient rule in the given expression:

$$\begin{aligned}\log_a a - \log_a b + \log_b b - \log_b a \\ = 1 - \log_a b + 1 - \log_b a\end{aligned}$$

Rearrange:

$$= 2 - (\log_a b + \log_b a)$$

Use the change of base reciprocal rule:

$$= 2 - \left(\log_a b + \frac{1}{\log_a b}\right)$$

Since $a \geq b \geq 1 \Rightarrow \log_a b > 0$, which is a positive real number and hence using the property $\frac{x}{y} + \frac{y}{x} \geq 2$ for positive real numbers:

$$\log_a b + \frac{1}{\log_a b} \geq 2$$

To maximize the expression, minimize the value of the logarithmic expression. Hence:

$$\text{Max} \left[2 - \left(\log_a b + \frac{1}{\log_a b} \right) \right] = 2 - 2 = 0$$

3.14: Minimum Value of an expression

The minimum value of $a^2 + b$ is:

$$b$$

This is because, by the trivial inequality:

$$a^2 \geq 0$$

Example 3.15: Completing the Square

Find the minimum and maximum value of $l^2 + w^2$ given that l and w are nonnegative real numbers and $l + w = 2$.

Let

$$X = l^2 + w^2$$

Substitute $l + w = 2 \Rightarrow l = 2 - w$ which is valid for $0 \leq l, w \leq 2$ and simplify:

$$(2 - w)^2 + w^2 = 4 - 4w + w^2 + w^2 = 2w^2 - 4w + 4 = 2(w^2 - 2w + 2)$$

Complete the square to get:

$$\begin{aligned}2[(w^2 - 1)^2 + 1] \\ \text{Min when: } w = 1 \Rightarrow X = 2 \\ \text{Max when: } w = 2 \Rightarrow X = 4\end{aligned}$$

Example 3.16: Change of Variable

Let $x \geq 0, y \geq 0$ be real numbers with $x + y = 2$. Prove that $x^2 y^2 (x^2 + y^2) \leq 2$. (Irish Math Olympiad 2000/P2/Q1)

We start with a change of variable that is not immediately recognizable as useful.
Let $x = 1 + t$. Then

$$x + y = 2 \Rightarrow 1 + t + y = 2 \Rightarrow y = 1 - t$$

Substitute $x = 1 + t, y = 1 - t$ in $x^2 y^2 (x^2 + y^2)$:

$$\begin{aligned} &= (1 + t)^2 (1 - t)^2 [(1 + t)^2 + (1 - t)^2] \\ &= (1 - t^2)^2 [2 + 2t^2] \\ &= 2(1 - t^2)(1 - t^2)[1 + t^2] \end{aligned}$$

Since $t^2, t^4 \geq 0$:

$$= 2 \underbrace{(1 - t^2)}_{\text{Between 0 and 1}} \underbrace{(1 - t^4)}_{\text{Between 0 and 1}}$$

B. AM-GM Inequality: Two Variables

3.17: Geometric Mean and Arithmetic Mean

$$\begin{aligned} GM(a_1, a_2, \dots, a_n) &= \sqrt[n]{a_1 \times a_2 \times \dots \times a_n} \\ AM(a_1, a_2, \dots, a_n) &= \frac{a_1 + a_2 + \dots + a_n}{n} \end{aligned}$$

The geometric mean is the n^{th} root of the product of n numbers.

Square roots (and geometric mean) are only defined for positive numbers.

Hence, you cannot apply the AM-GM inequality to expressions that can also be negative.

Example 3.18

Consider a rectangle with perimeter 20 and integer side lengths. Find the value of the side lengths that maximizes the area.

$$P = 2(l + w) = 20 \Rightarrow l + w = 10$$

Length	1	2	3	4	5	6	7	8	9
Width	9	8	7	6	5	4	3	2	1
Area	9	18	21	24	25	24	21	16	9

3.19: AM-GM Inequality

$$\underbrace{\frac{a + b}{2}}_{\text{Arithmetic Mean}} \geq \underbrace{\sqrt{ab}}_{\text{Geometric Mean}}$$

By the trivial inequality:

$$(a - b)^2 \geq 0$$

Square both sides:

$$a^2 - 2ab + b^2 \geq 0$$

Add $4ab$ to both sides:

$$a^2 + 2ab + b^2 \geq 4ab$$

Take the square root both sides:

$$a + b \geq 2\sqrt{ab}$$

Divide by 2 both sides:

$$\frac{a+b}{2} \geq \sqrt{ab}$$

3.20: AM-GM Equality

Equality of AM and GM is achieved *if and only if* $a = b$.

We first prove that $a = b \Rightarrow AM = GM$:

$$\begin{aligned} LHS &= \frac{a+a}{2} = \frac{2a}{2} = a \\ RHS &= \sqrt{aa} = \sqrt{a^2} = a \end{aligned}$$

Hence, equality is achieved when $a = b$.

Now we prove the converse. If $AM = GM \Rightarrow a = b$:

$$\frac{a+b}{2} = \sqrt{ab}$$

Eliminate fractions:

$$a+b = 2\sqrt{ab}$$

Square both sides:

$$(a+b)^2 = 4ab$$

Collate all terms on one side:

$$a^2 - 2ab + b^2 = 0$$

Factor:

$$(a-b)^2 = 0$$

Take the square root:

$$\begin{aligned} a-b &= 0 \\ a &= b \end{aligned}$$

Example 3.21

Find the minimum and maximum value of $l^2 + w^2$ given that l and w are nonnegative real numbers and $l + w = 2$.

Since l and w are nonnegative real numbers, we must have:

$$l \geq 0, w \geq 0 \Rightarrow \underline{lw \geq 0}$$

Inequality I

Use AM-GM on the given equality:

$$\sqrt{lw} \leq \frac{l+w}{2} = \frac{2}{2} = 1 \Rightarrow \underline{lw \leq 1}$$

Inequality II

Combine Inequality I and II:

$$0 \leq lw \leq 1$$

Complete the square on the expression we want to minimize:

$$l^2 + w^2 = (l+w)^2 - 2lw = 2^2 - 2lw = 4 - 2lw$$

This will be maximum when lw is minimum:

$$4 - 0 = 4$$

And it will be minimum when lw is maximum:

$$4 - 2lw = 4 - 2(1) = 2$$

3.22: Finding Minimum

Equality is achieved when

$$\frac{a+b}{2} \geq \sqrt{ab}$$
$$a = b$$

In the above inequality, if you are able to get a value on the RHS, it can be used to establish a lower bound for the LHS.

Example 3.23

- A. Two positive real numbers a and b have a product of 100. What is the minimum value of their sum?
- B. The area of a rectangle is 256 square meters. Find the minimum perimeter of the rectangle, in cm .

Part A

$$\frac{a+b}{2} \geq \sqrt{ab} = \sqrt{100} = 10 \Rightarrow a+b \geq 20$$

Hence, the minimum value of the sum is:

$$20. \text{ when } a = b = 10$$

Part B

$$\frac{l+w}{2} \geq \sqrt{lw} = \sqrt{256} = 16 \Rightarrow 2(l+w) = 64$$

Minimum perimeter is 64, when

$$l = w = 16$$

Example 3.24

Example 3.25

If $0 < x < \pi$, find the minimum value of $\frac{9x^2 \sin^2 x + 4}{x \sin x}$ (AIME 1983/9)

Use a change of variable. Let $y = x \sin x$:

$$\frac{9x^2 \sin^2 x + 4}{x \sin x} = \frac{9y^2 + 4}{y} = 9y + \frac{4}{y}$$

The domain gives us a clue. Both x and $\sin x$ are positive in the interval $0 < x < \pi$. So, when we take the product, the y 's will cancel.

Apply AM – GM

$$9y + \frac{4}{y} \geq 2 \sqrt{9y \times \frac{4}{y}} = 12$$

Equality holds when the two terms are equal:

$$9y = \frac{4}{y} \Rightarrow y = \frac{2}{3} \Rightarrow x \sin x = \frac{2}{3}$$

When $x = 0$

$$x = 0 \Rightarrow x \sin x = 0, \quad x = \frac{\pi}{2} \Rightarrow x \sin x = \frac{\pi}{2} \approx 1.57$$

$x \sin x$ is continuous.

Hence, it achieves $\frac{2}{3}$ between $x = 0$, and $x = \frac{\pi}{2}$

Example 3.26

Find the dimensions of a rectangle with perimeter p that maximize its area. Find that area.

Let

$$2(l + w) = p \Rightarrow w = \frac{p}{2} - l \Rightarrow \text{Area} = l\left(\frac{p}{2} - l\right)$$

By AM-GM Inequality:

$$\sqrt{l\left(\frac{p}{2} - l\right)} \leq \frac{l + \left(\frac{p}{2} - l\right)}{2} = \frac{p}{4}$$

Square both sides:

$$l\left(\frac{p}{2} - l\right) \leq \frac{p^2}{16}$$

The maximum area $\left(\frac{p^2}{16}\right)$ is achieved when:

$$l = \frac{p}{2} - l \Rightarrow l = \frac{p}{4}$$

In other words, for a given perimeter, the area of a rectangle is maximized when it is a square.

Example 3.27

If $\log_2 a + \log_2 b \geq \frac{1}{4}$, then the minimum value of $a + b$ can be written as $x^{\frac{y}{z}}$, where x is a prime number, and $\frac{y}{z}$ is an irreducible fraction. Find $x + y + z$.

Combine the logarithms using the product rule:

$$\log_2 ab \geq \frac{1}{4}$$

Convert from logarithmic form to exponential form:

$$ab \geq 2^{\frac{1}{4}}$$

From the AM-GM inequality, we know that:

$$a + b \geq 2\sqrt{ab} = 2\sqrt{2^{\frac{1}{4}}} = 2 \times 2^{\frac{1}{8}} = 2^{\frac{9}{8}}$$

Finally:

$$x + y + z = 2 + 9 + 8 = 19$$

Example 3.28

Let $x \geq 0, y \geq 0$ be real numbers with $x + y = 2$. Prove that $x^2 y^2 (x^2 + y^2) \leq 2$. (Irish Math Olympiad 2000/P2/Q1)²

Apply AM-GM on the given equality condition:

$$\sqrt{xy} \leq \frac{x + y}{2} = 1 \Rightarrow \sqrt{xy} \leq 1 \Rightarrow xy \leq 1 \Rightarrow \underbrace{(xy)^2 \leq xy \leq 1}_{\text{Inequality 1}}$$

The last inequality holds since the square of a number between 0 and 1 is less than the number (for example $0.9^2 = 0.81$).

Work with the LHS of the inequality to be proved. Complete the square:

$$LHS = x^2 y^2 (x^2 + y^2) = x^2 y^2 [(x + y)^2 - 2xy]$$

² Also see the [alternate solution](#) that does not use the AM-GM inequality earlier in this document.

Substitute $x + y = 2$, and factor:

$$= x^2 y^2 [(2)^2 - 2xy] = 2(xy)^2 [2 - xy]$$

Use Inequality I to create a chain of inequalities:

$$2(xy)^2 [2 - xy] \leq 2xy [2 - xy]$$

By AM-GM $ab \leq \left(\frac{a+b}{2}\right)^2$, which we apply to the last expression:

$$\underbrace{xy}_a \underbrace{[2 - xy]}_b \leq \left(\frac{xy + (2 - xy)}{2}\right)^2 = \left(\frac{2}{2}\right)^2 = (1)^2 = 1$$
$$2xy [2 - xy] \leq 2$$

Since $0 \leq xy \leq 1$, we must have $(xy)^2 \leq xy$:

$$2(xy)^2 [2 - xy] \leq 2$$

And finally, we conclude:

$$x^2 y^2 (x^2 + y^2) \leq 2$$

Example 3.29

The number of elements in the set below is: (JEE Main, July 29, 2022-II)

$$S = \left\{ x \in \mathbb{R} : 2 \cos\left(\frac{x^2 + x}{6}\right) = 4^x + 4^{-x} \right\}$$

Is:

$$\cos\left(\frac{x^2 + x}{6}\right) = \frac{4^x + 4^{-x}}{2}$$

By AM-GM, the RHS has minimum value 1 since:

$$\frac{4^x + 4^{-x}}{2} \geq \sqrt{4^x \cdot 4^{-x}} = \sqrt{4^{x-x}} = \sqrt{4^0} = \sqrt{1} = 1$$

This value is achieved when:

$$4^x = 4^{-x} \Rightarrow x = -x \Rightarrow x = 0$$

The LHS has maximum value 1, and hence the only solution for equation is when:

$$LHS = RHS = 1$$

The only solution that works for RHS is $x = 1$, and this also work for the LHS.

$$\cos\left(\frac{x^2 + x}{6}\right) = \cos\left(\frac{0^2 + 0}{6}\right) = \cos(0) = 1$$

Hence, there is exactly one element in S.

3.30: Cauchy Schwarz Inequality

$$\left(\sum_{i=1}^n u_i v_i\right)^2 \leq \left(\sum_{i=1}^n u_i^2\right) \left(\sum_{i=1}^n v_i^2\right)$$

- This is one of the most important and widely used inequalities in Maths.³
- It was first proved in the 19th century.

³ This inequality has a [Vector form](#), and an [integral form](#) as well.

Example 3.31: Proving the Two Variable Cauchy Schwarz⁴

- Substitute $n = 2$ in the Cauchy Schwarz equality to obtain the two-variable case of the inequality. Then make the substitution $u_1 = a, v_1 = b, u_2 = c, v_2 = d$ to make the variables easier to work with.
- Expand both sides of the inequality from Part A, and simplify, to get an equivalent inequality.
- Apply the $AM - GM$ inequality to prove Part B.

Part A

Substitute $n = 2$ to get:

$$(u_1 v_1 + u_2 v_2)^2 \leq (u_1^2 + u_2^2)(v_1^2 + v_2^2)$$

Substitute $u_1 = a, v_1 = b, u_2 = c, v_2 = d$

$$\Leftrightarrow (ab + cd)^2 \leq (a^2 + c^2)(b^2 + d^2)$$

Part B

$$\begin{aligned} \Leftrightarrow a^2 b^2 + 2abcd + c^2 d^2 &\leq a^2 b^2 + a^2 d^2 + c^2 b^2 + c^2 d^2 \\ \Leftrightarrow 2abcd &\leq a^2 d^2 + c^2 b^2 \end{aligned}$$

Part C

$$\begin{aligned} \sqrt{a^2 d^2 + c^2 b^2} &\leq \frac{a^2 d^2 + c^2 b^2}{2} \\ 2|abcd| &\leq a^2 d^2 + c^2 b^2 \\ 2abcd &\leq a^2 d^2 + c^2 b^2 \end{aligned}$$

C. AM-GM Inequality: Three Variables

3.32: AM-GM: Three Variables

For positive real numbers a, b and c :

$$\frac{a + b + c}{3} \geq \sqrt[3]{abc}$$

We assume the four-variable case here.⁵

Substitute $w = \frac{1}{3}(x + y + z)$ in $\frac{w+x+y+z}{4} \geq \sqrt[4]{wxyz}$:

$$LHS = \frac{\frac{1}{3}(x + y + z) + x + y + z}{4} = \frac{\frac{4}{3}(x + y + z)}{4} = \frac{x + y + z}{3}$$

And hence the inequality becomes:

$$\frac{x + y + z}{3} \geq \sqrt[4]{\frac{1}{3}(x + y + z)xyz}$$

Divide both sides by $\sqrt[4]{\frac{1}{3}(x + y + z)}$:

$$\left(\frac{x + y + z}{3}\right)^{\frac{3}{4}} \geq \sqrt[4]{xyz}$$

Raise both sides to the power $\frac{4}{3}$:

$$\frac{x + y + z}{3} \geq \sqrt[3]{xyz}$$

⁴ There are other ways of proving the general Cauchy Schwarz, which are not necessarily difficult. But our objective here is to *apply* the AM-GM inequality.

⁵ The four variable case is proved below.

Example 3.33

Let x , y , and z be positive real numbers such that $(x \cdot y) + z = (x + z) \cdot (y + z)$. What is the maximum possible value of xyz ? (HMMT 2005/Guts/2)

Expand the given expression on the RHS:

$$\begin{aligned}xy + z &= xy + xz + zy + z^2 \\z &= xz + zy + z^2\end{aligned}$$

Since $z \neq 0$, divide both sides by z :

$$x + y + z = 1$$

But we know that, by the three variable AM-GM Inequality:

$$\frac{x + y + z}{3} \geq \sqrt[3]{xyz}$$

Substitute $x + y + z = 1$:

$$\frac{1}{3} \geq \sqrt[3]{xyz}$$

Cube both sides:

$$\frac{1}{27} \geq xyz \Rightarrow xyz \leq \frac{1}{27} \Rightarrow \text{Max}(xyz) = \frac{1}{27}$$

Example 3.34

Let A , M , and C be nonnegative integers such that $A + M + C = 12$. What is the maximum value of $A \cdot M \cdot C + A \cdot M + M \cdot C + A \cdot C$? (AMC 12 2000/12)

Shortcut Method

Since the expression is symmetric, make the values of the variables equal:

$$A = M = C = \frac{12}{3} = 4$$

Formal Method

Note that:

$$(A + 1)(M + 1)(C + 1) = AMC + AM + MC + AC + A + M + C + 1$$

Substitute $A + M + C = 12$:

$$AMC + AM + MC + AC = (A + 1)(M + 1)(C + 1) - 13$$

Maximizing the LHS is the same as maximizing the RHS.

$$\sqrt[3]{(A + 1)(M + 1)(C + 1)} \leq \frac{(A + 1) + (M + 1) + (C + 1)}{3}$$

Cube both sides:

$$(A + 1)(M + 1)(C + 1) \leq \left(\frac{A + M + C + 3}{3}\right)^3 = \left(\frac{12 + 3}{3}\right)^3 = \left(\frac{15}{3}\right)^3 = 5^3 = 125$$

Equality is achieved when:

$$A + 1 = M + 1 = C + 1 = 5 \Rightarrow A = M = C = 4$$

3.35: Manipulating the Expression

- You may need to manipulate the expression to get into a form where AM-GM can be applied.
- This manipulation may not be obvious until you have experience with such expressions.

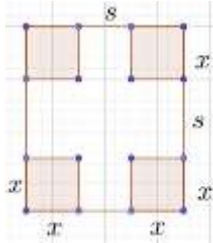
Example 3.36

A square sheet of metal of side length s has smaller squares of side length x cut at each of its corners, and the resulting “flaps” folded to form an open-top box.

- Show that the volume of the box is $V = (s - 2x)^2(x)$.
- Apply AM-GM on V .
- Apply AM-GM on $4V$. Find the value of x in terms of s .

Part A

$$V = (s - 2x)^2(x) = (s - 2x)(s - 2x)(x)$$



Part B

Apply AM-GM to the above:

$$\frac{(s - 2x) + (s - 2x) + x}{3} \geq \sqrt[3]{(s - 2x)^2(x)}$$

$$\frac{2s - 3x}{3} \geq \sqrt[3]{(s - 2x)^2(x)}$$

Now if we cube the expression, the LHS is the cube of a binomial making the expression very messy.

Part C

$$4V = (s - 2x)(s - 2x)(4x)$$

By AM-GM Inequality:

$$\frac{(s - 2x) + (s - 2x) + 4x}{3} \geq \sqrt[3]{(s - 2x)^2(4x)}$$

Simplify:

$$\frac{2s}{3} \geq \sqrt[3]{(s - 2x)^2(4x)}$$

Cube:

$$\frac{8s^3}{27} \geq (s - 2x)^2(4x)$$

Divide both sides by 4:

$$\frac{2s^3}{27} \geq (s - 2x)^2(x)$$

Hence, the maximum is:

$$\frac{2s^3}{27}$$

And this maximum is achieved when the three terms are equal:

$$s - 2x = 4x \Rightarrow x = \frac{s}{6}$$

D. AM-GM Inequality: General Case

3.37: AM-GM: Four Variables

$$\frac{w + x + y + z}{4} \geq \sqrt[4]{wxyz}$$

By AM-GM, we must have:

$$\frac{w + x}{2} \geq \sqrt{wx}, \quad \frac{y + z}{2} \geq \sqrt{yz}$$

Add the two, and then divide by two to get:

$$\frac{w + x + y + z}{4} \geq \frac{\sqrt{wx} + \sqrt{yz}}{2}$$

Inequality I

Apply AM-GM on the RHS one more time to get:

$$\frac{\sqrt{wx} + \sqrt{yz}}{2} \geq \sqrt[4]{wxyz}$$

Inequality III

And then combine Inequalities I and II to get:

$$\frac{w + x + y + z}{4} \geq \sqrt[4]{wxyz}$$

Example 3.38

Splitting terms

3.39: AM-GM: General Case

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

One way to prove the general case of the AM-GM inequality is by forward-backward induction.

- Prove the base case for $n = 2$. We did this above.
- Forward Step: Prove that if it is true for $n = k$, then it is true for $n = 2k$
- Backward: Prove that if it is true for $n = k$, then it is true for $n = k - 1$

3.3 $RMS \geq AM \geq GM \geq HM$ (In)equality

A. Relation between Means

3.40: AM, HM and GM: Two Variable Case

For two positive variables a and b :

$$GM^2 = AM \cdot HM$$

$$\begin{aligned} LHS &= GM^2 = (\sqrt{ab})^2 = ab \\ RHS &= AM \cdot HM = \frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab = LHS \end{aligned}$$

Example 3.41

The harmonic mean of two numbers is 4. Their arithmetic mean A and the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers. (JEE-A 1979)

Let the numbers be x and y . Since their harmonic mean is 4:

$$\frac{2xy}{x+y} = 4 \Rightarrow 2xy = 4x + 4y \Rightarrow \underline{xy = 2x + 2y} \quad \text{Equation I}$$

Since $2A + G^2 = 27$, we must have:

$$\frac{2(x+y)}{2} + (\sqrt{xy})^2 = 27 \Rightarrow x + y + xy = 27$$

Substitute the value of xy from Equation I in the above:

$$x + y + 2x + 2y = 27 \Rightarrow 3x + 3y = 27 \Rightarrow \underline{x + y = 9} \quad \text{Equation II}$$

$$xy = 2(x + y) = 18 \Rightarrow \underline{xy = 18} \quad \text{Equation III}$$

Solve the system of equations. Substitute $xy = 18 \Rightarrow y = \frac{18}{x}$ in Equation II to get a quadratic and solve:

$$x + \frac{18}{x} = 9 \Rightarrow x^2 - 9x + 18 = 0 \Rightarrow (x - 6)(x - 3) = 0 \Rightarrow x \in \{3, 6\}$$

Challenge 3.42

Mark all correct options

Let a, b be positive real numbers. If a, A_1, A_2, b are in arithmetic progression, a, G_1, G_2, b are in geometric progression and a, H_1, H_2, b are in harmonic progression, and let:

$$X = \frac{G_1 G_2}{H_1 H_2}, \quad Y = \frac{A_1 + A_2}{H_1 + H_2}, \quad Z = \frac{(2a + b)(a + 2b)}{9ab}$$

Then:

- A. $X = Y$
- B. $Y = Z$
- C. $X = Z$
- D. None of the above (JEE-A 2002, Adapted)

If the geometric progression is $a, G_1 = ar, G_2 = ar^2, b = ar^3$ for some common ratio r :

$$G_1 G_2 = (ar)(ar^2) = a^2 r^2 = a(ar^3) = ab$$

If the arithmetic progression is $a, A_1 = a + d, A_2 = a + 2d, b = a + 3d$ for some common difference then:

$$A_1 + A_2 = (a + d) + (a + 2d) = a + a + 3d = a + b$$

Finally, since a, H_1, H_2, b are in harmonic progression, their reciprocals are in arithmetic progression:

$$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{b} \Rightarrow \text{Common Difference} = D = \frac{1}{3} \left(\frac{1}{b} - \frac{1}{a} \right) = \frac{a - b}{3ab}$$

Then:

$$\begin{aligned} \frac{1}{H_1} &= \frac{1}{a} + D = \frac{1}{a} + \frac{a - b}{3ab} = \frac{3b + a - b}{3ab} = \frac{2b + a}{3ab} \Rightarrow H_1 = \frac{3ab}{2b + a} \\ \frac{1}{H_2} &= \frac{1}{a} + 2D = \frac{1}{a} + 2 \left(\frac{a - b}{3ab} \right) = \frac{3b + 2a - 2b}{3ab} = \frac{2a + b}{3ab} \Rightarrow H_2 = \frac{3ab}{2a + b} \end{aligned}$$

Hence:

$$\begin{aligned} H_1 H_2 &= \frac{3ab}{2b + a} \times \frac{3ab}{2a + b} = \frac{9a^2 b^2}{(2b + a)(2a + b)} \\ H_1 + H_2 &= \frac{3ab}{2b + a} + \frac{3ab}{2a + b} = \frac{(3ab)(2a + b) + (3ab)(2b + a)}{(2b + a)(2a + b)} = \frac{(9ab)(a + b)}{(2b + a)(2a + b)} \\ X &= \frac{G_1 G_2}{H_1 H_2} = \frac{ab}{\frac{9a^2 b^2}{(2b + a)(2a + b)}} = ab \cdot \frac{(2b + a)(2a + b)}{9a^2 b^2} = \frac{(2b + a)(2a + b)}{9ab} = Z \\ Y &= \frac{A_1 + A_2}{H_1 + H_2} = \frac{a + b}{\frac{(9ab)(a + b)}{(2b + a)(2a + b)}} = (a + b) \cdot \frac{(2b + a)(2a + b)}{(9ab)(a + b)} = \frac{(2b + a)(2a + b)}{9ab} = Z \end{aligned}$$

Hence:

$$X = Y = Z \Rightarrow \text{Options A, B, C}$$

B. $HM \leq GM \leq AM < RMS$ Inequality

3.43: HM-GM Inequality: Two Variable Case

$$\frac{2ab}{a + b} \leq \sqrt{ab}$$

Divide both sides by $\sqrt{ab}$:

$$\Leftrightarrow \frac{2\sqrt{ab}}{a + b} \leq 1$$

Multiply both sides by $a + b$:

$$\Leftrightarrow 2\sqrt{ab} \leq a + b$$

Square both sides:

$$\Leftrightarrow 4ab \leq a^2 + 2ab + b^2$$

Subtract $4ab$ from both sides:

$$\Leftrightarrow 0 \leq a^2 - 2ab + b^2$$

Factor the RHS:

$$\Leftrightarrow 0 \leq (a - b)^2$$

Which is true by the trivial inequality.

And since each of the above steps is reversible, the inequality above is true.

Example 3.44

Show that if the harmonic mean and the geometric mean of two numbers are equal, then the numbers are also equal.

$$\frac{2ab}{a+b} = \sqrt{ab}$$

Divide both sides by $\sqrt{ab}$:

$$\frac{2\sqrt{ab}}{a+b} = 1$$

Multiply both sides by $a+b$:

$$2\sqrt{ab} = a+b$$

Square both sides:

$$4ab = a^2 + 2ab + b^2$$

Subtract $4ab$ from both sides:

$$0 = a^2 - 2ab + b^2$$

Factor the RHS:

$$\begin{aligned} 0 &= (a-b)^2 \\ a-b &= 0 \\ a &= b \end{aligned}$$

Challenge 3.45

Choose exactly one statement per group

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, Let A_{n-1} and H_{n-1} have arithmetic, geometric, and harmonic means as A_n, G_n, H_n respectively.

Group I

- A.
- B.
- C.
- D. $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

Group II

- A.
- B.
- C.
- D. $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

Group III

- A.
- B.
- C.

D. $H_1 < H_3 < H_6 < \dots$ and $H_2 > H_4 > H_6 > \dots$ (JEE-A 2007)

Let the numbers be:

$$a \text{ and } b \Rightarrow a < H_1 < G_1 < A_1 < b$$

Group II

$A_2 = AM(H_1, A_1)$ lies between H_1 and A_1

Similarly, $A_3 = AM(H_2, A_2)$ lies between H_2 and A_2

In general:

$$A_1 > A_2 > A_3 \Rightarrow \text{Option A}$$

Group III

$H_2 = HM(H_1, A_1)$ lies between H_1 and A_1

Similarly, $H_3 = HM(H_2, A_2)$ lies between H_2 and A_2

In general:

$$H_1 < H_2 < H_3 \Rightarrow \text{Option B}$$

Group I

$$G_1^2 = ab = A_1 H_1$$

$$G_n^2 = A_n H_n$$

$$G_n^2 = A_{n-1} H_{n-1}$$

$$A_n H_n = A_{n-1} H_{n-1}$$

$$A_n H_n = A_{n-1} H_{n-1} = \dots = A_1 H_1 = ab$$

$$G_n^2 = A_n H_n = ab \Rightarrow \text{Option C}$$

3.46: Root Mean Square

The root mean square of n numbers is the square root of the arithmetic mean of the squares of the n numbers.

Example 3.47

Find

3.48: AM-RMS Inequality: Two Variable Case

$$\frac{a+b}{2} \leq \sqrt{\frac{a^2+b^2}{2}}$$

$$\begin{aligned} &\Leftrightarrow \frac{a^2 + 2ab + b^2}{4} \leq \frac{a^2 + b^2}{2} \\ &\Leftrightarrow a^2 + 2ab + b^2 \leq 2a^2 + 2b^2 \\ &\Leftrightarrow 0 \leq a^2 - 2ab + b^2 \\ &\Leftrightarrow 0 \leq (a-b)^2 \end{aligned}$$

Which is true by the trivial inequality.

And since each of the above steps is reversible, the inequality above is true.

3.49: Cascading Inequalities: $HM \leq GM \leq AM \leq RMS$

$$\frac{2ab}{a+b} \leq \sqrt{ab} \leq \frac{a+b}{2} \leq \sqrt{\frac{a^2+b^2}{2}}$$

Example 3.50

The harmonic mean of 75, 85, 65 is between

- A. 65 to 70
- B. 70 to 75
- C. 75 to 80
- D. 80 to 85

AM = middle value = 75.

HM has to be less than 75, and can be approximated to be not so far away from AM⁶.

option B

C. Minkowski Inequality

3.51: Minkowski Inequality: $n = 2, p = 2$

$$\sqrt{a_1^2 + a_2^2} + \sqrt{b_1^2 + b_2^2} \geq \sqrt{(a_1 + b_1)^2 + (a_2 + b_2)^2}$$

(Calculator) Example 3.52

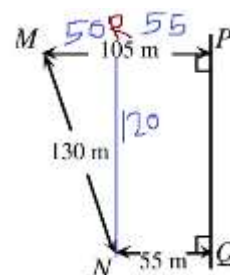
A main gas line runs through P and Q . From some point T on PQ , a supply line runs to a house at point M . A second supply line from T runs to a house at point N . What is the minimum total length of pipe required for the two supply lines? (CEMC Cayley 1999/22)

Drop a perpendicular from N to MP . Then,

$$MR = 105 - 55 = 50$$

By a Pythagorean Triplet

$$NR = 120$$



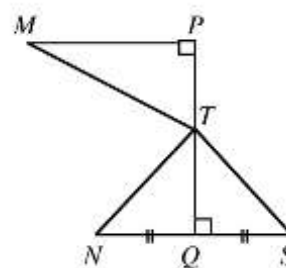
Reflect N across PQ to get S .

$$NQ = QS, \angle TQN = \angle TQS \Rightarrow \triangle TQN \cong \triangle TQS \Rightarrow TN = TS$$

We need to minimize

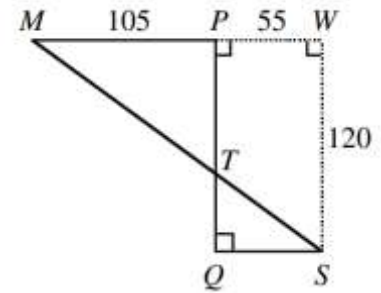
$$MT + TN = MT + TS$$

Which will get minimized when M, T and S are collinear.



⁶ Actual value of $HM(65, 75, 85) = \frac{3}{\frac{1}{65} + \frac{1}{75} + \frac{1}{85}} = 74 \frac{71}{671}$ is not a good expression to work with.

$$MS = \sqrt{(105 + 55)^2 + 120^2} = 200$$



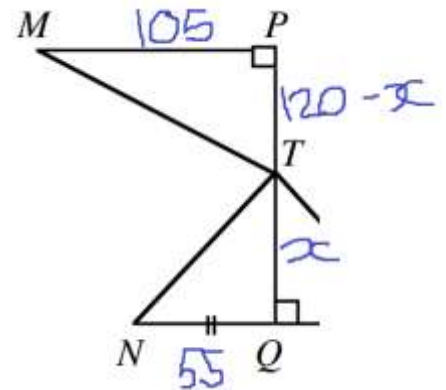
Example 3.53

A main gas line runs through P and Q . From some point T on PQ , a supply line runs to a house at point M . A second supply line from T runs to a house at point N . What is the minimum total length of pipe required for the two supply lines? (CEMC Cayley 1999/22)

$$\text{Pipe Length} = NT + MT = \sqrt{55^2 + x^2} + \sqrt{105^2 + (120 - x)^2}$$

$$\sqrt{a_1^2 + a_2^2} + \sqrt{b_1^2 + b_2^2} \geq \sqrt{(a_1 + b_1)^2 + (a_2 + b_2)^2}$$

$$\begin{aligned} \sqrt{(a_1 + b_1)^2 + (a_2 + b_2)^2} &= \sqrt{(55 + 105)^2 + (x + 120 - x)^2} \\ &= \sqrt{160^2 + 120^2} = 200 \end{aligned}$$



3.54: Minkowski Inequality: $p = 2$

$$\sqrt{\sum_{i=1}^n a_i^2} + \sqrt{\sum_{i=1}^n b_i^2} \geq \sqrt{\sum_{i=1}^n (a_i + b_i)^2}$$

3.55: Minkowski Inequality: General Case

$$\sqrt[p]{\sum_{i=1}^n a_i^p} + \sqrt[p]{\sum_{i=1}^n b_i^p} \geq \sqrt[p]{\sum_{i=1}^n (a_i + b_i)^p}$$

3.4 Jensen's Inequality

A. Statement

3.56: Jensen's Inequality

If $f(x)$ is a twice differentiable function that is concave down over an interval then for points $x_1, x_2, \dots, x_n$ in that interval:

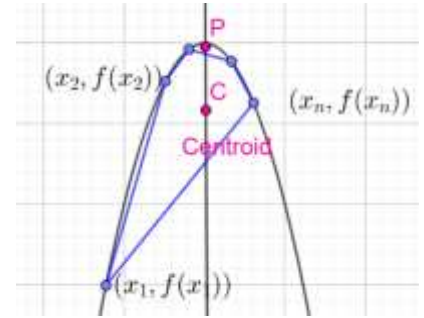
$$f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \geq \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}$$

Consider n points on a function with coordinates:

$$(x_1, f(x_1)), (x_2, f(x_2)), \dots, (x_n, f(x_n))$$

These points combine to form an n -sided polygon with centroid given by the average values of the x and the y coordinates:

$$\text{Centroid} = C = \left(\frac{x_1 + x_2 + \dots + x_n}{n}, \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n} \right)$$



In a concave down function, the line joining any two points lies strictly below that function.

Hence, the n points chosen, when connected, form a polygon, which lies completely below the function.

Hence, the centroid lies below the function.

The point on the function that lies exactly above the centroid has the same x coordinate but different y coordinate. Specifically, the y coordinate is given by the function evaluated at the x coordinate:

$$P = \left(\frac{x_1 + x_2 + \dots + x_n}{n}, f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \right)$$

From the diagram, the y coordinate of point P is greater than the y coordinate of point C .

$$f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \geq \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}$$

3.57: Jensen's Inequality: Expectation Version

$$f(\mathbb{E}(x_i)) \geq \mathbb{E}(f(x_i))$$

$$f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) = f(\mathbb{E}(x_i))$$

$$\frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n} = \mathbb{E}(f(x_i))$$

B. Products

3.58: Using Logs

You can use the log function to convert a product into a sum to then apply Jensen's Inequality

Example 3.59

If $A + B + C = \pi$, then find the maximum value of $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.

$$f(x) = \log\left(\sin \frac{x}{2}\right) \Rightarrow f' = \frac{1}{2} \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} = \frac{1}{2} \cot \frac{x}{2} \Rightarrow f'' = -\frac{1}{4} \csc^2 \frac{x}{2}$$

Note that $\csc^2 \frac{x}{2}$ is a perfect square and hence positive. This means that:

$$\csc^2 \frac{x}{2} = \left(\csc \frac{x}{2} \right)^2 > 0 \Rightarrow f'' = -\frac{1}{4} \csc^2 \frac{x}{2} < 0 \Rightarrow f \text{ is concave}$$

Hence, by Jensen's Inequality:

$$\frac{f\left(\frac{A}{2}\right) + f\left(\frac{B}{2}\right) + f\left(\frac{C}{2}\right)}{3} \leq f\left(\frac{\frac{A}{2} + \frac{B}{2} + \frac{C}{2}}{3}\right)$$

Apply $f(x) = \log\left(\sin \frac{x}{2}\right)$:

$$\frac{\log \sin\left(\frac{A}{2}\right) + \log \sin\left(\frac{B}{2}\right) + \log \sin\left(\frac{C}{2}\right)}{3} \leq \log\left(\sin \frac{A+B+C}{6}\right)$$

Combine the LHS using $\log a + \log b + \log c = \log abc$, and multiply both sides by 3:

$$\log\left(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}\right) \leq 3 \log\left(\sin \frac{\pi}{6}\right)$$

Since $RHS = 3 \log\left(\sin\left(\frac{\pi}{6}\right)\right) = 3 \log \frac{1}{2} = \log \frac{1}{8}$:

$$\log\left(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}\right) \leq \log \frac{1}{8}$$

Taking anti-logs gives us the maximum value:

$$\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$$

3.5 Further Topics

60 Examples